

BitFlip

CloudSherpa

Software Architecture Specifications

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1 | Introduction

The CloudSherpa Software Architecture Specification elaborates on the architectural and other technical decisions made throughout the development process. Its purpose is to serve as a reference for CloudSherpa's architectural direction. The document is intended for both developers and stakeholders and serves as a mechanism for maintaining alignment between them.

2 | Architectural Requirements

Terminology Note

In the source-code repository, the **Application Service** is implemented by the application named *service*. To distinguish this specific application from the general use of the term service within a service-oriented architecture, it is referred to throughout this document as the Application Service.

2.1 Architectural Patterns

2.1.1 Service Oriented Architecture

CloudSherpa follows the Service Oriented Architecture (SOA) pattern to promote decoupling between the subsystems responsible for cloud data ingestion, the subsystems responsible for processing and facilitating the visualization of this data and the subsystems used to perform intelligence operations. The CloudSherpa architecture identifies three distinct services:

- **Ingestion Service**
The ingestion service is responsible for connecting to the respective cloud provider interfaces for obtaining data from deployments.
- **Application Service**
The Service Service is responsible for the business logic of CloudSherpa, ex-

cutting tasks like data aggregation, authentication and communicating intelligence insights to the Frontend Service.

- **AI/ML Service**

The AI/ML Service is responsible for tasks related to CloudSherpa intelligence, such as Cost and Usage forecasting as well as optimization recommendations.

As part of our architecture we have a Dashboard presentation component that consumes the result of CloudSherpa processing by ways of the Application Service.

2.2 Architectural Diagram

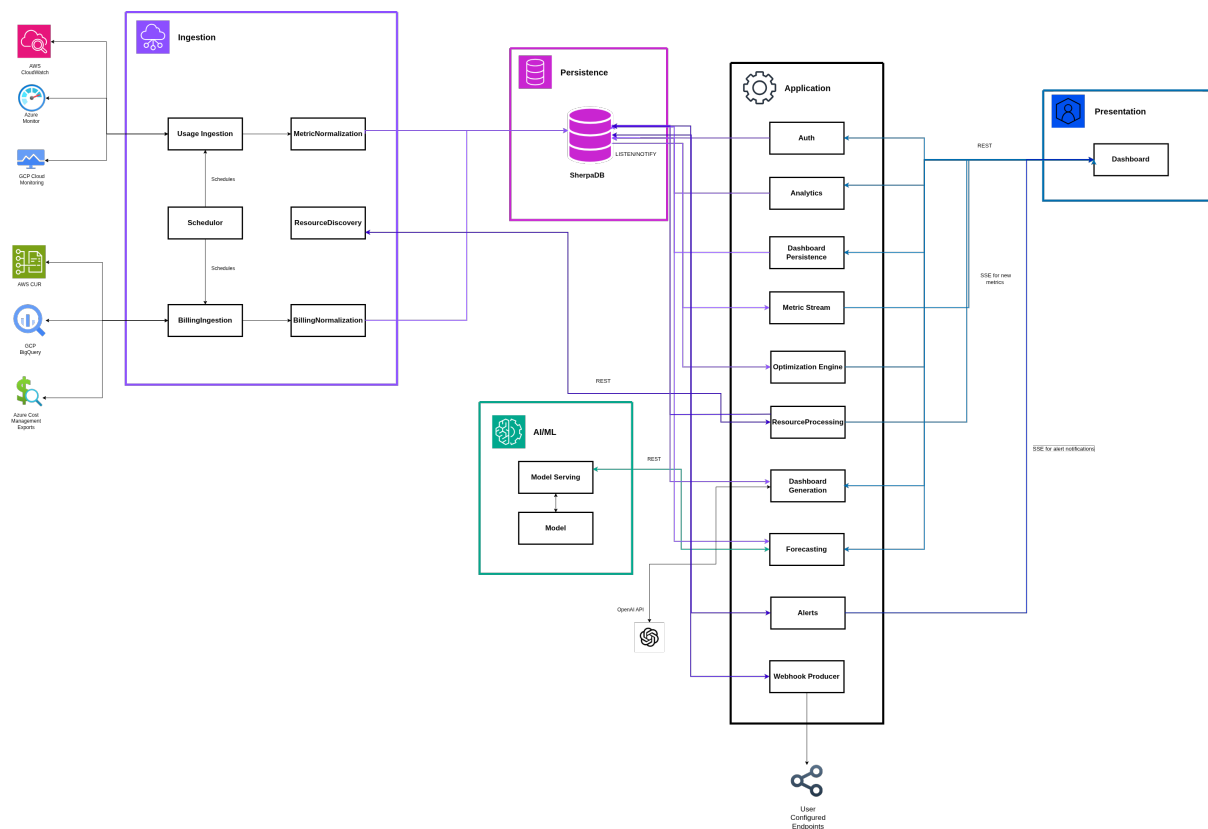


Figure 1: Architectural Diagram

2.3 Entity Relational Diagram

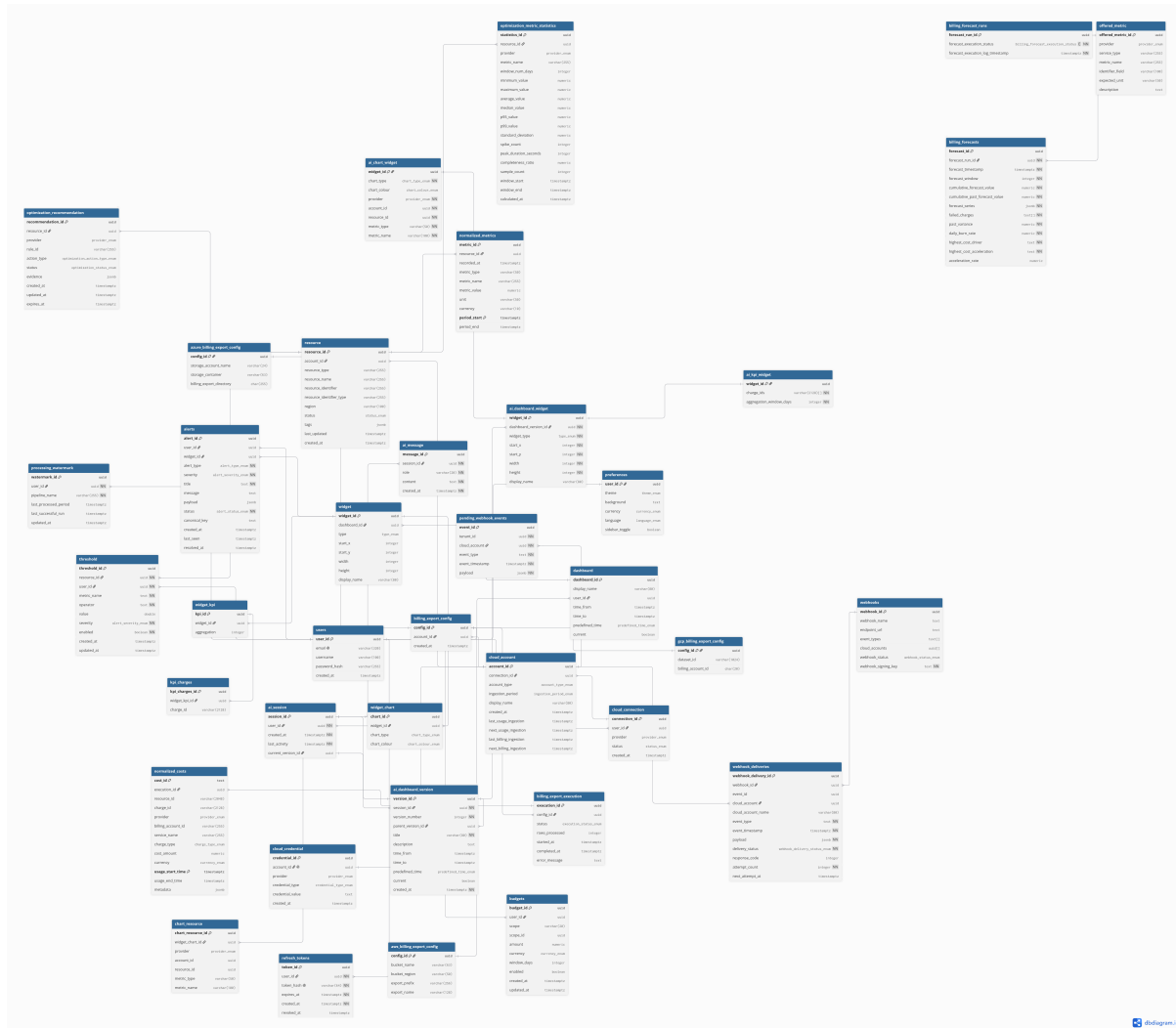


Figure 2: SherpaDB ERD

2.4 Design Patterns

2.4.1 Pipeline Design Pattern for AWS CUR Billing Ingestion

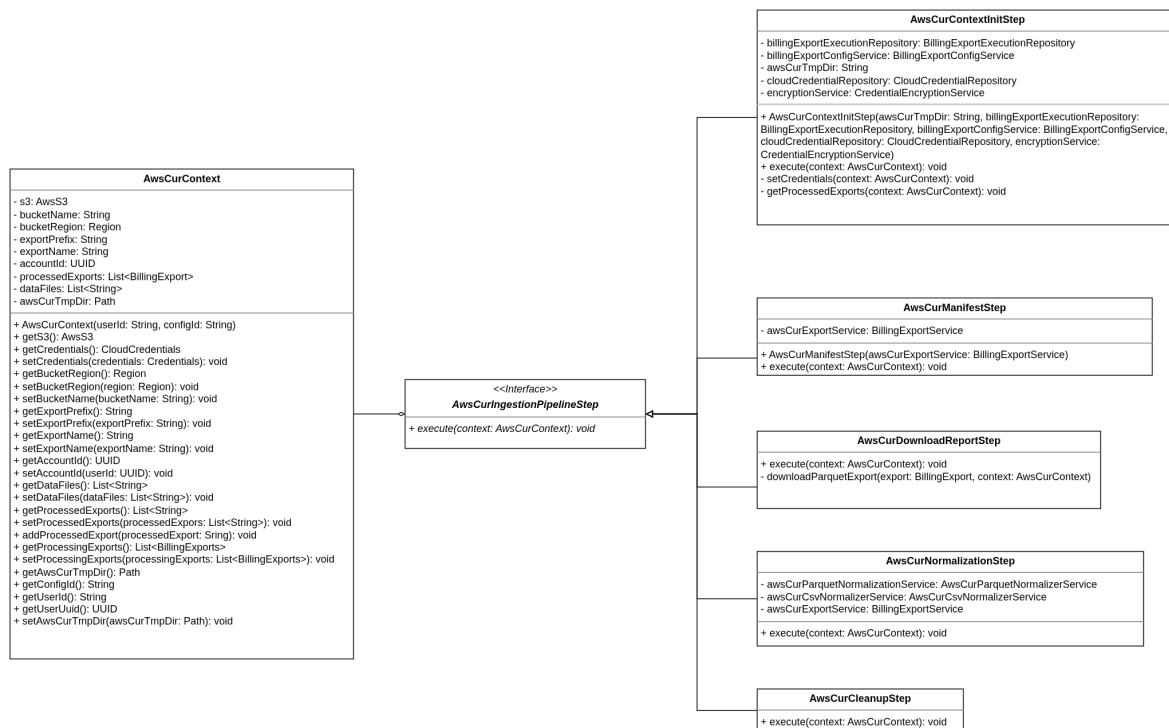


Figure 3: Pipeline Design Pattern For Billing Ingestion

Ingesting reports from a third party, such as a cloud provider, can introduce maintainability challenges. If the provider changes the API used to retrieve a report or modifies the report structure, the ingestion logic must be updated accordingly. Because the nature of future changes cannot be predicted, the ingestion process was implemented using the pipeline design pattern, which divides the process into a sequence of decoupled steps.

The pattern is reflected by the `AwsCurContext` object, which is passed through the pipeline and modified by each implementation of the `AwsCurIngestionPipelineStep` interface. Each step is responsible for a distinct part of the ingestion process.

This design improves maintainability by allowing individual steps to be modified, replaced, or extended without requiring substantial changes to the rest of the ingestion process. The modular structure also makes the billing ingestion process easier to test

and debug.

Note: Not all callers, associations, and aggregations are shown in the diagram.

2.4.2 Builder Design Pattern for Normalized Metrics

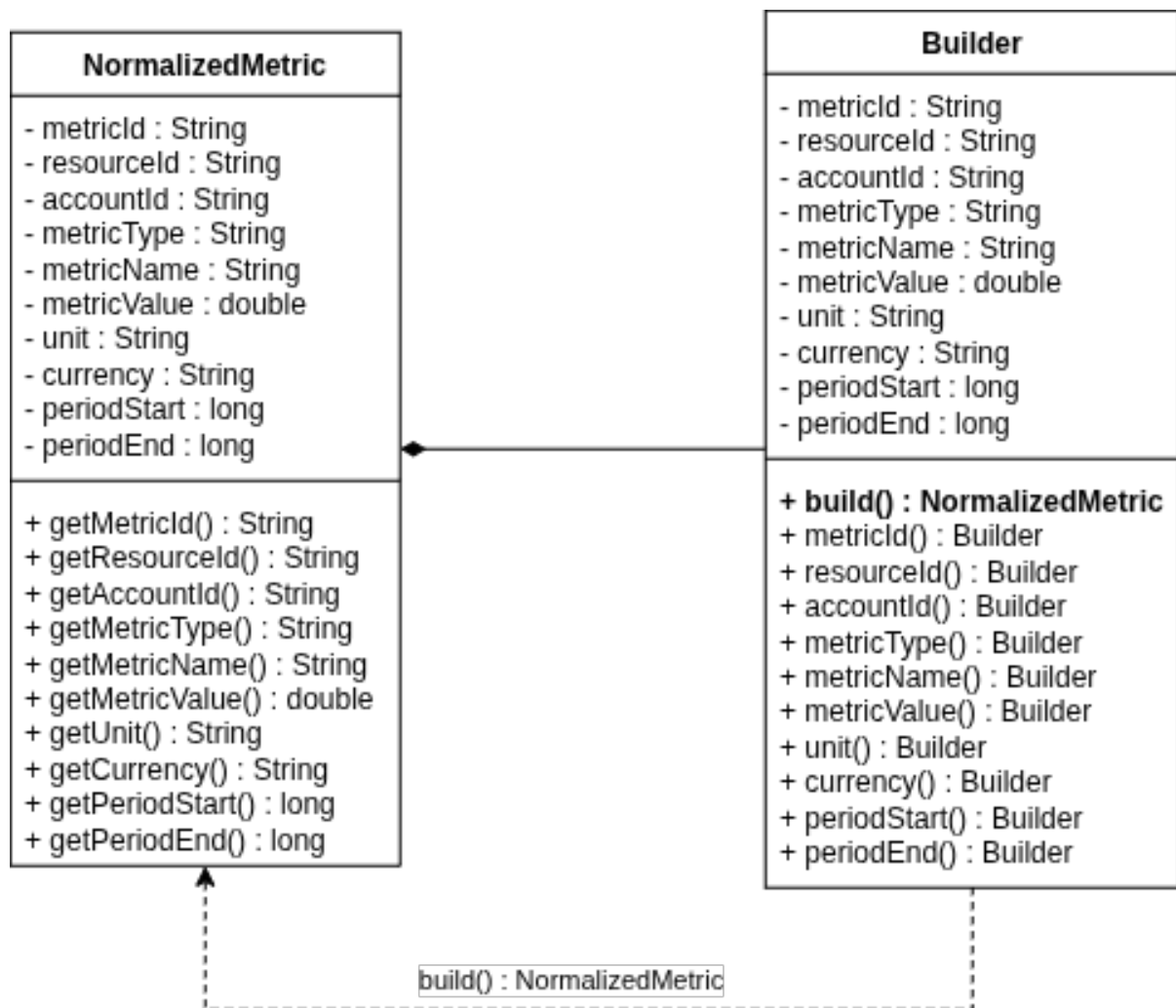


Figure 4: Builder Design Pattern for Normalized Metrics

The `NormalizedMetric` class uses a static nested `Builder` class to create objects using an interface. Each builder method sets a value and returns the builder, allowing methods to be chained together.

The `build()` method creates the final `NormalizedMetric` object using a private constructor. This makes the code easier to read and avoids the need for multiple constructors with different parameter combinations.

2.4.3 Strategy Design Pattern for Resource Discovery

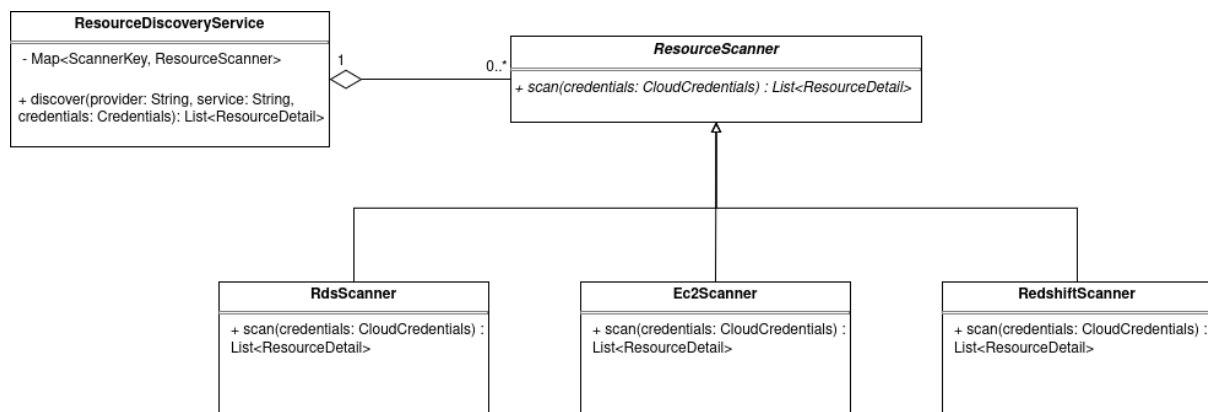


Figure 5: Strategy Design Pattern for Resource Discovery

The `ResourceDiscoveryService` class uses an interface class `ResourceScanner` to treat differing implementations of the `scan` method implemented by the concrete classes such as `Ec2Scanner` uniformly. This allows loose code coupling by avoiding case statements for each type of resource scanner based on the API request parameters.

2.4.4 Registry Design Pattern for scanner selection

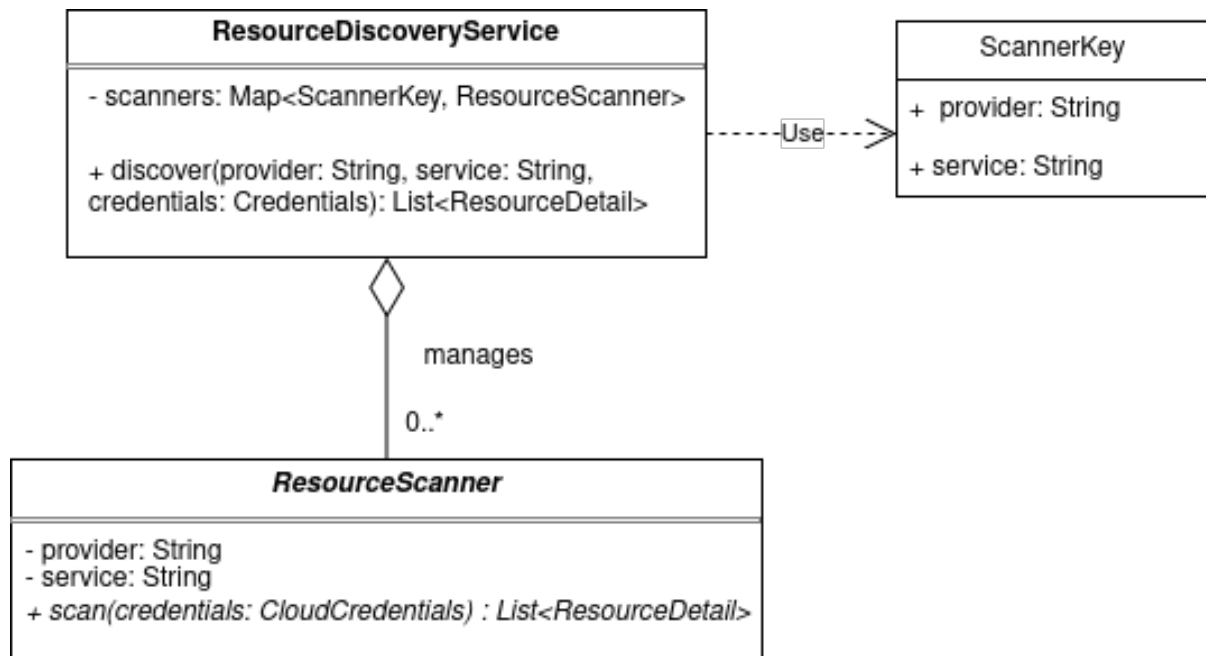


Figure 6: Registry Design Pattern for scanner selection

The `ResourceDiscoveryService` class acts as a registry, containing `<ScannerKey, ResourceScanner>` pairs which allows runtime selection of the Concrete class implementations of `ResourceScanner` interface. This is further enhanced by Springboot which automatically finds `ResourceScanner` concrete classes and adds them to the `ResourceDiscoveryService` scanners, allowing loose code coupling by allowing implementation of new scanners and services without modifying any existing files.

2.5 Constraints

2.5.1 Supported Cloud Providers

The system shall support AWS, Azure and GCP as cloud providers. Cloud providers not listed are outside of the scope of CloudSherpa.

2.5.2 Secure handling of cloud credentials

The system must follow the principle of least privilege when monitoring deployments across cloud providers.

2.5.3 Single Role System (No RBAC)

CloudSherpa does not make provision for role-based users. The developer focus is to be able to deliver a refined system that achieves its primary goal well, which is to serve as a FinOps platform to improve the visibility of and provide insights to cloud spend.

2.5.4 Use Provider APIs Instead of Pushing Data from User Infrastructure

To minimize the operational complexity and security risks associated with collecting usage metrics from CloudSherpa users, cloud provider APIs are used to retrieve data through dedicated IAM or system users, rather than relying on software installed on users' compute instances to push the data.

2.5.5 Cost & Usage Latency

Due to the architectural decision to use cloud provider APIs rather than pushing data from user infrastructure, the frequency and recency of available usage information are subject to provider-specific delays.

A similar constraint applies to billing ingestion. However, unlike usage metrics, billing data cannot be obtained by pushing information directly from user infrastructure.

2.6 Mapping Quality Requirements to Architectural Decisions

Table 1: Mapping Quality Requirements to Architectural Decisions

Quality Requirement (from SRS)	Architectural Decision
NFR1: Performance	
NFR1.1: The system shall retrieve historical metrics for visualization with a p95 request latency of less than 500 ms to ensure a responsive dashboard.	Spread the load of fetching historical metrics by using concurrent reads. Reduce the load of each historical metric series, in particular reducing message overhead by using a downsampling algorithm that reduces the amount of points that need to be included in the message while preserving visual integrity for dashboard visualization.

Quality Requirement (from SRS)	Architectural Decision
NFR1.2: After the initial billing forecast, the system shall retrieve subsequent forecasts within a 30-minute period with a p95 request latency of less than 200 ms to ensure responsive billing forecast pages.	Reuse the expensive forecast operation results by implementing a cache with a TTL of 1 hour. Since billing export ingestion are scheduled for every 6 hours, the potential one hour consistency consequence is accepted as a tradeoff due to how expensive the forecasting operation is.
NFR1.3: The usage ingestion pipeline shall normalize and persist cloud usage metric records at a throughput of at least 1000 records per second under normal operating conditions.	The architecture allows the ingestion service, which is responsible for processing and normalizing cloud billing and usage records, to be scaled independently according to ingestion volume and processing demand, hence improving performance by allocating sufficient resources. The scheduling subsystem allocates background jobs via JobRunr, which manages background asynchronous jobs, allowing multiple usage and ingestion jobs to be processed asynchronously. Our tenant per schema database allows those records to also be written to the database without conflict, due to the fact that separate tables are used for each user's usage and billing metrics. Therefore writes to one users' metric tables do not conflict with writes to other user's metric ingestion.
NFR1.4: The billing ingestion pipeline shall normalize and persist cloud billing records at a throughput of at least 1000 records per second under normal operating conditions.	The architecture allows the ingestion service, which is responsible for processing and normalizing cloud billing and usage records, to be scaled independently according to ingestion volume and processing demand, hence improving performance by scaling up resources. Spread the load by utilizing batch writes and set the tenant context once per multiple writes as opposed to on every write.
NFR2: Scalability	

Quality Requirement (from SRS)	Architectural Decision
NFR2.1: The system should successfully support 100 concurrent users, representing a stress-level load for CloudSherpa, for a period of 1 hour without experiencing more than a 10% degradation in the response times of the below endpoints compared to regular load.	
NFR2.1.1: Historical metric fetch response-time degradation	Data ingestion, processing, and normalization are performed by the independently deployable ingestion service, while dashboard requests are handled by the service layer. The service layer retrieves and aggregates historical data into a form optimized for dashboard rendering (<i>NFR 1.1.</i>), reducing the amount of data transferred and processed for each user request. Because the ingestion and user-facing services are decoupled and operate in isolated execution environments, increases in the number of connected cloud accounts primarily affect the ingestion service rather than the dashboard request path. Each service can therefore be scaled independently according to its workload, helping the system support additional cloud connections and concurrent dashboard users without a significant degradation in response time.
NFR2.1.2: Billing forecasting response-time degradation	Reduce the amount of work completed to limit response-time degradation, the decision made for <i>NFR 1.2</i> applies for <i>NFR 2.1.2</i> as well.
NFR2.1.3: Usage forecasting response-time degradation	Queue incoming forecasting requests to distribute the workload over time and prevent excessive VRAM pressure under concurrent load. (<i>Pending implementation</i>)
NFR3: Security	

Quality Requirement (from SRS)	Architectural Decision
NFR3.1: All sensitive user data and stored cloud provider credentials must be encrypted using TLS 1.3 in transit, and the system must enforce Multi-Factor Authentication (MFA) on 100% of accounts.	All external requests pass through a single controlled entry point before reaching the system's internal services. TLS 1.3 is enforced at this boundary, ensuring that communication between the dashboard and backend services is encrypted in transit.
NFR3.2: The system shall utilize secure, least-privilege identity access management (IAM) strategies, ensuring that CloudSherpa holds read-only access to client AWS, Azure, and GCP billing APIs and ensuring that none of CloudSherpa's requests require write permissions.	Third-party integrations are confined to a single service, allowing them to be implemented through consistent interfaces, governed by provider-specific security configurations, and monitored and audited from a central point. Permissions requests for IAM users include only necessary read-only permissions required for CloudSherpa to successfully discover and monitor selected services as authorized by the user.
NFR3.3: User metrics must remain separated such that no user gains access to resource details or cost and metric information from accounts not belonging to that user, as this information is potentially highly sensitive to individuals or corporations.	The architecture of our database, including our schema-per-tenant tables as well as strict session management and API authorization measures ensure that no user is capable of accessing any tenant specific information, including cost and resource metrics, not belonging to their accounts.
NFR3.4: The system shall have no known high or critical vulnerabilities in third-party dependencies on the main branch.	Automated vulnerability scanners running in the Continuous Integration pipeline
NFR4: Reliability	

Quality Requirement (from SRS)	Architectural Decision
NFR4.1: In the event of a third-party cloud provider API failure or timeout (e.g., an AWS CloudWatch outage), the ingestion pipeline shall automatically retry the connection and keep track of the failure period, ensuring 0% duplication or metric loss once the provider is back online.	The CloudSherpa ingestion scheduler uses a recurring database poll to find accounts due for ingestion, using database fields to keep track of the last successful ingestion as well as the next scheduled events. JobRunr background jobs are configured to retry requests three times upon failure, and database success fields are only updated upon success, ensuring that upon subsequent scans outstanding jobs will be handled when the external provider is back online.
NFR5: Maintainability	
NFR5.1: The codebase must maintain a minimum of 80% automated test coverage across all core components.	The modularity of our architecture improves the ease and pace at which tests can be developed, as large parts of our system remain loosely coupled. Separate CI workflows ensure that tests can be run for subsystems selectively.
NFR5.2: Static Analysis Should show zero maintainability issues for CloudSherpa services	Well defined domain boundaries ensure that code smells are isolated to modular components.
NFR6: Usability	
NFR6.2: The CloudSherpa dashboard shall render correctly on modern web browsers to ensure plain-language summaries remain highly accessible.	CloudSherpa was built using modern standards-compliant web technologies using features that are supported by all modern web browsers. Responsive CSS and the use of high quality component libraries such as Apache ECharts and Shadcn ensure that components render correctly on differing screen sizes and resolutions, and lighthouse reports are routinely run to ensure performance and accessibility. Playwright end-to-end tests verify that the dashboards functions on multiple web browsers.
NFR7: Availability	

Quality Requirement (from SRS)	Architectural Decision
NFR7.1: The dashboard shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure. If one service degrades or becomes unavailable, the remaining services can continue operating independently. The health of the services are continuously monitored during the availability periods.
NFR7.2: The application service, responsible for all functionality excluding ingestion and forecasting, shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure. If one service degrades or becomes unavailable, the remaining services can continue operating independently. The health of the services are continuously monitored during the availability periods.
NFR7.3: The ingestion service, responsible for ingesting cloud usage and billing data, shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure. If one service degrades or becomes unavailable, the remaining services can continue operating independently. The health of the services are continuously monitored during the availability periods.
NFR7.4: The intelligence service, responsible for forecasting functionality, shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure, however the intelligence service remains unaffected during database degradation. If one service degrades or becomes unavailable, the remaining services can continue operating independently.

2.7 NFR Traceability Matrix

Table 2: NFR Traceability Matrix

ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR1.1	Historical metric retrieval p95 latency at dashboard load.	Reduce load with efficient processing via LTTB downsampling	Grafana k6	< 1000ms / 272.2ms
NFR1.2	Usage ingestion throughput under normal load.	Asynchronous ingestion	Spring Boot Integration Test	>= 100 records/s / 333 records/s
NFR1.3	Billing ingestion throughput under normal load	Efficient processing via batching	Spring Boot Integration Test	pass (>= 1000 records/s) / GCP: pass, AWS: pass, Azure: pass,
NFR2.1: 50 concurrent users for 5 minutes with $\leq 10\%$ response-time degradation				
NFR2.1.1	Historical metric fetch response-time degradation	Reduce load with efficient processing via LTTB downsampling	Grafana k6	$\leq 10\%$ / 1245.90%
NFR2.1.2	Usage forecasting response-time degradation	No particular tactic, validated by test	Grafana k6	$\leq 10\%$ / 846.71%
NFR3.1	TLS 1.3 enforced at external ingress.	Encrypted client-to-ingress communication	openssl connect	TLS 1.3 / TLS 1.3, TLS 1.2 (fallback)
NFR3.2	Read-only cloud provider access	Least-privilege IAM policies	Read-only subsystems deny write	write action denied / TBD
NFR3.3	Cross-tenant data isolation	Schema-per-tenant isolation with authenticated access control	Spring Boot Integration Test	pass / pass

ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR3.4	High/critical dependency vulnerabilities	Automated dependency scanning and updates	GitHub dependabot, NPM audit	0 reported high or critical vulnerabilities / dependabot: 0, npm audit: 0
NFR4.1	Cloud provider API failure recovery without metric loss or duplication.	Retry failed interval with backoff after three failed attempts. Continuous retry and interval extension to cover failure period whilst limiting external API hits	Spring Boot Test	pass / pass
NFR5.1	Automated test coverage across core components	Automated testing enforced in CI	SonarQube & coverage reports	≥ 80% / Application: 46.10%, Ingestion: 71.70%, Intelligence: 73.70%
NFR5.2	Static-analysis maintainability issues	Automated static analysis in CI	SonarQube	0 issues / Ingestion: 0, Application: 45, Intelligence: 0, Dashboard: 2
NFR6.2	Cross-browser dashboard compatibility	Standards-compliant responsive web interface	Playwright E2E tests & Supported browsers pass	pass / pass
NFR7: Minimum 95% availability during the defined availability period				
NFR7.1	Dashboard availability	Independent service deployment and health monitoring	Uptime Kuma	≥ 95% / 99.57%
NFR7.2	Application service availability	Independent service deployment and health monitoring	Uptime Kuma	≥ 95% / 99.93%

ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR7.3	Ingestion service availability	Independent service deployment and health monitoring	Uptime Kuma	$\geq 95\%$ / 99.93%
NFR7.4	Intelligence service availability	Independent service deployment and health monitoring	Uptime Kuma	$\geq 95\%$ / 98.82%

2.7.1 NFR Testing Evidence

NFR1.1: Historical Metric Retrieval Latency

Test can be located in the repo at `nfr-tests/historical-metric/historical-metric-load.js` and is executed in the CloudSherpa Build and Deploy Github Actions workflow. Below is an example of a successful and unsuccessful run

```

1  Run grafana/run-k6-action@de51a7390bdf9ac85a3bef493691bd71d4c7c158
20 Validating test run files.
21 Found 1 valid k6 tests out of total 1 test files.
22 1. /home/runner/work/CloudSherpa/CloudSherpa/nfr-tests/historical-metric/historical-metric-load.js
23 Running test: k6 run --address= /home/runner/work/CloudSherpa/CloudSherpa/nfr-tests/historical-metric/historical-metric-load.js
24
25
26 THRESHOLDS
27
28 http_req_duration
29 ✓ 'p(95)<1000' p(95)=566.63ms
30
31 http_req_failed
32 ✓ 'rate<0.01' rate=0.00%
33
34
35 TOTAL RESULTS
36
37 checks_total.....: 54560 181.25937/s
38 checks_succeeded...: 100.00% 54560 out of 54560
39 checks_failed.....: 0.00% 0 out of 54560
40
41 ✓ status is 200
42 ✓ datapoints returned
43
44 HTTP
45 http_req_duration.....: avg=328.43ms min=21.16ms med=352.76ms max=1.55s p(90)=482.11ms p(95)=566.63ms
46 { expected_response:true }....: avg=328.43ms min=21.16ms med=352.76ms max=1.55s p(90)=482.11ms p(95)=566.63ms
47 http_req_failed.....: 0.00% 0 out of 27281
48 http_reqs.....: 27281 90.63907/s
49
50 EXECUTION
51 iteration_duration.....: avg=1.77s min=1.1s med=1.83s max=3.43s p(90)=2.09s p(95)=2.19s
52 iterations.....: 2728 9.062968/s
53 vus.....: 1 min=1 max=20
54 vus_max.....: 20 min=20 max=20
55

```

Figure 7: CloudSherpa Build and Deploy: Historical Metric Fetch Load Test

NFR1.2: Usage Ingestion Throughput The Usage Ingestion throughput for all sup-

ported providers (AWS, GCP and Azure) is measured via a Spring Boot integration test. It is not viable to perform an E2E test with connections established with the providers due to the financial liability it introduces, hence third party interactions are mocked while preserving the integrity of the test by populating the same provider SDK data structures used during regular ingestion.

The NFR tests can be found in the CloudSherpa repository under the following directory: `apps/ingestion/src/test/java/com/cloudsherpa/ingestion/nfr`

The tests are run during the Continuous Integration (CI) validation steps. The below figures are examples taken from Github Actions workflow runs.

```
895 2026-09-29T16:33:27.307Z INFO 2893 --- [ingestion] [          main] c.c.i.nfr.AwsUsageRecordIngestionTest :
896 Duration: 29.942634211s
897 AWS usage records processed: 10000
898 Records per second: 333.97195215130097
```

Figure 8: AWS Usage Throughput NFR Enforcement

```
1129 2026-09-29T16:34:20.535Z INFO 2893 --- [ingestion] [          main] c.c.i.nfr.GcpUsageRecordIngestionTest :
1130 Duration: 24.925265659s
1131 Gcp usage records processed: 10000
1132 Records per second: 401.1993347155843
```

Figure 9: GCP Usage Throughput NFR Enforcement

```
1221 2026-09-29T16:34:50.766Z INFO 2893 --- [ingestion] [          main] c.c.i.nfr.AzureUsageRecordIngestionTest :
1222 Duration: 24.815848002s
1223 Azure usage records processed: 10000
1224 Records per second: 402.96829667856053
```

Figure 10: Azure Usage Throughput NFR Enforcement

NFR1.3: Billing Ingestion Throughput

The Billing Ingestion throughput for all supported providers (AWS, GCP and Azure) is measured via a Spring Boot integration test. It is not viable to perform an E2E test with connections established with the providers due to the financial liability it introduces, hence third party interactions are mocked while preserving the integrity of the test by populating the same provider SDK data structures used during regular ingestion.

The NFR tests can be found in the CloudSherpa repository under the following directory: `apps/ingestion/src/test/java/com/cloudsherpa/ingestion/nfr`

The tests are run during the Continuous Integration (CI) validation steps. The below figures are examples taken from Github Actions workflow runs.

```
899 2026-09-28T20:56:33.512Z INFO 2836 --- [ingestion] [      main] c.c.i.nfr.AwsBillingRecordIngestionTest :
900 Duration: 9.397397195s
901 Records processed: 10000
902 Records per second: 1064.1244370644056
```

Figure 11: AWS Billing Throughput NFR Enforcement

```
818 2026-09-28T20:56:11.739Z WARN 2836 --- [ingestion] [      main] o.s.core.events.SpringDocAppInitializer : Springdoc /swagger-ui.html endpoint is e
production, set the property 'springdoc.swagger-ui.enabled=false'
819 2026-09-28T20:56:11.855Z INFO 2836 --- [ingestion] [      main] c.c.i.b.p.g.b.p.GcpBillingInitStep : GCP Billing Ingestion Dev Mode enabled
820 2026-09-28T20:56:18.880Z INFO 2836 --- [ingestion] [      main] c.c.i.nfr.GcpBillingRecordIngestionTest :
821 Duration: 7.025739696s
822 Records processed: 10000
823 Records per second: 1423.3376744221468
```

Figure 12: GCP Billing Throughput NFR Enforcement

```
974 2026-09-28T20:56:43.613Z INFO 2836 --- [ingestion] [      main] c.c.i.n.AzureBillingRecordIngestionTest :
975 Duration: 5.307251999s
976 Records processed: 10000
977 Records per second: 1884.214279232306
```

Figure 13: Azure Billing Throughput NFR Enforcement

NFR2.1.1: Historical Metric Fetch Response-Time Degradation

The k6 scalability test is run during the CI validation steps. It is not enforced since the requirement is not met, but it does continuously record the degradation as changes are made.

```
758 time="2026-09-27T22:03:12Z" level=info msg="Baseline p95: 184.52872884999985 ms" source=console
759 time="2026-09-27T22:03:12Z" level=info msg="Load p95: 2483.5645644999995 ms" source=console
760 time="2026-09-27T22:03:12Z" level=info msg="Degradation: 1245.90%" source=console
761
```

Figure 14: Example output of the Metric Fetch Degradation log in the CloudSherpa Build and Deploy workflow

NFR2.1.2: Usage Forecasting Response-Time Degradation

The k6 scalability test is run during the CI validation steps. It is not enforced since the requirement is not met, but it does continuously record the degradation as changes are made.

```
804 time="2026-09-27T22:09:39Z" level=info msg="Baseline p95: 3169.516564999999 ms" source=console
805 time="2026-09-27T22:09:39Z" level=info msg="Load p95: 30006.1423045 ms" source=console
806 time="2026-09-27T22:09:39Z" level=info msg="Degradation: 846.71%" source=console
```

Figure 15: Example output of the Usage Forecasting Degradation log in the CloudSherpa Build and Deploy workflow

NFR3.1: TLS at External Ingress

A TLS validation script is run against the target deployments after a deployment succeeded to provide feedback on whether the requirement is met.

```
✓ Verify TLS configuration after deployment
1 ▶ Run chmod +x nfr-tests/tls-test.sh
12 curl: (28) Failed to connect to cloudsherpa.gjjcs.org port 80 after 10002 ms: Timeout was reached
13 TLS test succeeded
```

Figure 16: Successful TLS validation after successful deployment

NFR3.2: Read-Only Cloud Provider Access

Evidence for NFR3.2 goes here.

NFR3.3: Cross-Tenant Data Isolation

```
[INFO] Running com.cloudsherpa.service.unit.SchemaMultiTenantConnectionProviderTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.187 s -- in com.cloudsherpa.service.unit.SchemaMultiTenantConnectionProviderTest
```

Figure 17: Successful integration testing results

NFR3.4: Dependency Vulnerabilities

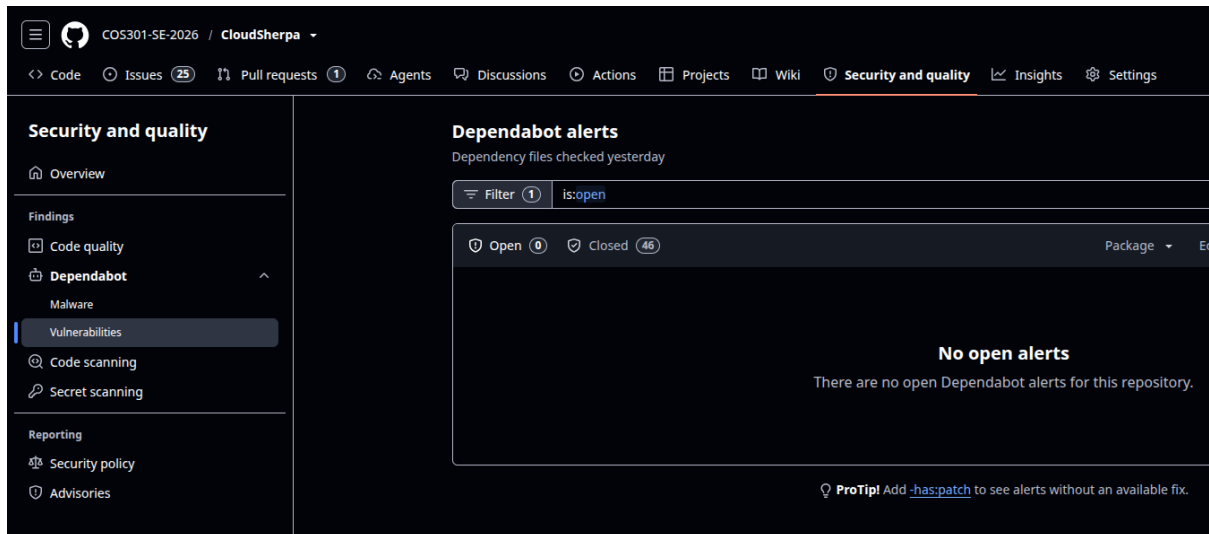


Figure 18: Dependabot

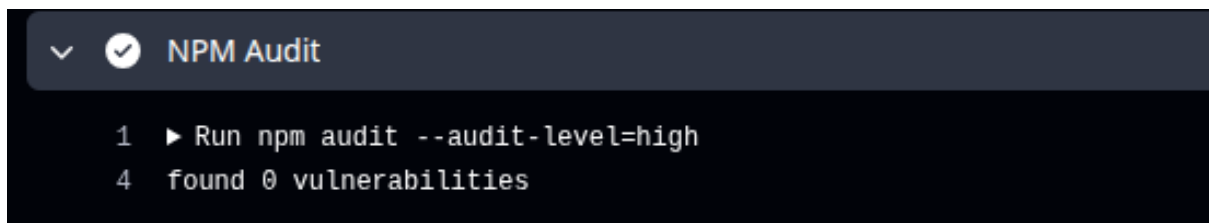


Figure 19: NPM audit

NFR4.1: Cloud Provider API Failure Recovery

The provider API failure recovery is measured via a Spring Boot unit test. The retry logic is handled via our JobRunr background runner in response to ingestion successes and failures. Three retries are attempted after which if no successful ingestion occurs, the task will be attempted again on the next pass.

The tests are run during the Continuous Integration (CI) validation steps. The below figures are examples taken from Github Actions workflow runs. Unfortunately individual unit test results are not visible in action runs, however the tests can be found in the CloudSherpa repository in the following file location:

```
apps/ingestion/src/test/java/com/cloudsherpa/ingestion/unit/scheduling/usage/UsageIngestionServiceTest.java
```

NFR5.1: Automated Test Coverage

Evidence for NFR5.1 goes here.

NFR5.2: Static Analysis

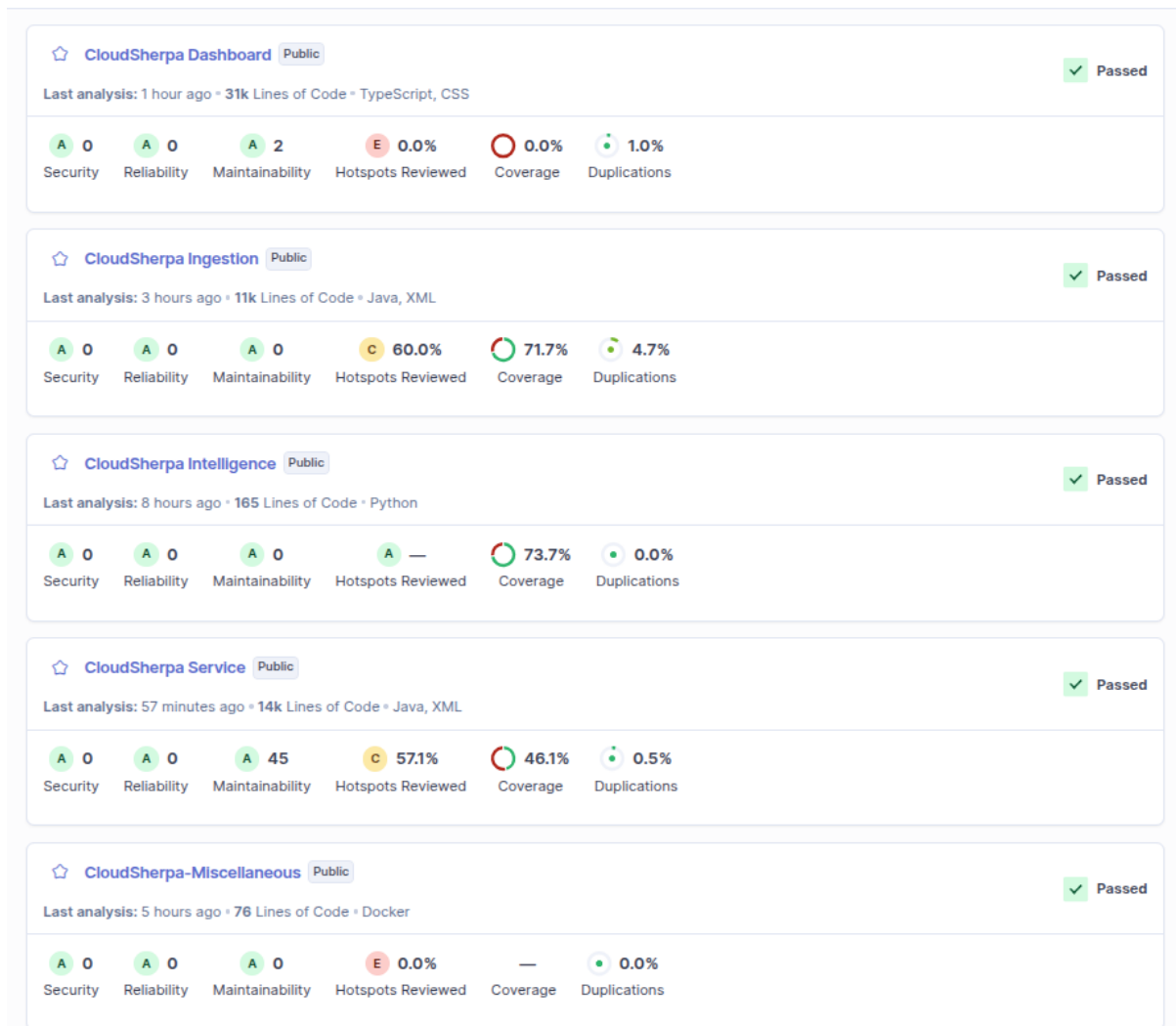


Figure 20: Self hosted SonarQube Results

NFR6.2: Cross-Browser Compatibility

Cross-browser compatibility is verified by running Playwright End-to-End (E2E) tests against different browsers. Playwright tests are run as a part of the CI suite before a

deployment.

A screenshot of a terminal window titled "Playwright E2E tests" with a status icon and a timer showing "1m 24s". The terminal output shows a sequence of commands and their results: line 1: "Run npm run test:e2e", line 10: blank, line 11: "> dashboard-frontend@0.1.0 test:e2e", line 12: "> playwright test", line 13: blank, line 14: blank, line 15: "Running 24 tests using 1 worker", line 16: ".....", line 17: "24 passed (1.4m)".

```
1 ▶ Run npm run test:e2e
10
11 > dashboard-frontend@0.1.0 test:e2e
12 > playwright test
13
14
15 Running 24 tests using 1 worker
16 .....
17 24 passed (1.4m)
```

Figure 21: Playwright E2E test step in the CloudSherpa Build and Deploy workflow

3 | Uptime Monitoring

Status Page

Uptime monitoring is achieved via the *Uptime Kuma* monitoring tool. The CloudSherpa status page can be viewed at <https://cloudsherpa-monitoring.gjjcs.org/status/cloudsherpa>

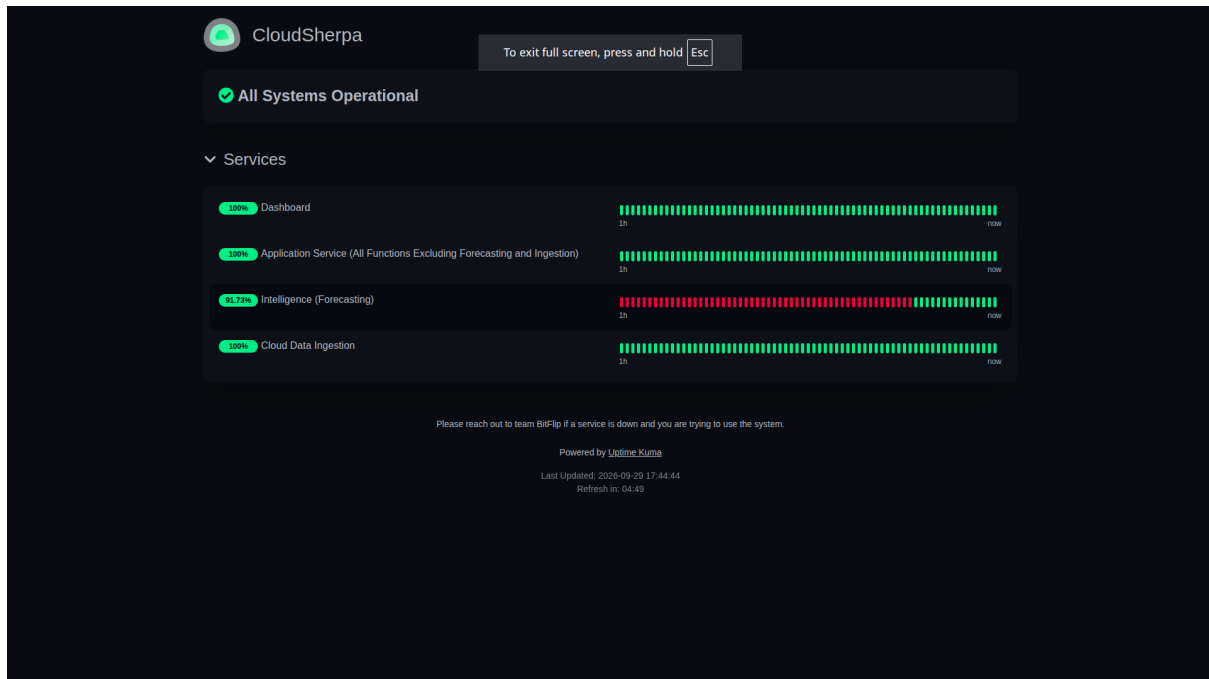


Figure 22: CloudSherpa Status Page Depicting an Outage

4 | Technology Requirements

Terminology Note

Please note that references such as **FR X** or **NFR X** refer to requirements documented in the CloudSherpa Software Requirement Specification documentation.

4.1 Next.js

We chose to use the Next.js full-stack framework for the CloudSherpa user-facing **Dashboard**.

4.1.1 Supporting functional requirements

- **FR 5. Web-Based Dashboard**

Next.js is a framework intended for web development and thus facilitates the development of our web-based Dashboard as elicited by **FR 5** in our SRS document. Reactive dashboard components are enabled by Next.js since it is built on top of the React framework.

- **FR 6. User and Account Management** Next.js middleware can inspect incoming HTTP requests and redirect users to the login page when an authorization token is absent.
- **FR 7. Alerts and Notifications** Next.js integrates with shadcn/ui, which provides reusable and accessible interface components for displaying alerts and notifications to users.

4.1.2 Supporting non-functional requirements

- **NFR 1. Performance**

Next.js Server Components can be leveraged for computationally intensive Dashboard tasks, limiting client-side computation and improving dashboard responsiveness.

- **NFR 3. Security**

Next.js Server Components allows developers to omit potentially sensitive information from the bundle that gets sent to the client browser. Next.js middleware allows code to be executed server-side before a request is completed, allowing developers to intercept and inspect incoming client requests and perform custom actions such as redirecting the user to the login page when the authorization token is omitted from the request.

- **NFR 6. Usability** Next.js integrates with shadcn/ui, our chosen component library, which provides reusable and accessible interface components that support the development of a consistent and user-friendly dashboard.

4.1.3 Ease of development

- **App Router**

The Next.js App Router simplifies navigation development by using the application's directory structure to define routes. It also provides conventions, such as route groups and private folders, that allow developers to organize related files and routes without necessarily affecting the resulting URL structure.

4.2 Spring Boot

We selected Spring Boot with Java 21 for the CloudSherpa Ingestion and Application services because of its performance benefits compared with the considered alternatives, such as NestJS, as well as Java's strong support for data structures and algorithms and the SDKs provided by the cloud providers that CloudsSherpa supports.

4.2.1 Supporting functional requirements

- **FR 1. Cloud Data Integration**

AWS, GCP, and Azure provide Java SDKs that CloudSherpa uses for resource discovery and the ingestion of usage metrics and billing data. The JobRunr Java library provides persistent, multithreaded background job processing, which we use to schedule and execute usage and billing ingestion tasks.

- **FR 2. Data storage and processing**

Spring Boot provides object-relational mapping support through JPA and Hibernate, allowing database records to be represented and accessed as Java objects. Hibernate also supports schema-based multitenancy, enabling tenant data to be isolated in separate database schemas. It also integrates with PostgreSQL's LISTEN/NOTIFY functionality, enabling event-driven processing in response to database changes.

- **FR 6. User and Account Management**

Utilizing Spring Security Filter Chain for managing authorization, users and user-related context can be obtained for each request as part of the Authentication Context facilitated by Spring Security and the JWT protocol.

4.2.2 Supporting non-functional requirements

- **NFR 1. Performance**

The spring boot multi-threaded architecture allows for competitive performance.

- **NFR 2. Scalability** Spring Batch supports the efficient processing of large data volumes through chunk-based processing and parallel execution.

- **NFR 3. Security**

Spring Security provides a pipe-and-filter architecture to ensure that unauthenticated or unauthorized requests never reach the controllers but gets filtered out at an earlier stage in the Spring Security Filter Chain.

- **NFR 4. Reliability**

Spring Boot Actuator exposes health and additional monitoring endpoints out of

the box, enabling health-checks and uptime monitoring. Java custom exception hierarchies allows developers handle exceptions gracefully.

- **NFR 5. Maintainability**

OOP and Java packages enables modular development making code easier to test, maintain and extend. Supporting testing frameworks such as *JUnit*, *Mockito* and *AssertJ* enable unit and integration testing.

4.2.3 Ease of development

- **Familiarity**

As team BitFlip is comprised of 5 university students with a lot of exposure to the Java programming language, we are apply to better apply learnt concepts during the development of the most technical parts of CloudSherpa.

4.3 Docker and Docker Compose

4.3.1 Supporting non-functional requirements

- **NFR 1. Performance**

Docker shares the operating system kernel which results in better performance than using traditional VMs.

- **NFR 2. Scalability**

Containers can easily be scaled up or down. Containers can be replicated to facilitate distributed runtime environments.

- **NFR 3. Security**

Our application runs as a set of isolated containers. Isolated containers have the benefit of increased security: when one part of the system is compromised it does not automatically mean that the entire system is compromised. Taking this into account, we explicitly handle all sensitive operations within a part of the system that runs in an isolated container and is less likely to be vulnerable to malicious exploits.

4.3.2 Ease of development

- **Portability**

Docker was chosen to improve application portability by reducing its dependence on the environment in which it is executed.

- **Reproducibility**

Containers isolate the application and its dependencies from the host environ-

ment. As a result, container behaviour should remain consistent across different environments, enabling more predictable and reproducible application execution.

4.4 PostgreSQL & TimescaleDB

We use **PostgreSQL** as our relational database management system (RDBMS), together with the TimescaleDB extension for storing and querying time-series data. A key reason for selecting PostgreSQL is that TimescaleDB is built on top of it, allowing time-series hypertables and conventional relational tables to coexist and be queried within a single database.

4.4.1 Supporting functional requirements

- **FR 2. Data Storage and Processing**

PostgreSQL schemas define the structure and organization of data stored at rest. The TimescaleDB extension is used to efficiently store and query time-series data, while PostgreSQL's `LISTEN/NOTIFY` functionality supports event-driven communication when new usage metric data becomes available. A schema-per-tenant architecture is used to logically isolate each tenant's data within the database.

4.4.2 Supporting non-functional requirements

- **NFR 1. Performance**

TimescaleDB improves the storage and query performance of time-series data through time-based partitioning and optimizations designed for time-series workloads.

4.5 FastAPI

FastAPI was chosen to facilitate communication between the forecasting model and the application service. Its Python runtime enables direct integration with the forecasting model (Chronos-2) and its supporting machine-learning libraries, while its performance and ease of development support the efficient implementation of the required API.

4.5.1 Supporting Functional Requirements

- **FR 3. Cost Prediction**

The FastAPI framework facilitates REST-based communication between the Application Service and the AI/ML Intelligence Service. The Application Service sends time-series data and a specified prediction horizon to the forecasting model

exposed through FastAPI, which returns predictions for the requested number of future timestamps.

4.5.2 Supporting Non-Functional Requirements

- **NFR 1. Performance** According to the official FastAPI documentation, its performance is comparable to that of high-performance frameworks built using technologies such as Node.js and Go. As FastAPI is primarily responsible for exposing the forecasting model through a REST API, its performance helps minimize the communication overhead between the Application Service and the AI/ML Intelligence Service.

4.5.3 Ease of Development

Built on Pydantic, FastAPI automates development tasks such as validation, serialization and endpoint documentation.

5 | Deployment

5.1 Deployment Requirements

- **Live, Accessible System:**
The system is accessible at <https://cloudsherpa.gjjcs.org>
- **Environment Parity:**
 - **Development Environment**
BitFlip members develop CloudSherpa on their local machines. To ensure reproducibility and consistency across the team, the development environment is containerized using Docker and Docker Compose. This setup closely mirrors the production deployment environment, particularly in how container images are built and services are orchestrated. Team members who do not have sufficient local computing resources to run the complete containerized environment can instead make use of the development servers provided by the frameworks that we use (refer to section 4).
 - **Staging Environment**
The CloudSherpa staging environment is hosted on an AWS EC2 instance, with deployment automated through GitHub Actions. It serves as a Quality

Assurance (QA) environment in which the team can test the platform under conditions that closely resemble the production environment. Nginx-enforced HTTP Basic Authentication restricts access to authorized team members and stakeholders, preventing unintended public access to the staging deployment. The automated deployment workflow is triggered by a push event to the `staging` branch.

- **Production Environment**

The CloudSherpa production environment is hosted on an AWS EC2 instance, with deployment automated through GitHub Actions. The production deployment workflow is triggered by a push event to the `main` branch.

- **Containerisation:**

Docker was selected as the containerisation platform for CloudSherpa, enabling the system to be developed and deployed as a set of isolated, portable services. Docker Compose is used to define and manage the platform's multi-container architecture. For deployment, AWS Elastic Container Registry (ECR) serves as a private, managed container registry that integrates seamlessly with the staging and production environments hosted on AWS.

- **Secrets Management:**

Local `.env` files are used to manage secrets and environment-specific configuration. To ensure consistency across environments and enable configuration validation, the project uses environment variable schemas. *Varlock* was selected to define and manage these schemas, together with a custom validation script that verifies that all required environment variables are present. The Varlock schema files can be safely committed to version control because they define only the expected configuration structure and do not contain any secret values.

- **Rollback Strategy:**

CloudSherpa container images are built and deployed automatically using GitHub Actions. Each image is tagged with the commit hash associated with the deployment, allowing deployments to be pinned to specific versions of the codebase. A deployment script accepts a commit hash as a command-line argument, pulls the corresponding images from the container registry, and deploys them. If a rollback is required, the commit hash of a previously stable deployment is supplied to the script, restoring the system to that version.

5.2 Deployment Diagram

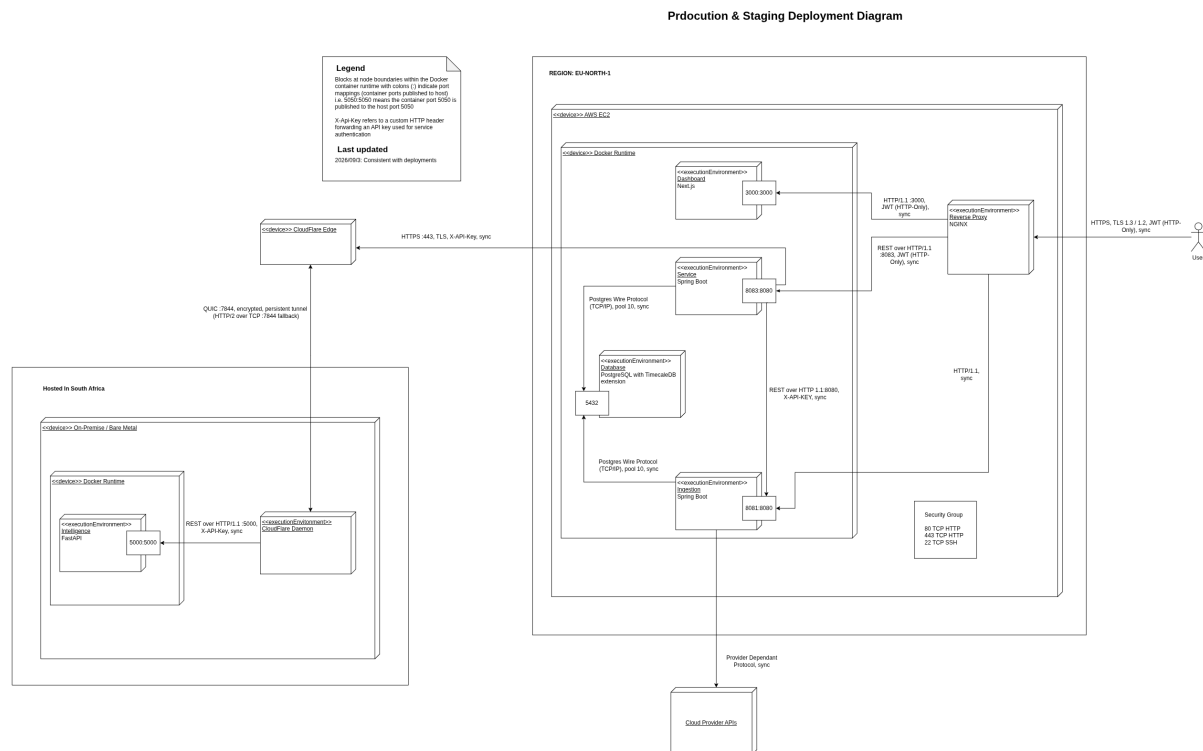


Figure 23: CloudSherpa Deployment Diagram (Production)

Note: Please zoom in when viewing the PDF to inspect diagram details.

5.3 CI/CD Pipeline Diagram

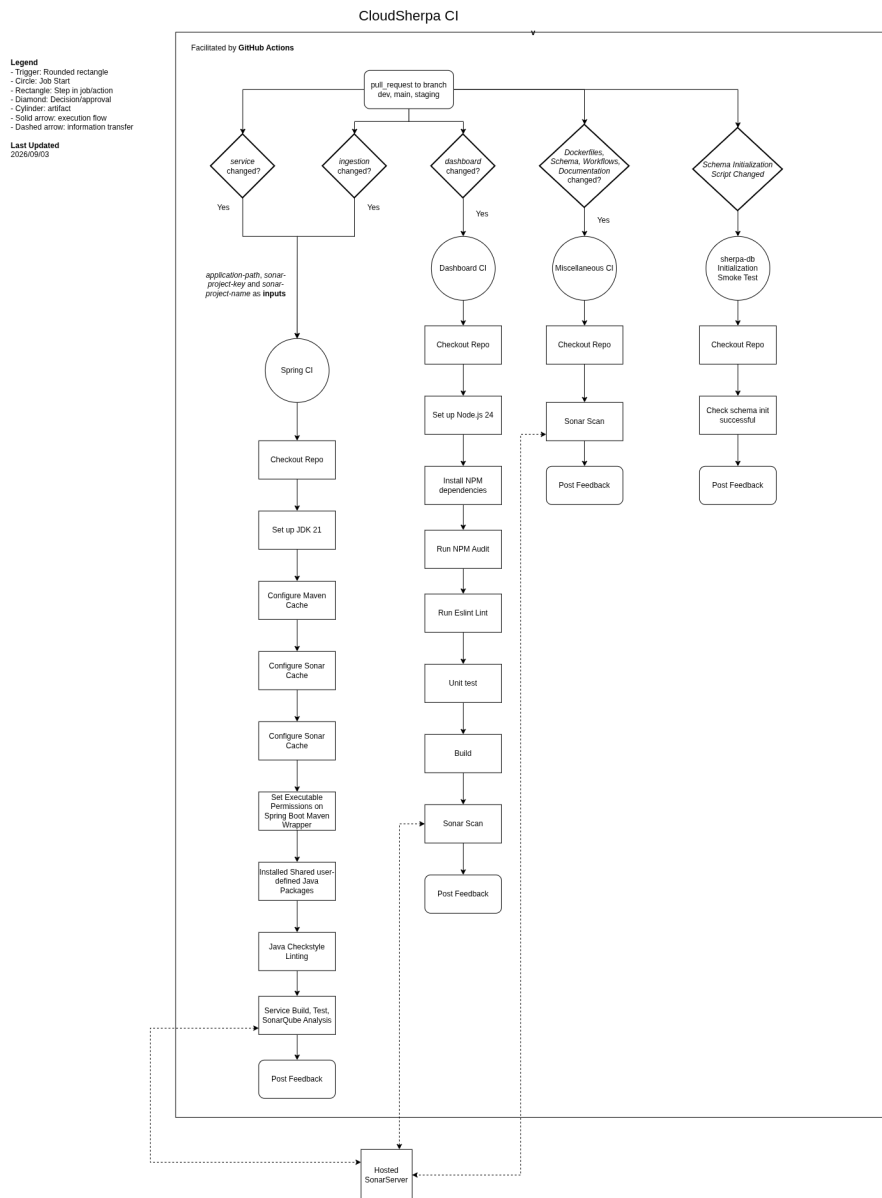
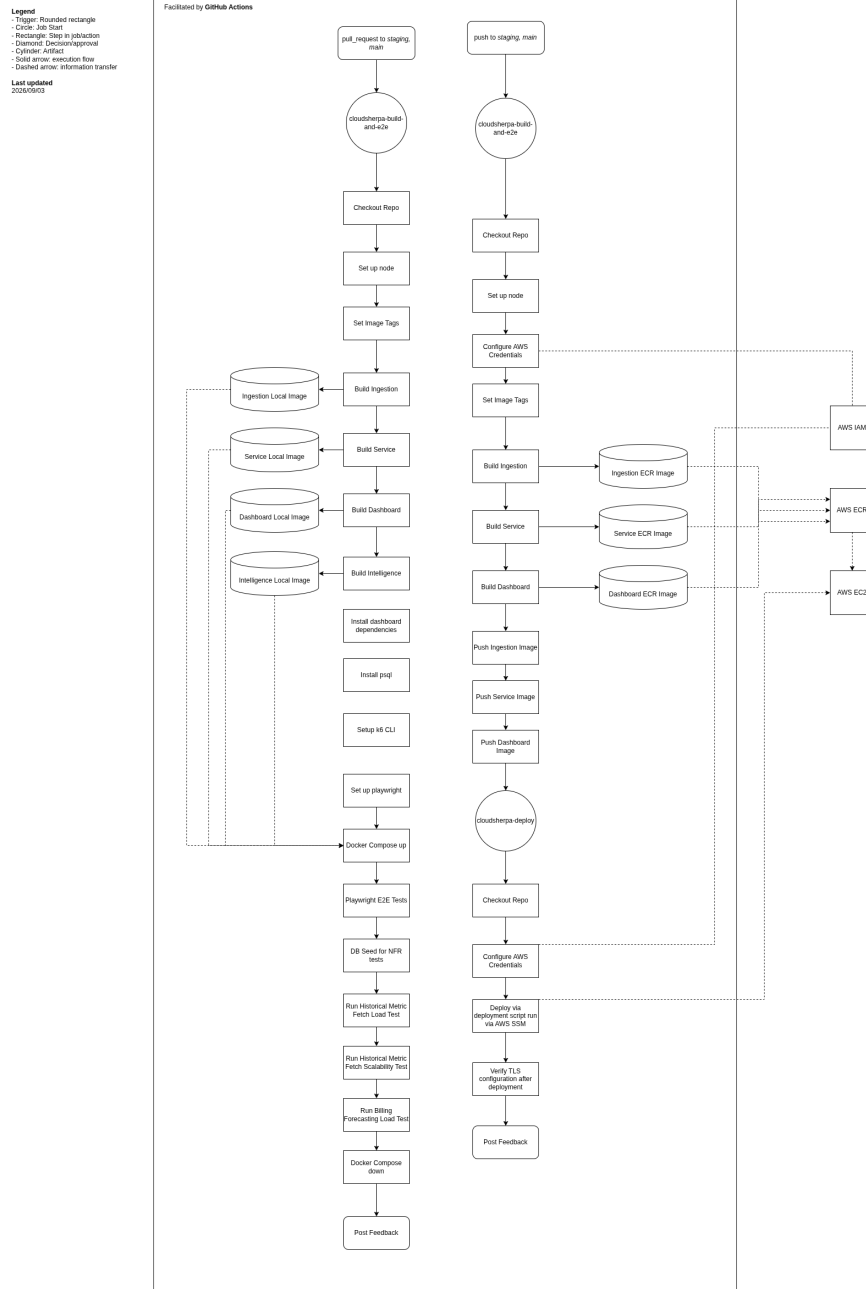


Figure 24: CI Pipeline Flow



A | Appendix A: Architecture Decision Records

A.1 ADR - 001

ADR - 001

Title: Ingestion Scheduler job management

Status: Accepted

Context: Ingestion jobs from multiple cloud accounts must be ingested at varying times dependent on user-selected intervals. These events must not interfere with the rest of the subsystem's functioning. Failure conditions must be handled to ensure resiliency and avoid data loss in the case of external outages.

Alternatives:

1. Use Postgress listen/notify to create one thread per due account (could lead to thread explosion and deployment costs, no outage protection, satisfies independent functioning, tightly coupled to database).
2. Use database polling with timestamped fields and a thread pool to queue due account jobs. (No thread explosion, outage protection, no retry logic, manual retry logic and exception handling required).
3. Use database polling with timestamped fields and a background job manager (JobRunr) to manage threads. (No thread explosion, outage protection, automatic retry attempts, detailed logs maintained, additional storage required, dependency)

Decision: Database polling with timestamped fields and JobRunr background job manager

Consequence:

- + Threads are managed by JobRunr to avoid thread explosion.
- + JobRunr manages retry logic automatically for resiliency.
- + Detailed logging is available through Jobrunr generated tables.
- + Polling allows jobs to be handled multiple times in case of failure
 - JobRunr handles retries uniformly, potentially limiting future custom handling
 - JobRunr adds overhead and logs may incur additional storage costs

A.2 ADR - 002

ADR - 002

Title: PostgreSQL LISTEN/NOTIFY for inter-service communication

Status: Accepted

Context: The ingestion subsystem must notify the service subsystem when metrics have been added. The communication mechanism should minimise deployment complexity while remaining sufficiently reliable for the expected workload.

Alternatives:

1. Use an event broker (e.g. Kafka or RabbitMQ). (Mature event-driven communication, durable messaging, scalable, increased operational complexity and additional infrastructure).
2. Use PostgreSQL LISTEN/NOTIFY. (Simple deployment, no additional infrastructure, lightweight, notifications are transient and less suitable for large-scale event streaming).

Decision: Use PostgreSQL LISTEN/NOTIFY for communication between the ingestion and service subsystems.

Consequence:

- + Reduces operational complexity by avoiding additional messaging infrastructure.
- + Reduces deployment size and simplifies maintenance.
- + Integrates naturally with the existing PostgreSQL database.
- Notifications are not durable and may be missed if consumers are unavailable.
- Less suitable than dedicated message brokers for future high-volume event streaming.

A.3 ADR - 003



ADR - 003

Title: Server-Sent Events for real-time dashboard updates

Status: Accepted

Context: Users should metric updates without manually refreshing the dashboard or generating unnecessary requests.

Alternatives:

1. Poll the server periodically. (Simple implementation, wasted requests when no updates exist, increased server load).
2. Use Server-Sent Events (SSE). (Real-time updates over a persistent HTTP connection, lower network overhead, persistent connections must be maintained).

Decision: Use Server-Sent Events (SSE) to push ingestion status updates to the dashboard.

Consequence:

- + Eliminates unnecessary polling requests.
- + Provides near real-time updates to users.
- Persistent HTTP connections consume server resources.
- Connections must be re-established after network interruptions.

A.4 ADR - 004



ADR - 004

Title: Cloud billing exports for cost analysis

Status: Accepted

Context: Cloud resource costs can either be approximated from usage metrics or obtained directly from provider billing exports. Accurate historical costs are required for reporting.

Alternatives:

1. Approximate costs using collected usage metrics. (Near real-time estimates, numerous unknown pricing variables reduce accuracy, metric specific logic implementation).
2. Use cloud provider billing exports. (Accurate provider-calculated costs, delayed availability due to billing export schedules).

Decision: Use provider billing exports as the source of cloud cost information.

Consequence:

- + Provides accurate provider-calculated billing values.
- + Avoids implementing and maintaining complex pricing logic.
- + Produces consistent historical reporting.
- Users accept latency before costs become visible.

A.5 ADR - 005



ADR - 005

Title: Use of foundational prediction models

Status: Accepted

Context: CloudSherpa provides projections of future metric values of cloud resource utilisation based on historical data. The prediction model could either be developed and trained specifically for the system or utilise an existing foundational model.

Alternatives:

1. Develop a custom prediction model. (Highly sophisticated models could achieve higher quality predictions if provided sufficient data, significant development effort, extensive training data required, skill requirement with parameter tuning or method selection).
2. Use an existing foundational language model. (High-quality predictions based on large datasets, rapid development, dependency on third-party model providers).

Decision: Use a foundational prediction model to predict future metric values based on historical data.

Consequence:

- + Significantly reduces development effort.
- + Produces high-quality predictions based on extensive training set.
- Creates dependency on external model providers.
- Less tuning and optimization according to specific customer profiles possible.

A.6 ADR - 006

ADR - 006

Title: Adopt a schema-per-tenant multi-tenancy architecture

Status: Accepted

Context: CloudSherpa must support a growing number of tenants while maintaining logical data isolation, efficient query performance, and manageable operational overhead. The chosen multi-tenancy strategy should balance scalability, maintainability, and performance as the number of tenants and stored historical cloud metrics increases.

Alternatives:

1. Database-per-tenant. (Strong physical isolation and independent administration, however managing potentially thousands of databases significantly increases operational complexity, resource consumption, backups, monitoring, and schema migrations.)
2. Shared schema with tenant identifiers. (Simplified database administration with a single schema, however all tenant data resides in the same tables, resulting in larger indexes, tenant filtering on every query, increased risk of accidental cross-tenant data access, and reduced query locality.)
3. Schema-per-tenant. (Provides logical isolation while sharing the same database instance, resulting in smaller tenant-specific tables, improved query performance, and significantly lower administrative overhead than maintaining separate databases.)

Decision: Store each tenant's resource, metric and billing data within a dedicated PostgreSQL schema while sharing a common database instance and maintaining shared public tables for common data.

Consequence:

- + Provides strong logical isolation between tenants.
- + Avoids the operational overhead of managing potentially thousands of databases.
- + Queries execute against smaller tenant-specific tables, improving index efficiency and query performance.
- + Simplifies backups, deployments, and infrastructure management compared to database-per-tenant architectures.
- + Reduces the likelihood of accidental cross-tenant queries.
- Database schema migrations must be applied to every tenant schema.
- All tenants share the same underlying database resources, requiring future horizontal scaling if database capacity is exceeded.

A.7 ADR - 007



ADR - 007

Title: Intelligence service obtaining historical context

Status: Proposed

Context: The forecasting model requires historical context to forecast values

Alternatives:

1. Open a direct connection between the intelligence service and the database.

Decision: Use the application service to query the required historical context using the existing database connection and communicate context with intelligence service via an HTTP protocol.

Consequence:

- + Reduces operational complexity, not needing to open DB connection from intelligence service.
- + Decouples the intelligence service from the database schema.
- + Reduces attack surface by not introducing additional paths to the DB that can potentially introduce high risk vulnerabilities.
- Network latency due to large payloads.
- API contract maintenance.

A.8 ADR - 008

ADR - 008

Title: Use of Spring Scheduler for Asynchronous Optimization Worker

Status: Accepted

Context: The Optimization Recommendation Engine uses an asynchronous, batch-processing architecture designed to decouple heavy analytical queries from synchronous user interactions. This Asynchronous Optimization Worker handles all heavy lifting, including computing heavy statistical summaries and saving them to SherpaDB. To track progress, the engine utilizes a watermark-based processing strategy, relying on a durable processing_watermark table to check exactly when it last successfully processed data for a specific tenant. Because state management and execution tracking are natively handled by this watermark table, the scheduling mechanism strictly needs to act as a lightweight job tracker.

Alternatives:

1. **JobRunr**

- **Advantages:** Provides distributed background job processing, and automatic retry mechanisms.
- **Disadvantages:** Introduces unnecessary database overhead and complexity.

2. **Spring Scheduler**

- **Advantages:** Lightweight, native to the Spring ecosystem, and easily configured to execute the worker once each day. It delegates tracking entirely to our database-native watermark strategy.
- **Disadvantages:** Lack of Built-In Retry and Failure Handling

Decision: We will use a standard Spring scheduler for the Optimization Worker. The worker will trigger once each day, at which point it will read the processing_watermark table to determine where it left off, pull the last 24 hours of unprocessed normalized metrics, run rule evaluations, update the recommendations, and go back to sleep.

Consequence:

- + Reduces overall architectural complexity by eliminating a heavy, third-party job tracking framework.
- + Maintains a single source of truth for processing state via the custom processing_watermark table.
- Introduces the risk of 24-hour data delays upon failure, as the scheduler lacks automatic retries for database or network errors.

A.9 ADR - 009



ADR - 009

Title: Dynamic billing ingestion bean registry via factory method vs simple switch-based dispatch logic

Status: Proposed

Context: The billing ingestion services for each provider subscribe to an interface that enforces the execute method contract that the scheduler uses to schedule ingestion.

Alternatives:

1. Use a factory pattern leveraging spring dependency injection to automatically load all services implementing this interface into a registry keyed by the provider enum. The scheduler code would then be unaware of the specific billing ingestion implementations and would be able to call the correct execute method with just the provider as context

Decision: Manually inject the billing ingestion services into the scheduler BillingIngestionClient and dispatch behavior via a switch statement on the provider enum and switch labeled rules for improved readability and explicit behavior for the limited scope of three Cloud Providers

Consequence:

- Adding a new billing ingestion service for a new provider requires manually injecting the service into the BillingIngestionClient

B | Appendix B: Service Contracts

B.1 Ingestion Service

```
1 {
2   "openapi": "3.1.0",
3   "info": {
4     "title": "OpenAPI definition",
5     "version": "v0"
6   },
7   "tags": [
8     {
9       "name": "Cloud Usage Ingestion",
10      "description": "Endpoints for ingesting cloud usage and
11                    billing data from cloud providers"
12    },
13    {
14      "name": "Cloud Resource Discovery",
15      "description": "Endpoints for discovering which services
16                    and resources are available from cloud providers"
17    }
18  ],
19  "paths": {
20    "/api/events/ingest": {
21      "post": {
22        "tags": [
23          "Cloud Usage Ingestion"
24        ],
25        "summary": "Ingest real cloud usage and billing data",
26        "description": "Retrieves cloud usage and/or billing data
27                      from configured cloud providers using the supplied
28                      credentials, account scopes, and time range.",
29        "operationId": "ingest",
30        "requestBody": {
31          "description": "Cloud ingestion request object",
32          "content": {
33            "application/json": {
34              "schema": {
35                "$ref": "#/components/schemas/
36                      IngestionRequestEvent"
```

```

32         }
33     },
34     },
35     "required": true
36 },
37 "responses": {
38     "200": {
39         "description": "Cloud data successfully ingested",
40         "content": {
41             "application/json": {
42                 "schema": {
43                     "$ref": "#/components/schemas/IngestionResult"
44                 }
45             }
46         }
47     }
48 }
49 },
50 },
51 "/api/events/ingest/mock": {
52     "post": {
53         "tags": [
54             "Cloud Usage Ingestion"
55         ],
56         "summary": "Ingest deterministic fixed-size mock cloud
                    data",
57         "description": "Generates and ingests deterministic mock
                        cloud usage datafor development and integration
                        testing.",
58         "operationId": "ingestMock",
59         "requestBody": {
60             "description": "Ingestion request object, credentials
                            and some other fields are not used",
61             "content": {
62                 "application/json": {
63                     "schema": {
64                         "$ref": "#/components/schemas/
                            IngestionRequestEvent"
65                     }
66                 }
67             },
68             "required": true
69         },
70         "responses": {

```

```

71         "200": {
72             "description": "Mock cloud data successfully ingested",
73             "content": {
74                 "application/json": {
75                     "schema": {
76                         "$ref": "#/components/schemas/IngestionResult"
77                     }
78                 }
79             }
80         }
81     },
82 },
83 "/api/events/ingest/mockNoise": {
84     "post": {
85         "tags": [
86             "Cloud Usage Ingestion"
87         ],
88         "summary": "Ingest mock cloud data with noise",
89         "description": "Generates and ingests mock cloud usage",
90         "data using Reinstein Uhlenbeck noise for analytics,",
91         "anomaly detection, and testing.",
92         "operationId": "ingestMockWithNoise",
93         "requestBody": {
94             "description": "Ingestion request object, credentials",
95             "and some other fields are not used",
96             "content": {
97                 "application/json": {
98                     "schema": {
99                         "$ref": "#/components/schemas/IngestionRequestEvent"
100                     }
101                 }
102             },
103             "required": true
104         },
105         "responses": {
106             "200": {
107                 "description": "Mock cloud data successfully ingested",
108                 "content": {
109                     "application/json": {
110                         "schema": {

```

```

109         "$ref": "#/components/schemas/IngestionResult"
110     }
111 }
112 }
113 }
114 }
115 }
116 },
117 "/api/events/ingest/gcp/billing/bigquery": {
118     "post": {
119         "tags": [
120             "Cloud Usage Ingestion"
121         ],
122         "summary": "Trigger GCP BigQuery billing ingestion manually",
123         "description": "Runs GCP BigQuery billing ingestion for testing and development. This endpoint is only available when the dev Spring profile is active.",
124         "operationId": "ingestGcpBigqueryBilling",
125         "responses": {
126             "200": {
127                 "description": "GCP BigQuery billing ingestion ran successfully"
128             },
129             "401": {
130                 "description": "Endpoint is only available in the dev profile"
131             }
132         }
133     }
134 },
135 "/api/events/ingest/azure/billing/{configId}": {
136     "post": {
137         "tags": [
138             "Cloud Usage Ingestion"
139         ],
140         "summary": "Trigger Azure billing ingestion manually for a particular user",
141         "description": "Runs Azure billing ingestion for testing and development. This endpoint is only available when the dev Spring profile is active.",
142         "operationId": "ingestAzureBilling",
143         "parameters": [
144             {

```

```

145         "name": "configId",
146         "in": "path",
147         "required": true,
148         "schema": {
149             "type": "string",
150             "format": "uuid"
151         }
152     },
153 ],
154 "responses": {
155     "200": {
156         "description": "GCP BigQuery billing ingestion ran
157             successfully"
158     }
159 },
160 },
161 "/api/events/ingest/aws/billing/cur": {
162     "post": {
163         "tags": [
164             "Cloud Usage Ingestion"
165         ],
166         "summary": "Trigger AWS CUR billing ingestion with
167             credentials stored for test user",
168         "description": "When database is seeded appropriately,
169             ingest a billing report for testing purposes",
170         "operationId": "ingestAwsBillingCur",
171         "responses": {
172             "200": {
173                 "description": "AWS CUR billing ingestion ran
174                     successfully",
175                 "content": {
176                     "application/json": {
177                         "schema": {
178                             "$ref": "#/components/schemas/AwsCurContext"
179                         }
180                     }
181                 }
182             }
183         }
184     },
185     "post": {

```

```

185     "tags": [
186         "Cloud Resource Discovery"
187     ],
188     "summary": "Get all supported services for a cloud
                  provider",
189     "description": "Retrieves a list of all services that can
                      be monitored by CloudSherpa for a given provider",
190     "operationId": "getAllOfferedServices",
191     "requestBody": {
192         "description": "Cloud provider name (e.g. aws, azure,
                          gcp)",
193         "content": {
194             "application/json": {
195                 "schema": {
196                     "type": "string"
197                 }
198             }
199         },
200         "required": true
201     },
202     "responses": {
203         "200": {
204             "description": "Cloud services successfully returned
                              ",
205             "content": {
206                 "application/json": {
207                     "schema": {
208                         "type": "string"
209                     }
210                 }
211             }
212         }
213     }
214 },
215 },
216 "/api/cloud-resources/resources/{provider}": {
217     "post": {
218         "tags": [
219             "Cloud Resource Discovery"
220         ],
221         "summary": "Get all resources within provided service
                      categories for a cloud provider",
222         "description": "Retrieves all discoverable resources
                          within a list of services of a cloud provider using

```

```
223         the supplied credentials",
224         "operationId": "getAllResources",
225         "parameters": [
226             {
227                 "name": "provider",
228                 "in": "path",
229                 "required": true,
230                 "schema": {
231                     "type": "string"
232                 }
233             },
234             "requestBody": {
235                 "content": {
236                     "application/json": {
237                         "schema": {
238                             "$ref": "#/components/schemas/
                                ResourceDiscoveryRequest"
239                         }
240                     }
241                 },
242                 "required": true
243             },
244             "responses": {
245                 "200": {
246                     "description": "Resources successfully returned",
247                     "content": {
248                         "application/json": {
249                             "schema": {
250                                 "$ref": "#/components/schemas/ResourceDetail"
251                             }
252                         }
253                     }
254                 }
255             }
256         },
257         "/api/cloud-resources/gcp/permissions": {
258             "post": {
259                 "tags": [
260                     "Cloud Resource Discovery"
261                 ],
262                 "summary": "Generate GCP permissions list",
```

```

264     "description": "Generates a least-privilege GCP
265         permissions list for the selected services",
266     "operationId": "generateGcpPermissionsPolicy",
267     "requestBody": {
268         "content": {
269             "application/json": {
270                 "schema": {
271                     "type": "array",
272                     "items": {
273                         "type": "string"
274                     }
275                 }
276             },
277             "required": true
278         },
279     "responses": {
280         "200": {
281             "description": "Permissions list successfully
282                 generated",
283             "content": {
284                 "*/*": {
285                     "schema": {
286                         "type": "array",
287                         "items": {
288                             "type": "string"
289                         },
290                         "uniqueItems": true
291                     }
292                 }
293             }
294         }
295     },
296 },
297 "/api/cloud-resources/azure/permissions": {
298     "post": {
299         "tags": [
300             "Cloud Resource Discovery"
301         ],
302         "summary": "Generate Azure permissions list",
303         "description": "Generates a least-privilege Azure
304             permissions list for the selected services",
305         "operationId": "generateAzurePermissionsPolicy",

```

```
305     "requestBody": {
306       "content": {
307         "application/json": {
308           "schema": {
309             "type": "array",
310             "items": {
311               "type": "string"
312             }
313           }
314         }
315       },
316       "required": true
317     },
318     "responses": {
319       "200": {
320         "description": "Permissions list successfully
321           generated",
322         "content": {
323           "*/*": {
324             "schema": {
325               "type": "array",
326               "items": {
327                 "type": "string"
328               },
329               "uniqueItems": true
330             }
331           }
332         }
333       }
334     },
335     "/api/cloud-resources/aws/permissions": {
336       "post": {
337         "tags": [
338           "Cloud Resource Discovery"
339         ],
340         "summary": "Generate AWS IAM policy",
341         "description": "Generates a least-privilege AWS IAM
342           policy for the selected services",
343         "operationId": "generateAwsPermissionsPolicy",
344         "requestBody": {
345           "content": {
346             "application/json": {
```

```
347         "schema": {
348             "type": "array",
349             "items": {
350                 "type": "string"
351             }
352         }
353     },
354     "required": true
355 },
356 "responses": {
357     "200": {
358         "description": "IAM policy successfully generated",
359         "content": {
360             "*/*": {
361                 "schema": {
362                     "type": "string"
363                 }
364             }
365         }
366     }
367 },
368 }
369 },
370 },
371 },
372 "components": {
373     "schemas": {
374         "AccountScope": {
375             "type": "object",
376             "properties": {
377                 "provider": {
378                     "type": "string"
379                 },
380                 "accountId": {
381                     "type": "string"
382                 },
383                 "subscriptionId": {
384                     "type": "string"
385                 },
386                 "projectId": {
387                     "type": "string"
388                 },
389                 "billingAccountId": {
390                     "type": "string"
```

```
391     },
392     "serviceScopes": {
393       "type": "array",
394       "items": {
395         "$ref": "#/components/schemas/ServiceScope"
396       }
397     }
398   },
399 },
400 "CloudCredentials": {
401   "type": "object",
402   "properties": {
403     "accessKeyId": {
404       "type": "string"
405     },
406     "secretAccessKey": {
407       "type": "string"
408     },
409     "subscriptionId": {
410       "type": "string"
411     },
412     "tenantId": {
413       "type": "string"
414     },
415     "clientId": {
416       "type": "string"
417     },
418     "clientSecret": {
419       "type": "string"
420     },
421     "projectId": {
422       "type": "string"
423     },
424     "serviceAccountJson": {
425       "type": "string"
426     }
427   }
428 },
429 "IngestionRequestEvent": {
430   "type": "object",
431   "properties": {
432     "scopes": {
433       "type": "array",
434       "items": {
```

```
435         "$ref": "#/components/schemas/AccountScope"
436     },
437 },
438 "credentials": {
439     "$ref": "#/components/schemas/CloudCredentials"
440 },
441 "from": {
442     "type": "string",
443     "format": "date-time"
444 },
445 "to": {
446     "type": "string",
447     "format": "date-time"
448 },
449 "period": {
450     "type": "integer",
451     "format": "int32"
452 },
453 "includeBilling": {
454     "type": "boolean"
455 },
456 "includeUsage": {
457     "type": "boolean"
458 },
459 "userId": {
460     "type": "string",
461     "format": "uuid"
462 }
463 },
464 },
465 "Instance": {
466     "type": "object",
467     "properties": {
468         "identifier": {
469             "type": "string"
470         },
471         "region": {
472             "type": "string"
473         }
474     }
475 },
476 "InstanceScope": {
477     "type": "object",
478     "properties": {
```

```
479         "identifierName": {
480             "type": "string"
481         },
482         "instances": {
483             "type": "array",
484             "items": {
485                 "$ref": "#/components/schemas/Instance"
486             }
487         }
488     },
489     "Metric": {
490         "type": "object",
491         "properties": {
492             "name": {
493                 "type": "string"
494             },
495             "unit": {
496                 "type": "string"
497             }
498         }
499     },
500     "ServiceScope": {
501         "type": "object",
502         "properties": {
503             "name": {
504                 "type": "string"
505             },
506             "instances": {
507                 "type": "array",
508                 "items": {
509                     "$ref": "#/components/schemas/InstanceScope"
510                 }
511             },
512             "metrics": {
513                 "type": "array",
514                 "items": {
515                     "$ref": "#/components/schemas/Metric"
516                 }
517             }
518         }
519     },
520     "BillingRecordModel": {
521         "type": "object",
```

```
523     "properties": {
524         "recordId": {
525             "type": "string",
526             "format": "uuid"
527         },
528         "provider": {
529             "type": "string"
530         },
531         "accountId": {
532             "type": "string"
533         },
534         "subscriptionId": {
535             "type": "string"
536         },
537         "projectId": {
538             "type": "string"
539         },
540         "billingAccountId": {
541             "type": "string"
542         },
543         "serviceName": {
544             "type": "string"
545         },
546         "resourceId": {
547             "type": "string"
548         },
549         "resourceType": {
550             "type": "string"
551         },
552         "region": {
553             "type": "string"
554         },
555         "cost": {
556             "type": "number",
557             "format": "double"
558         },
559         "usageQuantity": {
560             "type": "number",
561             "format": "double"
562         },
563         "unit": {
564             "type": "string"
565         },
566         "currency": {
```

```
567         "type": "string"
568     },
569     "pricingModel": {
570         "type": "string"
571     },
572     "usageStartTime": {
573         "type": "string",
574         "format": "date-time"
575     },
576     "usageEndTime": {
577         "type": "string",
578         "format": "date-time"
579     },
580     "billingPeriodStart": {
581         "type": "string",
582         "format": "date-time"
583     },
584     "billingPeriodEnd": {
585         "type": "string",
586         "format": "date-time"
587     },
588     "tags": {
589         "type": "object",
590         "additionalProperties": {
591             "type": "string"
592         }
593     },
594     "ingestionTimestamp": {
595         "type": "string",
596         "format": "date-time"
597     },
598     "ingestionId": {
599         "type": "string"
600     },
601     "source": {
602         "type": "string"
603     }
604 },
605 "IngestionResult": {
606     "type": "object",
607     "properties": {
608         "usage": {
609             "type": "array",
```

```
611         "items": {
612             "$ref": "#/components/schemas/UsageRecordModel"
613         }
614     },
615     "billing": {
616         "type": "array",
617         "items": {
618             "$ref": "#/components/schemas/BillingRecordModel"
619         }
620     }
621 },
622 {
623     "UsageRecordModel": {
624         "type": "object",
625         "properties": {
626             "recordId": {
627                 "type": "string",
628                 "format": "uuid"
629             },
630             "provider": {
631                 "type": "string"
632             },
633             "accountId": {
634                 "type": "string"
635             },
636             "subscriptionId": {
637                 "type": "string"
638             },
639             "projectId": {
640                 "type": "string"
641             },
642             "resourceId": {
643                 "type": "string"
644             },
645             "resourceType": {
646                 "type": "string"
647             },
648             "serviceName": {
649                 "type": "string"
650             },
651             "region": {
652                 "type": "string"
653             },
654             "metricName": {
```

```
655         "type": "string"
656     },
657     "value": {
658         "type": "number",
659         "format": "double"
660     },
661     "unit": {
662         "type": "string"
663     },
664     "timestamp": {
665         "type": "string",
666         "format": "date-time"
667     },
668     "periodStart": {
669         "type": "string",
670         "format": "date-time"
671     },
672     "periodEnd": {
673         "type": "string",
674         "format": "date-time"
675     },
676     "dimensions": {
677         "type": "object",
678         "additionalProperties": {
679             "type": "string"
680         }
681     },
682     "tags": {
683         "type": "object",
684         "additionalProperties": {
685             "type": "string"
686         }
687     },
688     "ingestionTimestamp": {
689         "type": "string",
690         "format": "date-time"
691     },
692     "ingestionId": {
693         "type": "string"
694     },
695     "source": {
696         "type": "string"
697     }
698 }
```

```
699     },
700     "AwsCurContext": {
701       "type": "object",
702       "properties": {
703         "bucketName": {
704           "type": "string"
705         },
706         "bucketRegion": {
707           "$ref": "#/components/schemas/Region"
708         },
709         "exportPrefix": {
710           "type": "string"
711         },
712         "exportName": {
713           "type": "string"
714         },
715         "accountId": {
716           "type": "string",
717           "format": "uuid"
718         },
719         "processedExports": {
720           "type": "array",
721           "items": {
722             "type": "string"
723           }
724         },
725         "configId": {
726           "type": "string"
727         },
728         "userId": {
729           "type": "string"
730         },
731         "dataFiles": {
732           "type": "array",
733           "items": {
734             "type": "string"
735           }
736         },
737         "awsCurTmpDir": {
738           "type": "object",
739           "properties": {
740             "absolute": {
741               "type": "boolean"
742             },
```

```
743     "parent": {
744       "type": "object",
745       "properties": {
746         "absolute": {
747           "type": "boolean"
748         },
749         "root": {
750           "type": "object",
751           "properties": {
752             "absolute": {
753               "type": "boolean"
754             },
755             "fileName": {
756               "type": "object",
757               "properties": {
758                 "absolute": {
759                   "type": "boolean"
760                 },
761                 "fileSystem": {
762                   "type": "object",
763                   "properties": {
764                     "open": {
765                       "type": "boolean"
766                     },
767                     "readOnly": {
768                       "type": "boolean"
769                     },
770                     "separator": {
771                       "type": "string"
772                     },
773                     "rootDirectories": {
774
775                     },
776                     "fileStores": {
777
778                     },
779                     "userPrincipalLookupService": {
780
781                     }
782                   }
783                 },
784                 "nameCount": {
785                   "type": "integer",
786                   "format": "int32"
```

```
787     }
788   }
789 },
790 "fileSystem": {
791   "type": "object",
792   "properties": {
793     "open": {
794       "type": "boolean"
795     },
796     "readOnly": {
797       "type": "boolean"
798     },
799     "separator": {
800       "type": "string"
801     },
802     "rootDirectories": {
803
804     },
805     "fileStores": {
806
807     },
808     "userPrincipalLookupService": {
809
810     }
811   }
812 },
813 "nameCount": {
814   "type": "integer",
815   "format": "int32"
816 }
817 },
818 },
819 "fileName": {
820   "type": "object",
821   "properties": {
822     "absolute": {
823       "type": "boolean"
824     },
825     "fileSystem": {
826       "type": "object",
827       "properties": {
828         "open": {
829           "type": "boolean"
830         },
```

```
831         "readOnly": {
832             "type": "boolean"
833         },
834         "separator": {
835             "type": "string"
836         },
837         "rootDirectories": {
838
839         },
840         "fileStores": {
841
842         },
843         "userPrincipalLookupService": {
844
845         }
846     },
847     "nameCount": {
848         "type": "integer",
849         "format": "int32"
850     }
851 },
852 },
853 },
854 "fileSystem": {
855     "type": "object",
856     "properties": {
857         "open": {
858             "type": "boolean"
859         },
860         "readOnly": {
861             "type": "boolean"
862         },
863         "separator": {
864             "type": "string"
865         },
866         "rootDirectories": {
867
868         },
869         "fileStores": {
870
871         },
872         "userPrincipalLookupService": {
873
874         }
```

```
875         }
876     },
877     "nameCount": {
878         "type": "integer",
879         "format": "int32"
880     }
881 }
882 },
883 "root": {
884     "type": "object",
885     "properties": {
886         "absolute": {
887             "type": "boolean"
888         },
889         "fileName": {
890             "type": "object",
891             "properties": {
892                 "absolute": {
893                     "type": "boolean"
894                 },
895                 "fileSystem": {
896                     "type": "object",
897                     "properties": {
898                         "open": {
899                             "type": "boolean"
900                         },
901                         "readOnly": {
902                             "type": "boolean"
903                         },
904                         "separator": {
905                             "type": "string"
906                         },
907                         "rootDirectories": {
908
909                         },
910                         "fileStores": {
911
912                         },
913                         "userPrincipalLookupService": {
914
915                         }
916                     }
917                 },
918                 "nameCount": {
```

```
919         "type": "integer",
920         "format": "int32"
921     }
922 }
923 },
924 "fileSystem": {
925     "type": "object",
926     "properties": {
927         "open": {
928             "type": "boolean"
929         },
930         "readOnly": {
931             "type": "boolean"
932         },
933         "separator": {
934             "type": "string"
935         },
936         "rootDirectories": {
937
938         },
939         "fileStores": {
940
941         },
942         "userPrincipalLookupService": {
943
944         }
945     }
946 },
947 "nameCount": {
948     "type": "integer",
949     "format": "int32"
950 }
951 }
952 },
953 "fileName": {
954     "type": "object",
955     "properties": {
956         "absolute": {
957             "type": "boolean"
958         },
959         "fileSystem": {
960             "type": "object",
961             "properties": {
962                 "open": {
```

```
963         "type": "boolean"
964     },
965     "readOnly": {
966         "type": "boolean"
967     },
968     "separator": {
969         "type": "string"
970     },
971     "rootDirectories": {
972
973     },
974     "fileStores": {
975
976     },
977     "userPrincipalLookupService": {
978
979     }
980 }
981 },
982 "nameCount": {
983     "type": "integer",
984     "format": "int32"
985 }
986 },
987 },
988 "fileSystem": {
989     "type": "object",
990     "properties": {
991         "open": {
992             "type": "boolean"
993         },
994         "readOnly": {
995             "type": "boolean"
996         },
997         "separator": {
998             "type": "string"
999         },
1000         "rootDirectories": {
1001
1002         },
1003         "fileStores": {
1004
1005         },
1006         "userPrincipalLookupService": {
```

```
1007         }
1008     },
1009     "nameCount": {
1010         "type": "integer",
1011         "format": "int32"
1012     },
1013     "userUuid": {
1014         "type": "string",
1015         "format": "uuid"
1016     },
1017     "Region": {
1018         "type": "object",
1019         "properties": {
1020             "globalRegion": {
1021                 "type": "boolean"
1022             }
1023         },
1024     },
1025     "ResourceDiscoveryRequest": {
1026         "type": "object",
1027         "properties": {
1028             "services": {
1029                 "type": "array",
1030                 "items": {
1031                     "type": "string"
1032                 }
1033             },
1034             "credentials": {
1035                 "$ref": "#/components/schemas/CloudCredentials"
1036             }
1037         },
1038     },
1039     "ResourceDetail": {
1040         "type": "object",
1041         "properties": {
1042             "resourceId": {
1043                 "type": "string"
1044             }
1045         },
1046     },
1047 }
```

```
1051     "name": {
1052         "type": "string"
1053     },
1054     "resourceType": {
1055         "type": "string"
1056     },
1057     "serviceCategory": {
1058         "type": "string"
1059     },
1060     "region": {
1061         "type": "string"
1062     },
1063     "tags": {
1064         "type": "object",
1065         "additionalProperties": {
1066             "type": "string"
1067         }
1068     }
1069 }
1070 }
1071 }
1072 }
1073 }
```

B.2 Service

```
1 {
2     "openapi": "3.1.0",
3     "info": {
4         "title": "OpenAPI definition",
5         "version": "v0"
6     },
7     "tags": [
8         {
9             "name": "Authentication",
10            "description": "Endpoints for user registration and login"
11        },
12        {
13            "name": "CloudSherpa Analytics",
14            "description": "CloudSherpa Analytics endpoints"
15        },
16        {
```

```
17     "name": "Optimization Recommendations",
18     "description": "Retrieve and manage cloud optimization
19         recommendations"
20 },
21 {
22     "name": "Intelligence",
23     "description": "CloudSherpa Forecasting Intelligence
24         Operations"
25 },
26 {
27     "name": "AI Dashboard",
28     "description": "Agentic Dashboard Construction Operations"
29 },
30 {
31     "name": "CloudSherpa Dashboard",
32     "description": "CloudSherpa Dashboard endpoints"
33 },
34 {
35     "name": "Webhooks",
36     "description": "CloudSherpa Webhooks CRUD operations"
37 },
38 {
39     "name": "Billing",
40     "description": "Billing KPI endpoints"
41 },
42 {
43     "name": "Budgets",
44     "description": "Create and manage billing budgets"
45 },
46 {
47     "name": "Alerts",
48     "description": "Operations for listing and managing alerts"
49 },
50 {
51     "name": "Cloud Resource Discovery",
52     "description": "Endpoints for discovering cloud services,
53         resources, and generating permissions policies"
54 },
55 {
56     "name": "Thresholds",
57     "description": "Manage resource-level threshold
58         configurations"
59 }
60 ],
```

```
57 "paths": {
58   "/dashboards/{dashboardId}/layout": {
59     "put": {
60       "tags": [
61         "CloudSherpa Dashboard"
62       ],
63       "summary": "Batch update the sizes and positions of all
64         widgets on a dashboard",
65       "operationId": "updateDashboardLayout",
66       "parameters": [
67         {
68           "name": "dashboardId",
69           "in": "path",
70           "required": true,
71           "schema": {
72             "type": "string",
73             "format": "uuid"
74           }
75         },
76       ],
77       "requestBody": {
78         "content": {
79           "application/json": {
80             "schema": {
81               "type": "array",
82               "items": {
83                 "$ref": "#/components/schemas/
84                   WidgetLayoutUpdatedDTO"
85               }
86             }
87           },
88           "required": true
89         },
90       ],
91       "responses": {
92         "200": {
93           "description": "OK",
94           "content": {
95             "*/*": {
96               "schema": {
97                 "type": "string",
98                 "additionalProperties": {
99                   "default": ""
100                 }
101               }
102             }
103           }
104         }
105       }
106     }
107   }
108 }
```

```

99         "default": ""
100     }
101 }
102 }
103 }
104 }
105 }
106 },
107 "/api/thresholds/{id}": {
108     "put": {
109         "tags": [
110             "Thresholds"
111         ],
112         "summary": "Update threshold",
113         "description": "Update an existing threshold.",
114         "operationId": "updateThreshold",
115         "parameters": [
116             {
117                 "name": "id",
118                 "in": "path",
119                 "description": "Threshold UUID",
120                 "required": true,
121                 "schema": {
122                     "type": "string",
123                     "default": ""
124                 }
125             }
126         ],
127         "requestBody": {
128             "content": {
129                 "application/json": {
130                     "schema": {
131                         "$ref": "#/components/schemas/UpdateThresholdRequest"
132                     }
133                 }
134             },
135             "required": true
136         },
137         "responses": {
138             "204": {
139                 "description": "Threshold updated"
140             },
141             "400": {

```

```
142         "description": "Invalid threshold request",
143         "content": {
144             "*/*": {
145                 "schema": {
146                     "type": "string",
147                     "additionalProperties": {
148                         "default": ""
149                     },
150                     "default": ""
151                 }
152             }
153         },
154     },
155     "404": {
156         "description": "Threshold not found",
157         "content": {
158             "*/*": {
159                 "schema": {
160                     "type": "string",
161                     "additionalProperties": {
162                         "default": ""
163                     },
164                     "default": ""
165                 }
166             }
167         }
168     },
169 },
170 "delete": {
171     "tags": [
172         "Thresholds"
173     ],
174     "summary": "Delete threshold",
175     "description": "Delete an existing threshold.",
176     "operationId": "deleteThreshold",
177     "parameters": [
178         {
179             "name": "id",
180             "in": "path",
181             "description": "Threshold UUID",
182             "required": true,
183             "schema": {
184                 "type": "string",
```

```
186         "default": ""
187     }
188 }
189 ],
190 "responses": {
191     "204": {
192         "description": "Threshold deleted"
193     },
194     "404": {
195         "description": "Threshold not found",
196         "content": {
197             "*/*": {
198                 "schema": {
199                     "type": "string",
200                     "additionalProperties": {
201                         "default": ""
202                     },
203                     "default": ""
204                 }
205             }
206         }
207     }
208 }
209 },
210 },
211 "/api/budgets/{id}": {
212     "put": {
213         "tags": [
214             "Budgets"
215         ],
216         "summary": "Update budget",
217         "description": "Update an existing budget record.",
218         "operationId": "updateBudget",
219         "parameters": [
220             {
221                 "name": "id",
222                 "in": "path",
223                 "description": "Budget UUID",
224                 "required": true,
225                 "schema": {
226                     "type": "string",
227                     "default": ""
228                 }
229             }
230         ]
231     }
232 }
```

```
230 ],
231 "requestBody": {
232   "content": {
233     "application/json": {
234       "schema": {
235         "$ref": "#/components/schemas/UpdateBudgetRequest
236       }
237     }
238   },
239   "required": true
240 },
241 "responses": {
242   "200": {
243     "description": "Budget updated",
244     "content": {
245       "*/*": {
246         "schema": {
247           "$ref": "#/components/schemas/BudgetResponse"
248         }
249       }
250     }
251   },
252   "400": {
253     "description": "Invalid budget request",
254     "content": {
255       "*/*": {
256         "schema": {
257           "type": "string",
258           "additionalProperties": {
259             "default": ""
260           },
261           "default": ""
262         }
263       }
264     }
265   },
266   "404": {
267     "description": "Budget not found",
268     "content": {
269       "*/*": {
270         "schema": {
271           "type": "string",
272           "additionalProperties": {
```

```
273         "default": ""
274     },
275     "default": ""
276 }
277 }
278 }
279 }
280 }
281 },
282 "delete": {
283     "tags": [
284         "Budgets"
285     ],
286     "summary": "Delete budget",
287     "description": "Delete or disable a budget.",
288     "operationId": "deleteBudget",
289     "parameters": [
290         {
291             "name": "id",
292             "in": "path",
293             "description": "Budget UUID",
294             "required": true,
295             "schema": {
296                 "type": "string",
297                 "default": ""
298             }
299         }
300     ],
301     "responses": {
302         "204": {
303             "description": "Deleted"
304         },
305         "404": {
306             "description": "Budget not found",
307             "content": {
308                 "*/*": {
309                     "schema": {
310                         "type": "string",
311                         "additionalProperties": {
312                             "default": ""
313                         },
314                         "default": ""
315                     }
316                 }
317             }
318         }
319     }
320 }
```

```
317     }
318   }
319 }
320 }
321 },
322 "/webhooks/edit/{webhookId}": {
323   "post": {
324     "tags": [
325       "Webhooks"
326     ],
327     "summary": "Edit existing webhook",
328     "operationId": "editWebhook",
329     "parameters": [
330       {
331         "name": "webhookId",
332         "in": "path",
333         "required": true,
334         "schema": {
335           "type": "string",
336           "format": "uuid"
337         }
338       }
339     ],
340     "requestBody": {
341       "description": "Edit webhook payload",
342       "content": {
343         "application/json": {
344           "schema": {
345             "$ref": "#/components/schemas/EditWebhookDto",
346             "additionalProperties": {
347               "default": ""
348             },
349             "default": ""
350           },
351           "examples": {
352             "Example payload": {
353               "description": "Example payload",
354               "value": {
355                 "name": "Billing update",
356                 "endpointUrl": "https://example.com/webhooks/billing-update",
357                 "eventTypes": [
358                   "billing.ingestion",
359                   "usage.threshold"
```

```
360         ],
361         "status": "PAUSED",
362         "cloudAccounts": [
363             "c2a49b36-55d2-4233-96dd-cc4e3ae97672"
364         ]
365     }
366 }
367 }
368 }
369 },
370 "required": true
371 },
372 "responses": {
373     "200": {
374         "description": "Succesfully added a new webhook"
375     },
376     "400": {
377         "description": "Bad request"
378     }
379 }
380 },
381 },
382 "/webhooks/dev": {
383     "post": {
384         "tags": [
385             "Webhooks"
386         ],
387         "operationId": "triggerDevEvent",
388         "responses": {
389             "200": {
390                 "description": "OK"
391             }
392         }
393     }
394 },
395 "/webhooks/add": {
396     "post": {
397         "tags": [
398             "Webhooks"
399         ],
400         "summary": "Add a new webhook",
401         "operationId": "addWebhook",
402         "requestBody": {
403             "description": "Add webhook payload",
```

```

404     "content": {
405         "application/json": {
406             "schema": {
407                 "$ref": "#/components/schemas/AddWebhookDto",
408                 "additionalProperties": {
409                     "default": ""
410                 },
411                 "default": ""
412             },
413             "examples": {
414                 "Example payload": {
415                     "description": "Example payload",
416                     "value": {
417                         "name": "Example Webhook",
418                         "endpointUrl": "https://example.com/api",
419                         "eventTypes": [
420                             "billing.ingestion",
421                             "usage.threshold"
422                         ],
423                         "cloudAccounts": [
424                             "c2a49b36-55d2-4233-96dd-cc4e3ae97672"
425                         ]
426                     }
427                 }
428             }
429         },
430     },
431     "required": true
432 },
433 "responses": {
434     "200": {
435         "description": "Successfully added a new webhook",
436         "content": {
437             "*/*": {
438                 "schema": {
439                     "$ref": "#/components/schemas/AddWebhookResponseDto",
440                     "additionalProperties": {
441                         "default": ""
442                     },
443                     "default": ""
444                 },
445                 "examples": {
446                     "Example secret": {

```

```

447         "description": "Example secret",
448         "value": {
449             "secret": "
                ceb3eebb7ad20566a86afc938ef49161d785791697acfa89321
                "
            }
450         }
451     }
452 }
453 }
454 }
455 },
456 "400": {
457     "description": "Bad request",
458     "content": {
459         "/*/*": {
460             "schema": {
461                 "type": "string",
462                 "additionalProperties": {
463                     "default": ""
464                 },
465                 "default": ""
466             }
467         }
468     }
469 }
470 }
471 }
472 },
473 "/preferences/theme": {
474     "get": {
475         "tags": [
476             "preferences-controller"
477         ],
478         "operationId": "getTheme",
479         "responses": {
480             "200": {
481                 "description": "OK",
482                 "content": {
483                     "/*/*": {
484                         "schema": {
485                             "type": "object",
486                             "additionalProperties": {
487                                 "type": "string"
488                             }

```

```
489         }
490     }
491 }
492 }
493 }
494 },
495 "post": {
496     "tags": [
497         "preferences-controller"
498     ],
499     "operationId": "updateTheme",
500     "requestBody": {
501         "content": {
502             "application/json": {
503                 "schema": {
504                     "$ref": "#/components/schemas/ThemeUpdateRequest"
505                 }
506             }
507         },
508         "required": true
509     },
510     "responses": {
511         "200": {
512             "description": "OK"
513         }
514     }
515 }
516 },
517 "/preferences/alert-notifications": {
518     "get": {
519         "tags": [
520             "preferences-controller"
521         ],
522         "operationId": "getAlertNotifications",
523         "responses": {
524             "200": {
525                 "description": "OK",
526                 "content": {
527                     "*/*": {
528                         "schema": {
529                             "type": "object",
530                             "additionalProperties": {
531                                 "type": "boolean"
532                             }
533                         }
534                     }
535                 }
536             }
537         }
538     }
539 }
```

```

533         }
534     }
535 }
536 }
537 }
538 },
539 "post": {
540     "tags": [
541         "preferences-controller"
542     ],
543     "operationId": "updateAlertNotifications",
544     "requestBody": {
545         "content": {
546             "application/json": {
547                 "schema": {
548                     "$ref": "#/components/schemas/AlertNotificationsUpdateRequest"
549                 }
550             }
551         },
552         "required": true
553     },
554     "responses": {
555         "200": {
556             "description": "OK"
557         }
558     }
559 },
560 },
561 "/optimization/recommendations/{recommendationId}/re-enable":
562 {
563     "post": {
564         "tags": [
565             "Optimization Recommendations"
566         ],
567         "operationId": "reEnableRecommendation",
568         "parameters": [
569             {
570                 "name": "recommendationId",
571                 "in": "path",
572                 "required": true,
573                 "schema": {
574                     "type": "string",
575                     "format": "uuid"

```

```

575     }
576   }
577 ],
578   "responses": {
579     "200": {
580       "description": "OK",
581       "content": {
582         "*/*": {
583           "schema": {
584             "type": "object",
585             "additionalProperties": {
586               }
587           }
588         }
589       }
590     }
591   }
592 }
593 },
594 },
595 "/optimization/recommendations/{recommendationId}/dismiss": {
596   "post": {
597     "tags": [
598       "Optimization Recommendations"
599     ],
600     "summary": "Dismiss a recommendation",
601     "description": "Marks an active recommendation as
        dismissed.",
602     "operationId": "dismissRecommendation",
603     "parameters": [
604       {
605         "name": "recommendationId",
606         "in": "path",
607         "description": "Recommendation UUID",
608         "required": true,
609         "schema": {
610           "type": "string",
611           "default": ""
612         }
613       }
614     ],
615     "responses": {
616       "200": {
617         "description": "OK",

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```

618         "content": {
619             "*/*": {
620                 "schema": {
621                     "type": "object",
622                     "additionalProperties": {
623
624                         }
625                     }
626                 }
627             }
628         }
629     }
630 },
631 "/optimization/recommendations/{recommendationId}/apply": {
632     "post": {
633         "tags": [
634             "Optimization Recommendations"
635         ],
636         "summary": "Apply a recommendation",
637         "description": "Marks an active recommendation as applied",
638         "operationId": "applyRecommendation",
639         "parameters": [
640             {
641                 "name": "recommendationId",
642                 "in": "path",
643                 "description": "Recommendation UUID",
644                 "required": true,
645                 "schema": {
646                     "type": "string",
647                     "default": ""
648                 }
649             }
650         ],
651         "responses": {
652             "200": {
653                 "description": "OK",
654                 "content": {
655                     "*/*": {
656                         "schema": {
657                             "type": "object",
658                             "additionalProperties": {

```

```

661     }
662   }
663 }
664 }
665 }
666 }
667 }
668 },
669 "/optimization/recommendations/{recommendationId}/acknowledge
    ": {
670   "post": {
671     "tags": [
672       "Optimization Recommendations"
673     ],
674     "summary": "Acknowledge a recommendation",
675     "description": "Marks an active recommendation as
        acknowledged.",
676     "operationId": "acknowledgeRecommendation",
677     "parameters": [
678       {
679         "name": "recommendationId",
680         "in": "path",
681         "description": "Recommendation UUID",
682         "required": true,
683         "schema": {
684           "type": "string",
685           "default": ""
686         }
687       }
688     ],
689     "responses": {
690       "200": {
691         "description": "OK",
692         "content": {
693           "*/*": {
694             "schema": {
695               "type": "object",
696               "additionalProperties": {
697
698             }
699           }
700         }
701       }
702     }

```

```

703     }
704   }
705 },
706 "/intelligence/forecasting/resource": {
707   "post": {
708     "tags": [
709       "Intelligence"
710     ],
711     "summary": "Forecast resource usage",
712     "description": "Generates predicted values and prediction
713                    intervals for a resource metric up to the requested
714                    forecast horizon",
715     "operationId": "resourceUsageForecast",
716     "requestBody": {
717       "description": "The resource, metric, and timestamp
718                     through which usage should be forecast",
719       "content": {
720         "application/json": {
721           "schema": {
722             "$ref": "#/components/schemas/
723                   ResourceUsageForecastRequestDto",
724             "additionalProperties": {
725               "default": ""
726             },
727             "default": ""
728           }
729         }
730       },
731       "required": true
732     },
733     "responses": {
734       "200": {
735         "description": "Resource usage forecast generated
736                       successfully",
737         "content": {
738           "application/json": {
739             "schema": {
740               "$ref": "#/components/schemas/
741                     ResourceUsageForecastResponseDto",
742             "additionalProperties": {
743               "default": ""
744             },
745             "default": ""
746           }
747         }
748       }
749     }
750   }

```

```
741     }
742   }
743 },
744 "400": {
745   "description": "The forecast request is malformed or
746     contains invalid values",
747   "content": {
748     "*/*": {
749       "schema": {
750         "type": "string",
751         "additionalProperties": {
752           "default": ""
753         },
754         "default": ""
755       }
756     }
757   },
758   "422": {
759     "description": "Insufficient historical data
760       available to make forecasting prediction",
761     "content": {
762       "*/*": {
763         "schema": {
764           "type": "string",
765           "additionalProperties": {
766             "default": ""
767           },
768           "default": ""
769         }
770       }
771     },
772     "404": {
773       "description": "Resource does not exist or MetricType
774         is not applicable to Resource",
775       "content": {
776         "*/*": {
777           "schema": {
778             "type": "string",
779             "additionalProperties": {
780               "default": ""
781             },
782             "default": ""

```

```

782     }
783   }
784 }
785 }
786 }
787 }
788 },
789 "/intelligence/forecasting/billing": {
790   "post": {
791     "tags": [
792       "Intelligence"
793     ],
794     "summary": "Forecast Billing for all non-credit charges",
795     "description": "Generates cumalative cost predection
                        value and analytics for all non-credit charges",
796     "operationId": "billingForecast",
797     "requestBody": {
798       "content": {
799         "application/json": {
800           "schema": {
801             "$ref": "#/components/schemas/
                        BillingForecastRequest"
802           }
803         }
804       },
805       "required": true
806     },
807     "responses": {
808       "200": {
809         "description": "Billing forecast generated
                        succesfully.",
810         "content": {
811           "application/json": {
812             "schema": {
813               "$ref": "#/components/schemas/
                        BillingForecastResponseDto",
814               "additionalProperties": {
815                 "default": ""
816               },
817               "default": ""
818             }
819           }
820         }
821       },

```

```

822     "422": {
823         "description": "Insufficient historical data
824             available to make any forecasting prediction",
825         "content": {
826             "*/*": {
827                 "schema": {
828                     "type": "string",
829                     "additionalProperties": {
830                         "default": ""
831                     },
832                     "default": ""
833                 }
834             }
835         },
836         "404": {
837             "description": "No non-credit charges found",
838             "content": {
839                 "*/*": {
840                     "schema": {
841                         "type": "string",
842                         "additionalProperties": {
843                             "default": ""
844                         },
845                         "default": ""
846                     }
847                 }
848             }
849         }
850     }
851 },
852 "/intelligence/forecasting/billing-charges": {
853     "post": {
854         "tags": [
855             "Intelligence"
856         ],
857         "summary": "Forecast Billing with Charges",
858         "description": "Generates cumalative cost predection
859             value and analytics for list of charges of which the
860             List can contain a single or multiple charges",
861         "operationId": "billingForecastWithCharges",
862         "requestBody": {
863             "content": {

```

```
863         "application/json": {
864             "schema": {
865                 "$ref": "#/components/schemas/
                        BillingForecastIndividualChargesRequestDto"
866             }
867         },
868     },
869     "required": true
870 },
871 "responses": {
872     "200": {
873         "description": "Billing forecast generated
                        succesfully.",
874         "content": {
875             "application/json": {
876                 "schema": {
877                     "$ref": "#/components/schemas/
                        BillingForecastResponseDto",
878                     "additionalProperties": {
879                         "default": ""
880                     },
881                     "default": ""
882                 }
883             }
884         },
885     },
886     "400": {
887         "description": "The billing forecast request is
                        malformed or contains invalid values",
888         "content": {
889             "*/*": {
890                 "schema": {
891                     "type": "string",
892                     "additionalProperties": {
893                         "default": ""
894                     },
895                     "default": ""
896                 }
897             }
898         },
899     },
900     "422": {
901         "description": "Insufficient historical data
                        available to make any forecasting prediction",
```

```

902         "content": {
903             "*/*": {
904                 "schema": {
905                     "type": "string",
906                     "additionalProperties": {
907                         "default": ""
908                     },
909                     "default": ""
910                 }
911             }
912         },
913     },
914     "404": {
915         "description": "None of the charges found",
916         "content": {
917             "*/*": {
918                 "schema": {
919                     "type": "string",
920                     "additionalProperties": {
921                         "default": ""
922                     },
923                     "default": ""
924                 }
925             }
926         }
927     },
928 },
929 },
930 },
931 "/gcp/connections": {
932     "get": {
933         "tags": [
934             "connection-query-controller"
935         ],
936         "operationId": "getAccountConnections",
937         "responses": {
938             "200": {
939                 "description": "OK",
940                 "content": {
941                     "*/*": {
942                         "schema": {
943                             "type": "array",
944                             "items": {
945                                 "$ref": "#/components/schemas/CloudAccount"

```

```

946         }
947     }
948 }
949 }
950 }
951 }
952 },
953 "post": {
954     "tags": [
955         "gcp-connection-persistence-controller"
956     ],
957     "operationId": "persistConnection",
958     "requestBody": {
959         "content": {
960             "application/json": {
961                 "schema": {
962                     "$ref": "#/components/schemas/
                        PersistGcpConnectionRequest"
963                 }
964             }
965         },
966         "required": true
967     },
968     "responses": {
969         "200": {
970             "description": "OK"
971         }
972     }
973 },
974 },
975 "/dashboards": {
976     "get": {
977         "tags": [
978             "CloudSherpa Dashboard"
979         ],
980         "summary": "Get all user dashboards with their widgets",
981         "operationId": "getUserDashboards",
982         "responses": {
983             "200": {
984                 "description": "OK",
985                 "content": {
986                     "*/*": {
987                         "schema": {
988                             "type": "array",

```

```

989         "additionalProperties": {
990             "default": ""
991         },
992         "contains": {
993             "default": ""
994         },
995         "items": {
996             "$ref": "#/components/schemas/DashboardDTO",
997             "default": ""
998         },
999         "unevaluatedItems": {
1000             "default": ""
1001         }
1002     }
1003 }
1004 }
1005 }
1006 }
1007 },
1008 "post": {
1009     "tags": [
1010         "CloudSherpa Dashboard"
1011     ],
1012     "summary": "Create a new blank dashboard",
1013     "operationId": "createDashboard",
1014     "requestBody": {
1015         "content": {
1016             "application/json": {
1017                 "schema": {
1018                     "$ref": "#/components/schemas/DashboardCreateDTO"
1019                 }
1020             }
1021         },
1022         "required": true
1023     },
1024     "responses": {
1025         "201": {
1026             "description": "Created",
1027             "content": {
1028                 "*/*": {
1029                     "schema": {
1030                         "$ref": "#/components/schemas/DashboardDTO",
1031                         "additionalProperties": {
1032                             "default": ""

```

```

1033         },
1034         "default": ""
1035     }
1036 }
1037 }
1038 }
1039 }
1040 }
1041 },
1042 "/dashboards/{dashboardId}/widgets": {
1043     "post": {
1044         "tags": [
1045             "CloudSherpa Dashboard"
1046         ],
1047         "summary": "Add a new widget to a dashboard",
1048         "operationId": "createChartWidget",
1049         "parameters": [
1050             {
1051                 "name": "dashboardId",
1052                 "in": "path",
1053                 "required": true,
1054                 "schema": {
1055                     "type": "string",
1056                     "format": "uuid"
1057                 }
1058             }
1059         ],
1060         "requestBody": {
1061             "content": {
1062                 "application/json": {
1063                     "schema": {
1064                         "oneOf": [
1065                             {
1066                                 "$ref": "#/components/schemas/ChartWidgetDTO"
1067                             },
1068                             {
1069                                 "$ref": "#/components/schemas/KpiWidgetDTO"
1070                             }
1071                         ]
1072                     }
1073                 }
1074             },
1075             "required": true
1076         },

```

```

1077     "responses": {
1078         "201": {
1079             "description": "Created",
1080             "content": {
1081                 "*/*": {
1082                     "schema": {
1083                         "$ref": "#/components/schemas/WidgetDTO",
1084                         "additionalProperties": {
1085                             "default": ""
1086                         },
1087                         "default": ""
1088                     }
1089                 }
1090             }
1091         }
1092     }
1093 },
1094 "/billing/kpis/preview": {
1095     "post": {
1096         "tags": [
1097             "Billing"
1098         ],
1099         "summary": "Preview billing KPI card",
1100         "description": "Returns a preview value for a KPI card
1101             using selected resources, date range, and aggregation
1102             ",
1103         "operationId": "previewKpi",
1104         "requestBody": {
1105             "content": {
1106                 "application/json": {
1107                     "schema": {
1108                         "$ref": "#/components/schemas/BillingKpiRequest"
1109                     }
1110                 },
1111                 "required": true
1112             },
1113             "responses": {
1114                 "200": {
1115                     "description": "KPI preview generated",
1116                     "content": {
1117                         "*/*": {
1118                             "schema": {

```

```
1119         "$ref": "#/components/schemas/  
1120             BillingKpiResponse",  
1121         "additionalProperties": {  
1122             "default": ""  
1123         },  
1124         "default": ""  
1125     }  
1126 }  
1127 },  
1128 "400": {  
1129     "description": "Invalid request",  
1130     "content": {  
1131         "*/*": {  
1132             "schema": {  
1133                 "$ref": "#/components/schemas/  
1134                     BillingKpiResponse"  
1135             }  
1136         }  
1137     }  
1138 }  
1139 },  
1140 },  
1141 "/billing/charges": {  
1142     "post": {  
1143         "tags": [  
1144             "Billing"  
1145         ],  
1146         "summary": "Get billing resources",  
1147         "operationId": "getResources",  
1148         "requestBody": {  
1149             "content": {  
1150                 "application/json": {  
1151                     "schema": {  
1152                         "$ref": "#/components/schemas/  
1153                             BillingKpiChargesRequest"  
1154                     }  
1155                 }  
1156             },  
1157             "required": true  
1158         },  
1159         "responses": {  
1160             "200": {
```

```

1160         "description": "Resources returned successfully",
1161         "content": {
1162             "*/*": {
1163                 "schema": {
1164                     "type": "array",
1165                     "additionalProperties": {
1166                         "default": ""
1167                     },
1168                     "contains": {
1169                         "default": ""
1170                     },
1171                     "items": {
1172                         "$ref": "#/components/schemas/
BillingChargeResponse",
1173                         "default": ""
1174                     },
1175                     "unevaluatedItems": {
1176                         "default": ""
1177                     }
1178                 }
1179             }
1180         }
1181     },
1182     "/azure/connections": {
1183         "get": {
1184             "tags": [
1185                 "connection-query-controller"
1186             ],
1187             "operationId": "getAccountConnections_1",
1188             "responses": {
1189                 "200": {
1190                     "description": "OK",
1191                     "content": {
1192                         "*/*": {
1193                             "schema": {
1194                                 "type": "array",
1195                                 "items": {
1196                                     "$ref": "#/components/schemas/CloudAccount"
1197                                 }
1198                             }
1199                         }
1200                     }
1201                 }
1202             }
1203         }
1204     }
1205 }

```

```

1203     }
1204   }
1205 }
1206 },
1207 "post": {
1208   "tags": [
1209     "azure-connection-persistence-controller"
1210   ],
1211   "operationId": "persistConnection_1",
1212   "requestBody": {
1213     "content": {
1214       "application/json": {
1215         "schema": {
1216           "$ref": "#/components/schemas/
             PersistAzureConnectionRequest"
1217         }
1218       }
1219     },
1220     "required": true
1221   },
1222   "responses": {
1223     "200": {
1224       "description": "OK"
1225     }
1226   }
1227 }
1228 },
1229 "/aws/connections": {
1230   "get": {
1231     "tags": [
1232       "connection-query-controller"
1233     ],
1234     "operationId": "getAccountConnections_2",
1235     "responses": {
1236       "200": {
1237         "description": "OK",
1238         "content": {
1239           "*/*": {
1240             "schema": {
1241               "type": "array",
1242               "items": {
1243                 "$ref": "#/components/schemas/CloudAccount"
1244               }
1245             }

```

```
1246         }
1247     }
1248 }
1249 },
1250 "post": {
1251     "tags": [
1252         "aws-connection-persistence-controller"
1253     ],
1254     "operationId": "persistConnection_2",
1255     "requestBody": {
1256         "content": {
1257             "application/json": {
1258                 "schema": {
1259                     "$ref": "#/components/schemas/
1260                     PersistAwsConnectionRequest"
1261                 }
1262             }
1263         },
1264         "required": true
1265     },
1266     "responses": {
1267         "200": {
1268             "description": "OK"
1269         }
1270     }
1271 },
1272 },
1273 "/auth/register": {
1274     "post": {
1275         "tags": [
1276             "Authentication"
1277         ],
1278         "summary": "Register a new user",
1279         "operationId": "register",
1280         "requestBody": {
1281             "content": {
1282                 "application/json": {
1283                     "schema": {
1284                         "$ref": "#/components/schemas/RegisterRequest"
1285                     }
1286                 }
1287             },
1288             "required": true
```

```
1289     },
1290     "responses": {
1291         "201": {
1292             "description": "User successfully registered",
1293             "content": {
1294                 "*/*": {
1295                     "schema": {
1296                         "type": "string",
1297                         "additionalProperties": {
1298                             "default": ""
1299                         },
1300                         "default": ""
1301                     }
1302                 }
1303             }
1304         },
1305         "400": {
1306             "description": "Invalid input"
1307         },
1308         "409": {
1309             "description": "Email already in use"
1310         }
1311     }
1312 },
1313 "/auth/refresh": {
1314     "post": {
1315         "tags": [
1316             "Authentication"
1317         ],
1318         "summary": "Refresh authentication tokens",
1319         "operationId": "refresh",
1320         "parameters": [
1321             {
1322                 "name": "refresh_token",
1323                 "in": "cookie",
1324                 "required": false,
1325                 "schema": {
1326                     "type": "string"
1327                 }
1328             }
1329         ],
1330         "responses": {
1331             "200": {
```

```
1333         "description": "Tokens successfully refreshed",
1334         "content": {
1335             "*/*": {
1336                 "schema": {
1337                     "$ref": "#/components/schemas/AuthUserResponse"
1338                 }
1339             }
1340         },
1341     },
1342     "401": {
1343         "description": "Invalid or expired refresh token",
1344         "content": {
1345             "*/*": {
1346                 "schema": {
1347                     "$ref": "#/components/schemas/AuthUserResponse"
1348                 }
1349             }
1350         }
1351     }
1352 }
1353 },
1354 "/auth/logout": {
1355     "post": {
1356         "tags": [
1357             "Authentication"
1358         ],
1359         "summary": "Logout user",
1360         "operationId": "logout",
1361         "parameters": [
1362             {
1363                 "name": "refresh_token",
1364                 "in": "cookie",
1365                 "required": false,
1366                 "schema": {
1367                     "type": "string"
1368                 }
1369             }
1370         ],
1371         "responses": {
1372             "200": {
1373                 "description": "Successfully logged out"
1374             }
1375         }
1376     }
```

```
1377     }
1378 },
1379 "/auth/login": {
1380   "post": {
1381     "tags": [
1382       "Authentication"
1383     ],
1384     "summary": "Login user",
1385     "operationId": "login",
1386     "requestBody": {
1387       "content": {
1388         "application/json": {
1389           "schema": {
1390             "$ref": "#/components/schemas/LoginRequest"
1391           }
1392         }
1393       },
1394       "required": true
1395     },
1396     "responses": {
1397       "200": {
1398         "description": "Successfully authenticated",
1399         "content": {
1400           "*/*": {
1401             "schema": {
1402               "$ref": "#/components/schemas/AuthUserResponse"
1403             },
1404             "additionalProperties": {
1405               "default": ""
1406             },
1407             "default": ""
1408           }
1409         }
1410       },
1411       "400": {
1412         "description": "Email or password is missing",
1413         "content": {
1414           "*/*": {
1415             "schema": {
1416               "$ref": "#/components/schemas/AuthUserResponse"
1417             }
1418           }
1419         }
1420       }
1421     }
1422   }
1423 }
```

```
1420     },
1421     "401": {
1422         "description": "Invalid credentials",
1423         "content": {
1424             "*/*": {
1425                 "schema": {
1426                     "$ref": "#/components/schemas/AuthUserResponse"
1427                 }
1428             }
1429         }
1430     }
1431 }
1432 },
1433 "/api/thresholds": {
1434     "get": {
1435         "tags": [
1436             "Thresholds"
1437         ],
1438         "summary": "List thresholds",
1439         "description": "List thresholds, optionally filtered by
1440             resource.",
1441         "operationId": "listThresholds",
1442         "parameters": [
1443             {
1444                 "name": "resourceId",
1445                 "in": "query",
1446                 "description": "Resource UUID to filter thresholds",
1447                 "required": false,
1448                 "schema": {
1449                     "type": "string",
1450                     "default": ""
1451                 }
1452             }
1453         ],
1454         "responses": {
1455             "200": {
1456                 "description": "List of thresholds",
1457                 "content": {
1458                     "application/json": {
1459                         "schema": {
1460                             "type": "array",
1461                             "additionalProperties": {
1462                                 "default": ""
```

```

1463         },
1464         "contains": {
1465             "default": ""
1466         },
1467         "items": {
1468             "$ref": "#/components/schemas/
                ThresholdResponse",
1469             "default": ""
1470         },
1471         "unevaluatedItems": {
1472             "default": ""
1473         }
1474     }
1475 }
1476 }
1477 }
1478 },
1479 },
1480 "post": {
1481     "tags": [
1482         "Thresholds"
1483     ],
1484     "summary": "Create threshold",
1485     "description": "Create a resource-level threshold.",
1486     "operationId": "createThreshold",
1487     "requestBody": {
1488         "description": "Threshold creation request",
1489         "content": {
1490             "application/json": {
1491                 "schema": {
1492                     "$ref": "#/components/schemas/
                        CreateThresholdRequest",
1493                     "additionalProperties": {
1494                         "default": ""
1495                     },
1496                     "default": ""
1497                 }
1498             }
1499         },
1500         "required": true
1501     },
1502     "responses": {
1503         "201": {
1504             "description": "Threshold created",

```

```

1505         "content": {
1506             "application/json": {
1507                 "schema": {
1508                     "$ref": "#/components/schemas/ThresholdResponse",
1509                     "additionalProperties": {
1510                         "default": ""
1511                     },
1512                     "default": ""
1513                 }
1514             }
1515         },
1516         "400": {
1517             "description": "Invalid threshold request",
1518             "content": {
1519                 "*/*": {
1520                     "schema": {
1521                         "type": "string",
1522                         "additionalProperties": {
1523                             "default": ""
1524                         },
1525                         "default": ""
1526                     }
1527                 }
1528             }
1529         }
1530     },
1531     },
1532 },
1533 "/api/cloud-resources/services": {
1534     "post": {
1535         "tags": [
1536             "Cloud Resource Discovery"
1537         ],
1538         "summary": "Get available cloud services",
1539         "description": "Retrieves a list of all supported",
1540             "services for a given cloud provider",
1541         "operationId": "getServices",
1542         "requestBody": {
1543             "description": "Cloud provider name (e.g. aws, azure, gcp)",
1544             "content": {
1545                 "application/json": {

```

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1546         "schema": {
1547             "type": "string",
1548             "additionalProperties": {
1549                 "default": ""
1550             },
1551             "default": ""
1552         }
1553     },
1554 },
1555     "required": true
1556 },
1557 "responses": {
1558     "200": {
1559         "description": "Services successfully returned",
1560         "content": {
1561             "application/json": {
1562                 "schema": {
1563                     "type": "string",
1564                     "additionalProperties": {
1565                         "default": ""
1566                     },
1567                     "default": ""
1568                 }
1569             }
1570         }
1571     }
1572 },
1573 },
1574 },
1575 "/api/cloud-resources/resources/{provider}": {
1576     "post": {
1577         "tags": [
1578             "Cloud Resource Discovery"
1579         ],
1580         "summary": "Get cloud resources",
1581         "description": "Retrieves all discoverable resources for
                        the specified cloud provider using the supplied
                        credentials",
1582         "operationId": "getResources_1",
1583         "parameters": [
1584             {
1585                 "name": "provider",
1586                 "in": "path",
1587                 "required": true,

```

```

1588         "schema": {
1589             "type": "string"
1590         }
1591     },
1592 ],
1593 "requestBody": {
1594     "content": {
1595         "application/json": {
1596             "schema": {
1597                 "$ref": "#/components/schemas/
                    ResourceDiscoveryDto"
1598             }
1599         }
1600     },
1601     "required": true
1602 },
1603 "responses": {
1604     "200": {
1605         "description": "Resources successfully returned",
1606         "content": {
1607             "application/json": {
1608                 "schema": {
1609                     "$ref": "#/components/schemas/ResourceDetailDto
                        ",
1610                     "additionalProperties": {
1611                         "default": ""
1612                     },
1613                     "default": ""
1614                 }
1615             }
1616         }
1617     }
1618 },
1619 },
1620 },
1621 "/api/cloud-resources/gcp/permissions": {
1622     "post": {
1623         "tags": [
1624             "Cloud Resource Discovery"
1625         ],
1626         "summary": "Generate GCP permissions list",
1627         "description": "Generates a least-privilege GCP
                    permissions list for the selected GCP services",
1628         "operationId": "generateGcpPermissions",

```

```

1629     "requestBody": {
1630         "content": {
1631             "application/json": {
1632                 "schema": {
1633                     "type": "array",
1634                     "items": {
1635                         "type": "string"
1636                     }
1637                 }
1638             }
1639         },
1640         "required": true
1641     },
1642     "responses": {
1643         "200": {
1644             "description": "Permissions list successfully
1645                             generated",
1646             "content": {
1647                 "application/json": {
1648                     "schema": {
1649                         "type": "string",
1650                         "additionalProperties": {
1651                             "default": ""
1652                         },
1653                         "default": ""
1654                     }
1655                 }
1656             }
1657         }
1658     },
1659     "/api/cloud-resources/azure/permissions": {
1660         "post": {
1661             "tags": [
1662                 "Cloud Resource Discovery"
1663             ],
1664             "summary": "Generate Azure permissions list",
1665             "description": "Generates a least-privilege Azure
1666                             permissions list for the selected Azure services",
1667             "operationId": "generateAzurePermissions",
1668             "requestBody": {
1669                 "content": {
1670                     "application/json": {

```

```
1671         "schema": {
1672             "type": "array",
1673             "items": {
1674                 "type": "string"
1675             }
1676         }
1677     },
1678 },
1679     "required": true
1680 },
1681     "responses": {
1682         "200": {
1683             "description": "Permissions list successfully
1684                 generated",
1685             "content": {
1686                 "application/json": {
1687                     "schema": {
1688                         "type": "string",
1689                         "additionalProperties": {
1690                             "default": ""
1691                         },
1692                         "default": ""
1693                     }
1694                 }
1695             }
1696         }
1697     },
1698 },
1699     "/api/cloud-resources/aws/permissions": {
1700         "post": {
1701             "tags": [
1702                 "Cloud Resource Discovery"
1703             ],
1704             "summary": "Generate AWS IAM policy",
1705             "description": "Generates a least-privilege AWS IAM
1706                 policy for the selected AWS services",
1707             "operationId": "generateAwsPermissions",
1708             "requestBody": {
1709                 "content": {
1710                     "application/json": {
1711                         "schema": {
1712                             "type": "array",
1713                             "items": {
```

```
1713         "type": "string"
1714     }
1715 }
1716 }
1717 },
1718 "required": true
1719 },
1720 "responses": {
1721     "200": {
1722         "description": "IAM policy successfully generated",
1723         "content": {
1724             "application/json": {
1725                 "schema": {
1726                     "type": "string",
1727                     "additionalProperties": {
1728                         "default": ""
1729                     },
1730                     "default": ""
1731                 }
1732             }
1733         }
1734     }
1735 }
1736 },
1737 "/api/budgets": {
1738     "get": {
1739         "tags": [
1740             "Budgets"
1741         ],
1742         "summary": "List budgets",
1743         "description": "List budgets for the current tenant; optional scope filter.",
1744         "operationId": "listBudgets",
1745         "parameters": [
1746             {
1747                 "name": "scope",
1748                 "in": "query",
1749                 "description": "Scope (TENANT|ACCOUNT|RESOURCE)",
1750                 "required": false,
1751                 "schema": {
1752                     "type": "string",
1753                     "default": ""
1754                 }
1755             }
1756         ]
1757     }
1758 }
```

```
1756     },
1757     {
1758         "name": "scopeId",
1759         "in": "query",
1760         "description": "Scope id (UUID) when scope is ACCOUNT
                           or RESOURCE",
1761         "required": false,
1762         "schema": {
1763             "type": "string",
1764             "default": ""
1765         }
1766     }
1767 ],
1768 "responses": {
1769     "200": {
1770         "description": "List of budgets",
1771         "content": {
1772             "application/json": {
1773                 "schema": {
1774                     "type": "array",
1775                     "additionalProperties": {
1776                         "default": ""
1777                     },
1778                     "contains": {
1779                         "default": ""
1780                     },
1781                     "items": {
1782                         "$ref": "#/components/schemas/BudgetResponse",
1783                         "default": ""
1784                     },
1785                     "unevaluatedItems": {
1786                         "default": ""
1787                     }
1788                 }
1789             }
1790         }
1791     }
1792 },
1793 },
1794 "post": {
1795     "tags": [
1796         "Budgets"
1797     ],
```

```
1798     "summary": "Create budget",
1799     "description": "Create a tenant/account/resource scoped
1800         budget.",
1801     "operationId": "createBudget",
1802     "requestBody": {
1803         "description": "Budget creation request",
1804         "content": {
1805             "application/json": {
1806                 "schema": {
1807                     "$ref": "#/components/schemas/CreateBudgetRequest",
1808                     "additionalProperties": {
1809                         "default": ""
1810                     },
1811                     "default": ""
1812                 }
1813             },
1814             "required": true
1815         },
1816         "responses": {
1817             "201": {
1818                 "description": "Budget created",
1819                 "content": {
1820                     "application/json": {
1821                         "schema": {
1822                             "$ref": "#/components/schemas/BudgetResponse",
1823                             "additionalProperties": {
1824                                 "default": ""
1825                             },
1826                             "default": ""
1827                         }
1828                     }
1829                 },
1830             },
1831             "400": {
1832                 "description": "Invalid budget request",
1833                 "content": {
1834                     "*/*": {
1835                         "schema": {
1836                             "type": "string",
1837                             "additionalProperties": {
1838                                 "default": ""
1839                             },
```

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1840         "default": ""
1841     }
1842 }
1843 }
1844 }
1845 }
1846 }
1847 },
1848 "/api/alerts/{id}/notification-silence": {
1849     "post": {
1850         "tags": [
1851             "Alerts"
1852         ],
1853         "operationId": "updateNotificationSilence",
1854         "parameters": [
1855             {
1856                 "name": "id",
1857                 "in": "path",
1858                 "required": true,
1859                 "schema": {
1860                     "type": "string",
1861                     "format": "uuid"
1862                 }
1863             }
1864         ],
1865         "requestBody": {
1866             "content": {
1867                 "application/json": {
1868                     "schema": {
1869                         "$ref": "#/components/schemas/
AlertNotificationSilenceRequest"
1870                     }
1871                 }
1872             },
1873             "required": true
1874         },
1875         "responses": {
1876             "200": {
1877                 "description": "OK"
1878             }
1879         }
1880     }
1881 },
1882 "/api/alerts/{id}/enable": {

```

```
1883     "post": {
1884         "tags": [
1885             "Alerts"
1886         ],
1887         "operationId": "enable",
1888         "parameters": [
1889             {
1890                 "name": "id",
1891                 "in": "path",
1892                 "description": "Alert UUID",
1893                 "required": true,
1894                 "schema": {
1895                     "type": "string",
1896                     "default": ""
1897                 }
1898             }
1899         ],
1900         "responses": {
1901             "200": {
1902                 "description": "OK"
1903             }
1904         }
1905     },
1906 },
1907 "/api/alerts/{id}/disable": {
1908     "post": {
1909         "tags": [
1910             "Alerts"
1911         ],
1912         "operationId": "disable",
1913         "parameters": [
1914             {
1915                 "name": "id",
1916                 "in": "path",
1917                 "description": "Alert UUID",
1918                 "required": true,
1919                 "schema": {
1920                     "type": "string",
1921                     "default": ""
1922                 }
1923             }
1924         ],
1925         "responses": {
1926             "200": {
```

```

1927         "description": "OK"
1928     }
1929 }
1930 }
1931 },
1932 "/analytics/historical-resource-metric": {
1933     "post": {
1934         "tags": [
1935             "CloudSherpa Analytics"
1936         ],
1937         "summary": "Get historical metric data for a resource",
1938         "operationId": "postMethodName",
1939         "requestBody": {
1940             "description": "Resource metric query parameters",
1941             "content": {
1942                 "application/json": {
1943                     "schema": {
1944                         "$ref": "#/components/schemas/
1945                             ResourceMetricHistoricalRequestDto",
1946                         "additionalProperties": {
1947                             "default": ""
1948                         },
1949                         "default": ""
1950                     }
1951                 },
1952                 "required": true
1953             },
1954             "responses": {
1955                 "200": {
1956                     "description": "Return metric values and timestamps
1957                         for the resource",
1958                     "content": {
1959                         "*/*": {
1960                             "schema": {
1961                                 "$ref": "#/components/schemas/
1962                                     ResourceMetricHistoricalResponseDto",
1963                                 "additionalProperties": {
1964                                     "default": ""
1965                                 },
1966                                 "default": ""
1967                             }
1968                         }
1969                     }
1970                 }
1971             }
1972         }
1973     }
1974 }

```

```

1968     },
1969     "400": {
1970         "description": "Invalid request",
1971         "content": {
1972             "*/*": {
1973                 "schema": {
1974                     "type": "string",
1975                     "additionalProperties": {
1976                         "default": ""
1977                     },
1978                     "default": ""
1979                 }
1980             }
1981         }
1982     },
1983     "404": {
1984         "description": "No resource metrics found",
1985         "content": {
1986             "*/*": {
1987                 "schema": {
1988                     "type": "string",
1989                     "additionalProperties": {
1990                         "default": ""
1991                     },
1992                     "default": ""
1993                 }
1994             }
1995         }
1996     }
1997 },
1998 },
1999 },
2000 "/analytics/downsampled-historical-series": {
2001     "post": {
2002         "tags": [
2003             "CloudSherpa Analytics"
2004         ],
2005         "summary": "Get downsampled historical metric series for
a specific resource and metric",
2006         "operationId": "postMethodName_1",
2007         "requestBody": {
2008             "description": "POST parameters for downsampled
historical metric series",
2009             "content": {

```

```
2010         "application/json": {
2011             "schema": {
2012                 "$ref": "#/components/schemas/
2013                     DownsampedSeriesRequestDto",
2014                 "additionalProperties": {
2015                     "default": ""
2016                 },
2017                 "default": ""
2018             }
2019         },
2020         "required": true
2021     },
2022     "responses": {
2023         "200": {
2024             "description": "Succesfully returns downsamped
2025                 historical metric series",
2026             "content": {
2027                 "*/*": {
2028                     "schema": {
2029                         "type": "array",
2030                         "additionalProperties": {
2031                             "default": ""
2032                         },
2033                         "contains": {
2034                             "default": ""
2035                         },
2036                         "items": {
2037                             "$ref": "#/components/schemas/
2038                                 NormalizedMetrics",
2039                             "default": ""
2040                         },
2041                         "unevaluatedItems": {
2042                             "default": ""
2043                         }
2044                     }
2045                 }
2046             }
2047         }
2048     },
2049     "/ai/sessions": {
2050         "post": {
```

```

2051     "tags": [
2052         "AI Dashboard"
2053     ],
2054     "summary": "Create an AI dashboard session",
2055     "description": "Creates a new isolated AI dashboard
2056         construction session for the authenticated user",
2057     "operationId": "createSession",
2058     "responses": {
2059         "201": {
2060             "description": "AI dashboard session created
2061                 successfully",
2062             "content": {
2063                 "application/json": {
2064                     "schema": {
2065                         "$ref": "#/components/schemas/
2066                             AiSessionResponseDto",
2067                         "additionalProperties": {
2068                             "default": ""
2069                         },
2070                         "default": ""
2071                     },
2072                     "default": ""
2073                 },
2074             },
2075         },
2076         "401": {
2077             "description": "User is not authenticated",
2078             "content": {
2079                 "*/*": {
2080                     "schema": {
2081                         "type": "string",
2082                         "additionalProperties": {
2083                             "default": ""
2084                         },
2085                         "default": ""
2086                     },
2087                     "default": ""
2088                 },
2089             },
2090         },
2091     },
2092     "/ai/sessions/{sessionId}/versions/{versionId}/apply": {
2093         "post": {
2094             "tags": [

```

```
2092     "AI Dashboard"
2093 ],
2094 "summary": "Apply an AI dashboard version",
2095 "description": "Applies a staged AI dashboard version by
                replacing the dashboard the session started on or
                creating a new dashboard",
2096 "operationId": "applyVersion",
2097 "parameters": [
2098     {
2099         "name": "sessionId",
2100         "in": "path",
2101         "required": true,
2102         "schema": {
2103             "type": "string",
2104             "format": "uuid"
2105         }
2106     },
2107     {
2108         "name": "versionId",
2109         "in": "path",
2110         "required": true,
2111         "schema": {
2112             "type": "string",
2113             "format": "uuid"
2114         }
2115     }
2116 ],
2117 "requestBody": {
2118     "content": {
2119         "application/json": {
2120             "schema": {
2121                 "$ref": "#/components/schemas/
                AiDashboardApplyRequestDto"
2122             }
2123         }
2124     },
2125     "required": true
2126 },
2127 "responses": {
2128     "200": {
2129         "description": "Dashboard version applied
                successfully",
2130         "content": {
2131             "application/json": {
```

```
2132         "schema": {
2133             "type": "string"
2134         }
2135     }
2136 },
2137 "400": {
2138     "description": "The apply request is invalid",
2139     "content": {
2140         "*/*": {
2141             "schema": {
2142                 "type": "string",
2143                 "additionalProperties": {
2144                     "default": ""
2145                 },
2146                 "default": ""
2147             }
2148         }
2149     }
2150 },
2151 "404": {
2152     "description": "AI session, dashboard version, or
2153         target dashboard does not exist or does not belong
2154         to the authenticated user",
2155     "content": {
2156         "*/*": {
2157             "schema": {
2158                 "type": "string",
2159                 "additionalProperties": {
2160                     "default": ""
2161                 },
2162                 "default": ""
2163             }
2164         }
2165     }
2166 },
2167 },
2168 "/ai/sessions/{sessionId}/versions/{versionId}/active": {
2169     "post": {
2170         "tags": [
2171             "AI Dashboard"
2172         ],
2173     },
```

```
2174     "summary": "Activate an AI dashboard version",
2175     "description": "Marks the selected dashboard version as
        active for the AI session and makes it the parent
        context for the next generated version",
2176     "operationId": "activateVersion",
2177     "parameters": [
2178         {
2179             "name": "sessionId",
2180             "in": "path",
2181             "required": true,
2182             "schema": {
2183                 "type": "string",
2184                 "format": "uuid"
2185             }
2186         },
2187         {
2188             "name": "versionId",
2189             "in": "path",
2190             "required": true,
2191             "schema": {
2192                 "type": "string",
2193                 "format": "uuid"
2194             }
2195         }
2196     ],
2197     "responses": {
2198         "200": {
2199             "description": "Dashboard version activated
        successfully",
2200             "content": {
2201                 "application/json": {
2202                     "schema": {
2203                         "$ref": "#/components/schemas/
        AiVersionResponseDto",
2204                         "additionalProperties": {
2205                             "default": ""
2206                         },
2207                         "default": ""
2208                     }
2209                 }
2210             }
2211         },
2212         "404": {
```

```
2213         "description": "AI session or dashboard version does
           not exist or does not belong to the authenticated
           user",
2214         "content": {
2215             "*/*": {
2216                 "schema": {
2217                     "type": "string",
2218                     "additionalProperties": {
2219                         "default": ""
2220                     },
2221                     "default": ""
2222                 }
2223             }
2224         }
2225     },
2226 },
2227 },
2228 },
2229 "/ai/dashboard/plan": {
2230     "post": {
2231         "tags": [
2232             "AI Dashboard"
2233         ],
2234         "summary": "Generate an AI dashboard plan",
2235         "description": "Processes a natural-language dashboard
           request and creates a new staged dashboard version",
2236         "operationId": "createDashboardPlan",
2237         "requestBody": {
2238             "description": "Natural-language dashboard instruction
           and the AI session in which it should be processed",
2239             "content": {
2240                 "application/json": {
2241                     "schema": {
2242                         "$ref": "#/components/schemas/
           AiDashboardPlanRequestDto",
2243                         "additionalProperties": {
2244                             "default": ""
2245                         },
2246                         "default": ""
2247                     }
2248                 }
2249             },
2250             "required": true
2251         },
```

```
2252     "responses": {
2253       "200": {
2254         "description": "Dashboard plan generated successfully",
2255         "content": {
2256           "application/json": {
2257             "schema": {
2258               "$ref": "#/components/schemas/AiDashboardPlanResponseDto",
2259               "additionalProperties": {
2260                 "default": ""
2261               },
2262               "default": ""
2263             }
2264           }
2265         },
2266       },
2267       "400": {
2268         "description": "The dashboard request is malformed",
2269         "content": {
2270           "*/*": {
2271             "schema": {
2272               "type": "string",
2273               "additionalProperties": {
2274                 "default": ""
2275               },
2276               "default": ""
2277             }
2278           }
2279         },
2280       },
2281       "404": {
2282         "description": "The AI session does not exist or does not belong to the authenticated user",
2283         "content": {
2284           "*/*": {
2285             "schema": {
2286               "type": "string",
2287               "additionalProperties": {
2288                 "default": ""
2289               },
2290               "default": ""
2291             }
2292           }
2293         }
2294       }
2295     }
2296   }
2297 }
```

```

2293     }
2294   }
2295 }
2296 }
2297 },
2298 "/dashboards/{dashboardId}/window": {
2299   "patch": {
2300     "tags": [
2301       "CloudSherpa Dashboard"
2302     ],
2303     "operationId": "updateDashboardWindow",
2304     "parameters": [
2305       {
2306         "name": "dashboardId",
2307         "in": "path",
2308         "required": true,
2309         "schema": {
2310           "type": "string",
2311           "format": "uuid"
2312         }
2313       }
2314     ],
2315     "requestBody": {
2316       "content": {
2317         "application/json": {
2318           "schema": {
2319             "$ref": "#/components/schemas/
DashboardWindowUpdatedDTO"
2320           }
2321         }
2322       },
2323       "required": true
2324     },
2325     "responses": {
2326       "200": {
2327         "description": "OK",
2328         "content": {
2329           "*/*": {
2330             "schema": {
2331               "type": "string",
2332               "additionalProperties": {
2333                 "default": ""
2334               },
2335               "default": ""

```

```

2336         }
2337     }
2338 }
2339 }
2340 }
2341 }
2342 },
2343 "/dashboards/widgets/{widgetId}/config": {
2344     "patch": {
2345         "tags": [
2346             "CloudSherpa Dashboard"
2347         ],
2348         "summary": "Update a widget's visual or data
                configuration",
2349         "operationId": "updateWidgetConfig",
2350         "parameters": [
2351             {
2352                 "name": "widgetId",
2353                 "in": "path",
2354                 "required": true,
2355                 "schema": {
2356                     "type": "string",
2357                     "format": "uuid"
2358                 }
2359             }
2360         ],
2361         "requestBody": {
2362             "content": {
2363                 "application/json": {
2364                     "schema": {
2365                         "oneOf": [
2366                             {
2367                                 "$ref": "#/components/schemas/
                                    ChartWidgetConfigUpdatedDTO"
2368                             },
2369                             {
2370                                 "$ref": "#/components/schemas/
                                    KpiWidgetConfigUpdatedDTO"
2371                             }
2372                         ]
2373                     }
2374                 }
2375             },
2376             "required": true

```

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2377     },
2378     "responses": {
2379         "200": {
2380             "description": "OK",
2381             "content": {
2382                 "*/*": {
2383                     "schema": {
2384                         "type": "string",
2385                         "additionalProperties": {
2386                             "default": ""
2387                         },
2388                         "default": ""
2389                     }
2390                 }
2391             }
2392         }
2393     }
2394 },
2395 "/aws/resources/{resourceId}/status": {
2396     "patch": {
2397         "tags": [
2398             "aws-connection-persistence-controller"
2399         ],
2400         "operationId": "setResourceStatus",
2401         "parameters": [
2402             {
2403                 "name": "resourceId",
2404                 "in": "path",
2405                 "required": true,
2406                 "schema": {
2407                     "type": "string",
2408                     "format": "uuid"
2409                 }
2410             }
2411         ],
2412         "requestBody": {
2413             "content": {
2414                 "application/json": {
2415                     "schema": {
2416                         "$ref": "#/components/schemas/
2417                             UpdateResourceStatusRequest"
2418                     }
2419                 }

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2420     },
2421     "required": true
2422   },
2423   "responses": {
2424     "200": {
2425       "description": "OK"
2426     }
2427   }
2428 },
2429 {
2430   "/aws/connections/{accountId}/name": {
2431     "patch": {
2432       "tags": [
2433         "aws-connection-persistence-controller"
2434       ],
2435       "operationId": "setAccountDisplayName",
2436       "parameters": [
2437         {
2438           "name": "accountId",
2439           "in": "path",
2440           "required": true,
2441           "schema": {
2442             "type": "string",
2443             "format": "uuid"
2444           }
2445         }
2446       ],
2447       "requestBody": {
2448         "content": {
2449           "application/json": {
2450             "schema": {
2451               "$ref": "#/components/schemas/UpdateAccountNameRequest"
2452             }
2453           }
2454         },
2455         "required": true
2456       },
2457       "responses": {
2458         "200": {
2459           "description": "OK"
2460         }
2461       }
2462     }
2463   }
```

```
2463 },
2464 "/webhooks": {
2465   "get": {
2466     "tags": [
2467       "Webhooks"
2468     ],
2469     "summary": "Get all webhooks for the current user",
2470     "operationId": "getWebhooks",
2471     "responses": {
2472       "200": {
2473         "description": "Succesfully returned all webhooks",
2474         "content": {
2475           "*/*": {
2476             "schema": {
2477               "type": "array",
2478               "additionalProperties": {
2479                 "default": ""
2480               },
2481               "contains": {
2482                 "default": ""
2483               },
2484               "items": {
2485                 "$ref": "#/components/schemas/WebhookResponse",
2486                 "default": ""
2487               },
2488               "unevaluatedItems": {
2489                 "default": ""
2490               }
2491             },
2492             "examples": {
2493               "Webhook list": {
2494                 "description": "Webhook list",
2495                 "value": [
2496                   {
2497                     "id": "c2a49b36-55d2-4233-96dd-cc4e3ae97671",
2498                     "name": "General alerts",
2499                     "endpointUrl": "https://example.com/webhooks/general",
2500                     "eventTypes": [
2501                       "usage.threshold",
2502                       "resource.discovery"
2503                     ],
```

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2504         "webhookStatus": "ACTIVE",
2505         "cloudAccounts": [
2506             "c2a49b36-55d2-4233-96dd-cc4e3ae97672"
2507         ]
2508     },
2509     {
2510         "id": "c2a49b36-55d2-4233-96dd-cc4e3ae97673",
2511         "name": "Billing update",
2512         "endpointUrl": "https://example.com/webhooks/billing-update",
2513         "eventTypes": [
2514             "billing.ingestion"
2515         ],
2516         "webhookStatus": "PAUSED",
2517         "cloudAccounts": [
2518             "c2a49b36-55d2-4233-96dd-cc4e3ae97672"
2519         ]
2520     }
2521 ]
2522 }
2523 }
2524 }
2525 }
2526 }
2527 }
2528 }
2529 },
2530 "/webhooks/events": {
2531     "get": {
2532         "tags": [
2533             "Webhooks"
2534         ],
2535         "summary": "Get all supported webhook events",
2536         "operationId": "getEvents",
2537         "responses": {
2538             "200": {
2539                 "description": "Succesfully returned all supported
2540                     webhook events",
2541                 "content": {
2542                     "*/*": {
2543                         "schema": {
2544                             "type": "string",

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```
2545         "default": ""
2546     },
2547     "default": ""
2548 },
2549 "examples": {
2550     "Webhook events by category": {
2551         "description": "Webhook events by category",
2552         "value": {
2553             "Dev": {
2554                 "Dev Event": {
2555                     "type": "dev.event",
2556                     "timestamp": "1970-01-01T00:00:00Z",
2557                     "account": {
2558                         "name": "Development account",
2559                         "provider": "AZURE"
2560                     },
2561                     "data": {
2562                         "devString": "Dev Event String",
2563                         "devDecimal": 20,
2564                         "devTime": "1970-01-01T00:00:00Z"
2565                     }
2566                 }
2567             },
2568             "Usage": {
2569                 "Usage Threshold Alert": {
2570                     "type": "usage.threshold",
2571                     "timestamp": "2026-09-16T08:32:15Z",
2572                     "account": {
2573                         "name": "Azure production",
2574                         "provider": "AZURE"
2575                     },
2576                     "data": {
2577                         "metric": "cost",
2578                         "threshold": 500,
2579                         "currentValue": 612.45
2580                     }
2581                 }
2582             }
2583         }
2584     }
2585 }
2586 }
2587 }
2588 }
```

```
2589     }
2590   }
2591 },
2592 "/webhooks/deliveries": {
2593   "get": {
2594     "tags": [
2595       "Webhooks"
2596     ],
2597     "summary": "Get all webhook deliveries for the current
                user",
2598     "operationId": "getDeliveries",
2599     "parameters": [
2600       {
2601         "name": "page",
2602         "in": "query",
2603         "required": true,
2604         "schema": {
2605           "type": "integer",
2606           "format": "int32"
2607         }
2608       },
2609       {
2610         "name": "pageSize",
2611         "in": "query",
2612         "required": true,
2613         "schema": {
2614           "type": "integer",
2615           "format": "int32"
2616         }
2617       },
2618       {
2619         "name": "search",
2620         "in": "query",
2621         "required": false,
2622         "schema": {
2623           "type": "string"
2624         }
2625       },
2626       {
2627         "name": "webhook",
2628         "in": "query",
2629         "required": false,
2630         "schema": {
2631           "type": "string",
```

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2632         "format": "uuid"
2633     }
2634 },
2635 {
2636     "name": "status",
2637     "in": "query",
2638     "required": false,
2639     "schema": {
2640         "type": "string",
2641         "enum": [
2642             "PENDING",
2643             "PROCESSING",
2644             "DELIVERED",
2645             "FAILED"
2646         ]
2647     }
2648 },
2649 ],
2650 "responses": {
2651     "200": {
2652         "description": "Succesfully returned all webhook
2653             deliveries",
2654         "content": {
2655             "*/*": {
2656                 "schema": {
2657                     "$ref": "#/components/schemas/
2658                         PagedWebhookDeliveryResponse",
2659                     "additionalProperties": {
2660                         "default": ""
2661                     },
2662                     "default": ""
2663                 },
2664                 "examples": {
2665                     "Webhook delivery list": {
2666                         "description": "Webhook delivery list",
2667                         "value": {
2668                             "deliveries": [
2669                                 {
2670                                     "deliveryId": "7d5905cf-3518-42c0-8f23-
2671                                     7a71a7cc5d58",

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```
2672         "cloudAccountId": "c2a49b36-55d2
2673             -4233-96dd-cc4e3ae97672",
2674         "cloudAccountName": "Some Account",
2675         "result": "DELIVERED",
2676         "responseCode": 200
2677     },
2678     {
2679         "deliveryId": "8c307a33-6d4a-4c35-bf14-
2680             c643d6b1f3c6",
2681         "timestamp": "2026-09-16T08:45:02Z",
2682         "webhookId": "c2a49b36-55d2-4233-96dd-
2683             cc4e3ae97673",
2684         "eventType": "billing.ingestion",
2685         "cloudAccountId": "c2a49b36-55d2
2686             -4233-96dd-cc4e3ae97672",
2687         "cloudAccountName": "Some Account",
2688         "result": "FAILED",
2689         "responseCode": 503
2690     }
2691 ],
2692 "page": 0,
2693 "size": 10,
2694 "totalElements": 2,
2695 "totalPages": 1
2696 }
2697 }
2698 }
2699 }
2700 },
2701 "/stream": {
2702     "get": {
2703         "tags": [
2704             "sse-controller"
2705         ],
2706         "operationId": "subscribe",
2707         "responses": {
2708             "200": {
2709                 "description": "OK",
2710                 "content": {
2711                     "text/event-stream": {
```

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2712         "schema": {
2713             "$ref": "#/components/schemas/SseEmitter"
2714         }
2715     }
2716 }
2717 }
2718 }
2719 },
2720 },
2721 "/optimization/recommendations": {
2722     "get": {
2723         "tags": [
2724             "Optimization Recommendations"
2725         ],
2726         "summary": "List optimization recommendations",
2727         "description": "Returns optimization recommendations
2728             matching the supplied filters.",
2729         "operationId": "getRecommendations",
2730         "parameters": [
2731             {
2732                 "name": "status",
2733                 "in": "query",
2734                 "description": "Recommendation lifecycle status",
2735                 "required": false,
2736                 "schema": {
2737                     "type": "string",
2738                     "default": ""
2739                 },
2740                 "example": "ACTIVE"
2741             },
2742             {
2743                 "name": "provider",
2744                 "in": "query",
2745                 "description": "Cloud provider",
2746                 "required": false,
2747                 "schema": {
2748                     "type": "string",
2749                     "default": ""
2750                 },
2751                 "example": "AWS"
2752             },
2753             {
2754                 "name": "resourceType",
2755                 "in": "query",

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```
2755     "description": "Cloud resource type",
2756     "required": false,
2757     "schema": {
2758         "type": "string",
2759         "default": ""
2760     },
2761     "example": "compute_instance"
2762 },
2763 {
2764     "name": "actionType",
2765     "in": "query",
2766     "description": "Recommended action type",
2767     "required": false,
2768     "schema": {
2769         "type": "string",
2770         "default": ""
2771     },
2772     "example": "DOWNSIZE"
2773 },
2774 {
2775     "name": "resourceId",
2776     "in": "query",
2777     "description": "Resource UUID",
2778     "required": false,
2779     "schema": {
2780         "type": "string",
2781         "default": ""
2782     }
2783 }
2784 ],
2785 "responses": {
2786     "200": {
2787         "description": "OK",
2788         "content": {
2789             "*/*": {
2790                 "schema": {
2791                     "type": "array",
2792                     "items": {
2793                         "type": "object",
2794                         "additionalProperties": {
2795
2796                         }
2797                     }
2798                 }
```

```

2799         }
2800     }
2801 }
2802 }
2803 }
2804 },
2805 "/optimization/recommendations/{recommendationId}": {
2806     "get": {
2807         "tags": [
2808             "Optimization Recommendations"
2809         ],
2810         "summary": "Get an optimization recommendation",
2811         "description": "Returns a single optimization
2812             recommendation by its identifier.",
2813         "operationId": "getRecommendation",
2814         "parameters": [
2815             {
2816                 "name": "recommendationId",
2817                 "in": "path",
2818                 "description": "Recommendation UUID",
2819                 "required": true,
2820                 "schema": {
2821                     "type": "string",
2822                     "default": ""
2823                 }
2824             },
2825             {
2826                 "name": "recommendationId",
2827                 "in": "path",
2828                 "description": "Recommendation UUID",
2829                 "required": true,
2830                 "schema": {
2831                     "type": "string",
2832                     "default": ""
2833                 }
2834             }
2835         ],
2836         "responses": {
2837             "200": {
2838                 "description": "OK",
2839                 "content": {
2840                     "*/*": {
2841                         "schema": {

```

```
2842     "/optimization/recommendations/summary": {
2843         "get": {
2844             "tags": [
2845                 "Optimization Recommendations"
2846             ],
2847             "summary": "Get recommendation summary",
2848             "description": "Returns recommendation counts grouped by
                status and action type.",
2849             "operationId": "getRecommendationSummary",
2850             "responses": {
2851                 "200": {
2852                     "description": "OK",
2853                     "content": {
2854                         "*/*": {
2855                             "schema": {
2856                                 "type": "object",
2857                                 "additionalProperties": {
2858
2859                                 }
2860                             }
2861                         }
2862                     }
2863                 }
2864             }
2865         },
2866     },
2867     "/mock": {
2868         "get": {
2869             "tags": [
2870                 "mock-controller"
2871             ],
2872             "operationId": "getMethodName",
2873             "responses": {
2874                 "200": {
2875                     "description": "OK",
2876                     "content": {
2877                         "*/*": {
2878                             "schema": {
2879                                 "type": "string"
2880                             }
2881                         }
2882                     }
2883                 }
2884             }
2885         }
2886     }
```

```

2885     }
2886 },
2887 "/intelligence/mock/usage-sample": {
2888     "get": {
2889         "tags": [
2890             "intelligence-mock-controller"
2891         ],
2892         "operationId": "sampleUsage",
2893         "responses": {
2894             "200": {
2895                 "description": "OK",
2896                 "content": {
2897                     "*/*": {
2898                         "schema": {
2899                             "type": "array",
2900                             "items": {
2901                                 "$ref": "#/components/schemas/
2902                                     TimestampedNumericDataPoint"
2903                             }
2904                         }
2905                     }
2906                 }
2907             }
2908         }
2909 },
2910 "/azure/accounts/{accountId}/resources/count": {
2911     "get": {
2912         "tags": [
2913             "connection-query-controller"
2914         ],
2915         "operationId": "getResourceCount",
2916         "parameters": [
2917             {
2918                 "name": "accountId",
2919                 "in": "path",
2920                 "required": true,
2921                 "schema": {
2922                     "type": "string",
2923                     "format": "uuid"
2924                 }
2925             }
2926         ],
2927         "responses": {

```

```

2928         "200": {
2929             "description": "OK",
2930             "content": {
2931                 "*/*": {
2932                     "schema": {
2933                         "$ref": "#/components/schemas/
                                ResourceCountResponse"
2934                     }
2935                 }
2936             }
2937         }
2938     },
2939 },
2940 "/gcp/accounts/{accountId}/resources/count": {
2941     "get": {
2942         "tags": [
2943             "connection-query-controller"
2944         ],
2945         "operationId": "getResourceCount_1",
2946         "parameters": [
2947             {
2948                 "name": "accountId",
2949                 "in": "path",
2950                 "required": true,
2951                 "schema": {
2952                     "type": "string",
2953                     "format": "uuid"
2954                 }
2955             }
2956         ],
2957         "responses": {
2958             "200": {
2959                 "description": "OK",
2960                 "content": {
2961                     "*/*": {
2962                         "schema": {
2963                             "$ref": "#/components/schemas/
                                ResourceCountResponse"
2964                         }
2965                     }
2966                 }
2967             }
2968         }
2969     }

```

```
2970     }
2971 },
2972 "/aws/accounts/{accountId}/resources/count": {
2973     "get": {
2974         "tags": [
2975             "connection-query-controller"
2976         ],
2977         "operationId": "getResourceCount_2",
2978         "parameters": [
2979             {
2980                 "name": "accountId",
2981                 "in": "path",
2982                 "required": true,
2983                 "schema": {
2984                     "type": "string",
2985                     "format": "uuid"
2986                 }
2987             }
2988         ],
2989         "responses": {
2990             "200": {
2991                 "description": "OK",
2992                 "content": {
2993                     "*/*": {
2994                         "schema": {
2995                             "$ref": "#/components/schemas/ResourceCountResponse"
2996                         }
2997                     }
2998                 }
2999             }
3000         }
3001     },
3002 },
3003 "/aws/accounts/{accountId}/resources": {
3004     "get": {
3005         "tags": [
3006             "connection-query-controller"
3007         ],
3008         "operationId": "getResourcesForAccount",
3009         "parameters": [
3010             {
3011                 "name": "accountId",
3012                 "in": "path",
```

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3013         "required": true,
3014         "schema": {
3015             "type": "string",
3016             "format": "uuid"
3017         }
3018     },
3019 ],
3020 "responses": {
3021     "200": {
3022         "description": "OK",
3023         "content": {
3024             "*/*": {
3025                 "schema": {
3026                     "type": "array",
3027                     "items": {
3028                         "$ref": "#/components/schemas/Resource"
3029                     }
3030                 }
3031             }
3032         }
3033     }
3034 },
3035 },
3036 },
3037 "/gcp/accounts/{accountId}/resources": {
3038     "get": {
3039         "tags": [
3040             "connection-query-controller"
3041         ],
3042         "operationId": "getResourcesForAccount_1",
3043         "parameters": [
3044             {
3045                 "name": "accountId",
3046                 "in": "path",
3047                 "required": true,
3048                 "schema": {
3049                     "type": "string",
3050                     "format": "uuid"
3051                 }
3052             }
3053         ],
3054         "responses": {
3055             "200": {
3056                 "description": "OK",
```

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3057         "content": {
3058             "*/*": {
3059                 "schema": {
3060                     "type": "array",
3061                     "items": {
3062                         "$ref": "#/components/schemas/Resource"
3063                     }
3064                 }
3065             }
3066         }
3067     },
3068     },
3069 },
3070 {
3071     "/azure/accounts/{accountId}/resources": {
3072         "get": {
3073             "tags": [
3074                 "connection-query-controller"
3075             ],
3076             "operationId": "getResourcesForAccount_2",
3077             "parameters": [
3078                 {
3079                     "name": "accountId",
3080                     "in": "path",
3081                     "required": true,
3082                     "schema": {
3083                         "type": "string",
3084                         "format": "uuid"
3085                     }
3086                 }
3087             ],
3088             "responses": {
3089                 "200": {
3090                     "description": "OK",
3091                     "content": {
3092                         "*/*": {
3093                             "schema": {
3094                                 "type": "array",
3095                                 "items": {
3096                                     "$ref": "#/components/schemas/Resource"
3097                                 }
3098                             }
3099                         }
3100                     }
3101                 }
3102             }
3103         }
3104     }
3105 }

```

```

3101     }
3102   }
3103 }
3104 },
3105 "/azure/accounts/{accountId}": {
3106   "get": {
3107     "tags": [
3108       "connection-query-controller"
3109     ],
3110     "operationId": "getAccount",
3111     "parameters": [
3112       {
3113         "name": "accountId",
3114         "in": "path",
3115         "required": true,
3116         "schema": {
3117           "type": "string",
3118           "format": "uuid"
3119         }
3120       }
3121     ],
3122     "responses": {
3123       "200": {
3124         "description": "OK",
3125         "content": {
3126           "*/*": {
3127             "schema": {
3128               "$ref": "#/components/schemas/CloudAccountDetailsResponse"
3129             }
3130           }
3131         }
3132       }
3133     }
3134   },
3135 },
3136 "/gcp/accounts/{accountId}": {
3137   "get": {
3138     "tags": [
3139       "connection-query-controller"
3140     ],
3141     "operationId": "getAccount_1",
3142     "parameters": [
3143       {

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3144         "name": "accountId",
3145         "in": "path",
3146         "required": true,
3147         "schema": {
3148             "type": "string",
3149             "format": "uuid"
3150         }
3151     },
3152 ],
3153     "responses": {
3154         "200": {
3155             "description": "OK",
3156             "content": {
3157                 "*/*": {
3158                     "schema": {
3159                         "$ref": "#/components/schemas/
CloudAccountDetailsResponse"
3160                     }
3161                 }
3162             }
3163         }
3164     }
3165 },
3166 "/aws/accounts/{accountId}": {
3167     "get": {
3168         "tags": [
3169             "connection-query-controller"
3170         ],
3171         "operationId": "getAccount_2",
3172         "parameters": [
3173             {
3174                 "name": "accountId",
3175                 "in": "path",
3176                 "required": true,
3177                 "schema": {
3178                     "type": "string",
3179                     "format": "uuid"
3180                 }
3181             }
3182         ],
3183         "responses": {
3184             "200": {
3185                 "description": "OK",

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```
3187         "content": {
3188             "*/*": {
3189                 "schema": {
3190                     "$ref": "#/components/schemas/
                        CloudAccountDetailsResponse"
3191                 }
3192             }
3193         }
3194     }
3195 }
3196 },
3197 "/auth/me": {
3198     "get": {
3199         "tags": [
3200             "Authentication"
3201         ],
3202         "summary": "Get current authenticated user",
3203         "operationId": "me",
3204         "responses": {
3205             "200": {
3206                 "description": "Current authenticated user",
3207                 "content": {
3208                     "*/*": {
3209                         "schema": {
3210                             "$ref": "#/components/schemas/AuthUserResponse
3211                             ",
3212                             "additionalProperties": {
3213                                 "default": ""
3214                             },
3215                             "default": ""
3216                         }
3217                     }
3218                 }
3219             },
3220             "401": {
3221                 "description": "Unauthenticated",
3222                 "content": {
3223                     "*/*": {
3224                         "schema": {
3225                             "$ref": "#/components/schemas/AuthUserResponse"
3226                         }
3227                     }
3228                 }
3229             }
3230         }
3231     }
3232 }
```

```
3229     }
3230   }
3231 }
3232 },
3233 "/api/alerts": {
3234   "get": {
3235     "tags": [
3236       "Alerts"
3237     ],
3238     "summary": "List alerts",
3239     "description": "Return alerts for the current tenant,
3240       optionally filtered by alert type.",
3241     "operationId": "getAlerts",
3242     "parameters": [
3243       {
3244         "name": "alertType",
3245         "in": "query",
3246         "required": false,
3247         "schema": {
3248           "type": "string",
3249           "enum": [
3250             "THRESHOLD",
3251             "BUDGET",
3252             "ANOMALY"
3253           ]
3254         }
3255       }
3256     ],
3257     "responses": {
3258       "200": {
3259         "description": "Alerts returned",
3260         "content": {
3261           "application/json": {
3262             "schema": {
3263               "type": "array",
3264               "additionalProperties": {
3265                 "default": ""
3266               },
3267               "contains": {
3268                 "default": ""
3269               },
3270               "items": {
3271                 "$ref": "#/components/schemas/AlertResponse",
3272                 "default": ""
3273               }
3274             }
3275           }
3276         }
3277       }
3278     }
3279   }
3280 }
```

```

3272         },
3273         "unevaluatedItems": {
3274             "default": ""
3275         }
3276     }
3277 }
3278 }
3279 }
3280 }
3281 }
3282 },
3283 "/api/alerts/{id}": {
3284     "get": {
3285         "tags": [
3286             "Alerts"
3287         ],
3288         "summary": "Get alert",
3289         "description": "Return details for a single alert.",
3290         "operationId": "getAlert",
3291         "parameters": [
3292             {
3293                 "name": "id",
3294                 "in": "path",
3295                 "description": "Alert UUID",
3296                 "required": true,
3297                 "schema": {
3298                     "type": "string",
3299                     "default": ""
3300                 }
3301             }
3302         ],
3303         "responses": {
3304             "200": {
3305                 "description": "Alert found",
3306                 "content": {
3307                     "application/json": {
3308                         "schema": {
3309                             "$ref": "#/components/schemas/AlertResponse",
3310                             "additionalProperties": {
3311                                 "default": ""
3312                             },
3313                             "default": ""
3314                         }
3315                     }

```

```
3316     }
3317 },
3318 "404": {
3319     "description": "Alert not found",
3320     "content": {
3321         "*/*": {
3322             "schema": {
3323                 "type": "string",
3324                 "additionalProperties": {
3325                     "default": ""
3326                 },
3327                 "default": ""
3328             }
3329         }
3330     }
3331 },
3332 },
3333 },
3334 "delete": {
3335     "tags": [
3336         "Alerts"
3337     ],
3338     "summary": "Delete alert",
3339     "description": "Delete an existing alert.",
3340     "operationId": "deleteAlert",
3341     "parameters": [
3342         {
3343             "name": "id",
3344             "in": "path",
3345             "description": "Alert UUID",
3346             "required": true,
3347             "schema": {
3348                 "type": "string",
3349                 "default": ""
3350             }
3351         }
3352     ],
3353     "responses": {
3354         "204": {
3355             "description": "Alert deleted"
3356         },
3357         "404": {
3358             "description": "Alert not found",
3359             "content": {
```

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3360         "*/*": {
3361             "schema": {
3362                 "type": "string",
3363                 "additionalProperties": {
3364                     "default": ""
3365                 },
3366                 "default": ""
3367             }
3368         }
3369     }
3370 }
3371 }
3372 }
3373 },
3374 "/analytics/resource-names": {
3375     "get": {
3376         "tags": [
3377             "CloudSherpa Analytics"
3378         ],
3379         "operationId": "getResourceNames",
3380         "responses": {
3381             "200": {
3382                 "description": "OK",
3383                 "content": {
3384                     "*/*": {
3385                         "schema": {
3386                             "type": "array",
3387                             "items": {
3388                                 "$ref": "#/components/schemas/ResourceNameDto"
3389                             }
3390                         }
3391                     }
3392                 }
3393             }
3394         }
3395     }
3396 },
3397 "/analytics/resource-metrics": {
3398     "get": {
3399         "tags": [
3400             "CloudSherpa Analytics"
3401         ],

```

```
3402     "summary": "For each resource return available/recorded
3403         metrics",
3404     "operationId": "getResourceMetrics",
3405     "responses": {
3406         "200": {
3407             "description": "Succesfully fetched available metrics
3408                 for each resource",
3409             "content": {
3410                 "*/*": {
3411                     "schema": {
3412                         "type": "array",
3413                         "additionalProperties": {
3414                             "default": ""
3415                         },
3416                         "contains": {
3417                             "default": ""
3418                         },
3419                         "items": {
3420                             "$ref": "#/components/schemas/
3421                                 ResourceMetricsGroupDto",
3422                             "default": ""
3423                         },
3424                         "unevaluatedItems": {
3425                             "default": ""
3426                         }
3427                     }
3428                 }
3429             }
3430         },
3431         "/analytics/historical": {
3432             "get": {
3433                 "tags": [
3434                     "CloudSherpa Analytics"
3435                 ],
3436                 "summary": "Get authoratative metric data",
3437                 "operationId": "getHistoricalData",
3438                 "parameters": [
3439                     {
3440                         "name": "from",
3441                         "in": "query",
3442                         "required": true,
```

```
3443         "schema": {
3444             "type": "string"
3445         }
3446     },
3447     {
3448         "name": "to",
3449         "in": "query",
3450         "required": true,
3451         "schema": {
3452             "type": "string"
3453         }
3454     },
3455     {
3456         "name": "interval",
3457         "in": "query",
3458         "required": false,
3459         "schema": {
3460             "type": "string",
3461             "default": "daily"
3462         }
3463     }
3464 ],
3465 "responses": {
3466     "204": {
3467         "description": "No metrics found for selected time
3468             window",
3469         "content": {
3470             "*/*": {
3471                 "schema": {
3472                     "type": "array",
3473                     "items": {
3474                         "$ref": "#/components/schemas/MetricDto"
3475                     }
3476                 }
3477             }
3478         },
3479         "200": {
3480             "description": "Return metrics found",
3481             "content": {
3482                 "*/*": {
3483                     "schema": {
3484                         "type": "array",
3485                         "additionalProperties": {
```

```

3486         "default": ""
3487     },
3488     "contains": {
3489         "default": ""
3490     },
3491     "items": {
3492         "$ref": "#/components/schemas/
            AggregatedMetric",
3493         "default": ""
3494     },
3495     "unevaluatedItems": {
3496         "default": ""
3497     }
3498 }
3499 }
3500 }
3501 }
3502 }
3503 }
3504 },
3505 "/ai/sessions/{sessionId}/versions": {
3506     "get": {
3507         "tags": [
3508             "AI Dashboard"
3509         ],
3510         "summary": "Get AI dashboard versions",
3511         "description": "Returns all staged dashboard versions
            belonging to the authenticated user's AI session",
3512         "operationId": "getVersions",
3513         "parameters": [
3514             {
3515                 "name": "sessionId",
3516                 "in": "path",
3517                 "required": true,
3518                 "schema": {
3519                     "type": "string",
3520                     "format": "uuid"
3521                 }
3522             }
3523         ],
3524         "responses": {
3525             "200": {
3526                 "description": "Dashboard versions retrieved
                    successfully",

```

```
3527         "content": {
3528             "application/json": {
3529                 "schema": {
3530                     "$ref": "#/components/schemas/
3531                         AiVersionSummaryDto",
3532                     "additionalProperties": {
3533                         "default": ""
3534                     },
3535                     "default": ""
3536                 }
3537             }
3538         },
3539         "404": {
3540             "description": "AI session does not exist or does not
3541                 belong to the authenticated user",
3542             "content": {
3543                 "*/*": {
3544                     "schema": {
3545                         "type": "string",
3546                         "additionalProperties": {
3547                             "default": ""
3548                         },
3549                         "default": ""
3550                     }
3551                 }
3552             }
3553         }
3554     },
3555     "/ai/sessions/{sessionId}/versions/{versionId}": {
3556         "get": {
3557             "tags": [
3558                 "AI Dashboard"
3559             ],
3560             "summary": "Get an AI dashboard version",
3561             "description": "Returns the complete staged dashboard
3562                 configuration for a specific dashboard version",
3563             "operationId": "getVersion",
3564             "parameters": [
3565                 {
3566                     "name": "sessionId",
3567                     "in": "path",
```

```
3568         "required": true,
3569         "schema": {
3570             "type": "string",
3571             "format": "uuid"
3572         }
3573     },
3574     {
3575         "name": "versionId",
3576         "in": "path",
3577         "required": true,
3578         "schema": {
3579             "type": "string",
3580             "format": "uuid"
3581         }
3582     }
3583 ],
3584 "responses": {
3585     "200": {
3586         "description": "Dashboard version retrieved successfully",
3587         "content": {
3588             "application/json": {
3589                 "schema": {
3590                     "$ref": "#/components/schemas/AiVersionResponseDto",
3591                     "additionalProperties": {
3592                         "default": ""
3593                     },
3594                     "default": ""
3595                 }
3596             }
3597         },
3598     },
3599     "404": {
3600         "description": "Dashboard version does not exist or does not belong to the authenticated user's session",
3601         "content": {
3602             "*/*": {
3603                 "schema": {
3604                     "type": "string",
3605                     "additionalProperties": {
3606                         "default": ""
3607                     },

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3608         "default": ""
3609     }
3610 }
3611 }
3612 }
3613 }
3614 }
3615 },
3616 "/webhooks/delete/{webhookId}": {
3617     "delete": {
3618         "tags": [
3619             "Webhooks"
3620         ],
3621         "summary": "Delete webhook",
3622         "operationId": "deleteWebhook",
3623         "parameters": [
3624             {
3625                 "name": "webhookId",
3626                 "in": "path",
3627                 "required": true,
3628                 "schema": {
3629                     "type": "string",
3630                     "format": "uuid"
3631                 }
3632             }
3633         ],
3634         "responses": {
3635             "204": {
3636                 "description": "Deleted webhook / not, generic
                                response code protects against information
                                gathering via response codes"
3637             }
3638         }
3639     }
3640 },
3641 "/dashboards/{dashboardId}": {
3642     "delete": {
3643         "tags": [
3644             "CloudSherpa Dashboard"
3645         ],
3646         "summary": "Delete a dashboard",
3647         "operationId": "deleteDashboard",
3648         "parameters": [
3649             {

```

```

3650         "name": "dashboardId",
3651         "in": "path",
3652         "required": true,
3653         "schema": {
3654             "type": "string",
3655             "format": "uuid"
3656         }
3657     },
3658 ],
3659 "responses": {
3660     "204": {
3661         "description": "No Content",
3662         "content": {
3663             "*/*": {
3664                 "schema": {
3665                     "type": "string",
3666                     "additionalProperties": {
3667                         "default": ""
3668                     },
3669                     "default": ""
3670                 }
3671             }
3672         }
3673     }
3674 }
3675 },
3676 },
3677 "/dashboards/widgets/{widgetId}": {
3678     "delete": {
3679         "tags": [
3680             "CloudSherpa Dashboard"
3681         ],
3682         "summary": "Delete a widget from a dashboard",
3683         "operationId": "deleteWidget",
3684         "parameters": [
3685             {
3686                 "name": "widgetId",
3687                 "in": "path",
3688                 "required": true,
3689                 "schema": {
3690                     "type": "string",
3691                     "format": "uuid"
3692                 }
3693             }

```

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3694     ],
3695     "responses": {
3696         "204": {
3697             "description": "No Content",
3698             "content": {
3699                 "*/*": {
3700                     "schema": {
3701                         "type": "string",
3702                         "additionalProperties": {
3703                             "default": ""
3704                         },
3705                         "default": ""
3706                     }
3707                 }
3708             }
3709         }
3710     }
3711 },
3712 "/aws/connections/{accountId}": {
3713     "delete": {
3714         "tags": [
3715             "aws-connection-persistence-controller"
3716         ],
3717         "operationId": "deleteAccount",
3718         "parameters": [
3719             {
3720                 "name": "accountId",
3721                 "in": "path",
3722                 "required": true,
3723                 "schema": {
3724                     "type": "string",
3725                     "format": "uuid"
3726                 }
3727             }
3728         ],
3729         "responses": {
3730             "200": {
3731                 "description": "OK"
3732             }
3733         }
3734     }
3735 },
3736 "/ai/sessions/{sessionId}": {
3737

```

```

3738     "delete": {
3739         "tags": [
3740             "AI Dashboard"
3741         ],
3742         "summary": "Delete an AI dashboard session",
3743         "description": "Deletes the AI session and its staged
                        data belonging to the authenticated user",
3744         "operationId": "deleteSession",
3745         "parameters": [
3746             {
3747                 "name": "sessionId",
3748                 "in": "path",
3749                 "required": true,
3750                 "schema": {
3751                     "type": "string",
3752                     "format": "uuid"
3753                 }
3754             }
3755         ],
3756         "responses": {
3757             "204": {
3758                 "description": "AI dashboard session deleted
                                successfully"
3759             },
3760             "404": {
3761                 "description": "AI session does not exist or does not
                                belong to the authenticated user",
3762                 "content": {
3763                     "*/*": {
3764                         "schema": {
3765                             "type": "string",
3766                             "additionalProperties": {
3767                                 "default": ""
3768                             },
3769                             "default": ""
3770                         }
3771                     }
3772                 }
3773             }
3774         }
3775     }
3776 },
3777 "components": {
3778

```

```
3779     "schemas": {
3780       "WidgetLayoutUpdatedDTO": {
3781         "type": "object",
3782         "properties": {
3783           "id": {
3784             "type": "string",
3785             "format": "uuid"
3786           },
3787           "x": {
3788             "type": "integer",
3789             "format": "int32"
3790           },
3791           "y": {
3792             "type": "integer",
3793             "format": "int32"
3794           },
3795           "w": {
3796             "type": "integer",
3797             "format": "int32"
3798           },
3799           "h": {
3800             "type": "integer",
3801             "format": "int32"
3802           }
3803         }
3804       },
3805       "UpdateThresholdRequest": {
3806         "type": "object",
3807         "properties": {
3808           "metric_name": {
3809             "type": "string"
3810           },
3811           "operator": {
3812             "type": "string"
3813           },
3814           "value": {
3815             "type": "number",
3816             "format": "double"
3817           },
3818           "severity": {
3819             "type": "string",
3820             "enum": [
3821               "WARNING",
3822               "CRITICAL"
```

```
3823         ]
3824     },
3825     "enabled": {
3826         "type": "boolean"
3827     }
3828 }
3829 },
3830 "UpdateBudgetRequest": {
3831     "type": "object",
3832     "properties": {
3833         "amount": {
3834             "type": "number"
3835         },
3836         "currency": {
3837             "type": "string",
3838             "enum": [
3839                 "USD",
3840                 "EUR",
3841                 "ZAR"
3842             ]
3843         },
3844         "window_days": {
3845             "type": "integer",
3846             "format": "int32"
3847         },
3848         "enabled": {
3849             "type": "boolean"
3850         }
3851     }
3852 },
3853 "BudgetResponse": {
3854     "type": "object",
3855     "properties": {
3856         "budget_id": {
3857             "type": "string",
3858             "format": "uuid"
3859         },
3860         "user_id": {
3861             "type": "string",
3862             "format": "uuid"
3863         },
3864         "scope": {
3865             "type": "string"
3866         },

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```
3867     "scope_id": {
3868         "type": "string",
3869         "format": "uuid"
3870     },
3871     "amount": {
3872         "type": "number"
3873     },
3874     "currency": {
3875         "type": "string",
3876         "enum": [
3877             "USD",
3878             "EUR",
3879             "ZAR"
3880         ]
3881     },
3882     "window_days": {
3883         "type": "integer",
3884         "format": "int32"
3885     },
3886     "enabled": {
3887         "type": "boolean"
3888     },
3889     "created_at": {
3890         "type": "string",
3891         "format": "date-time"
3892     },
3893     "updated_at": {
3894         "type": "string",
3895         "format": "date-time"
3896     }
3897 },
3898 {
3899     "EditWebhookDto": {
3900         "type": "object",
3901         "properties": {
3902             "name": {
3903                 "type": "string"
3904             },
3905             "endpointUrl": {
3906                 "type": "string"
3907             },
3908             "eventTypes": {
3909                 "type": "array",
3910                 "items": {
```

```
3911         "type": "string"
3912     },
3913 },
3914 "status": {
3915     "type": "string",
3916     "enum": [
3917         "ACTIVE",
3918         "PAUSED"
3919     ]
3920 },
3921 "cloudAccounts": {
3922     "type": "array",
3923     "items": {
3924         "type": "string",
3925         "format": "uuid"
3926     }
3927 }
3928 },
3929 },
3930 "AddWebhookDto": {
3931     "type": "object",
3932     "properties": {
3933         "name": {
3934             "type": "string"
3935         },
3936         "endpointUrl": {
3937             "type": "string"
3938         },
3939         "eventTypes": {
3940             "type": "array",
3941             "items": {
3942                 "type": "string"
3943             }
3944         },
3945         "cloudAccounts": {
3946             "type": "array",
3947             "items": {
3948                 "type": "string",
3949                 "format": "uuid"
3950             }
3951         }
3952     }
3953 },
3954 "AddWebhookResponseDto": {
```

```
3955     "type": "object",
3956     "properties": {
3957         "secret": {
3958             "type": "string"
3959         }
3960     }
3961 },
3962 "ThemeUpdateRequest": {
3963     "type": "object",
3964     "properties": {
3965         "theme": {
3966             "type": "string"
3967         }
3968     }
3969 },
3970 "AlertNotificationsUpdateRequest": {
3971     "type": "object",
3972     "properties": {
3973         "enabled": {
3974             "type": "boolean"
3975         }
3976     }
3977 },
3978 "ResourceUsageForecastRequestDto": {
3979     "type": "object",
3980     "properties": {
3981         "resourceId": {
3982             "type": "string",
3983             "format": "uuid"
3984         },
3985         "metricType": {
3986             "type": "string"
3987         },
3988         "forecastHorizon": {
3989             "type": "string",
3990             "format": "date-time"
3991         }
3992     }
3993 },
3994 "ResourceUsageForecastResponseDto": {
3995     "type": "object",
3996     "properties": {
3997         "horizonTimestamps": {
3998             "type": "array",
```

```
3999         "items": {
4000             "type": "string",
4001             "format": "date-time"
4002         }
4003     },
4004     "predictedValues": {
4005         "type": "array",
4006         "items": {
4007             "type": "number"
4008         }
4009     },
4010     "q1Values": {
4011         "type": "array",
4012         "items": {
4013             "type": "number"
4014         }
4015     },
4016     "q3Values": {
4017         "type": "array",
4018         "items": {
4019             "type": "number"
4020         }
4021     }
4022 },
4023 "BillingForecastRequest": {
4024     "type": "object",
4025     "properties": {
4026         "forecastSteps": {
4027             "type": "integer",
4028             "format": "int32"
4029         }
4030     }
4031 },
4032 },
4033 "BillingForecastResponseDto": {
4034     "type": "object",
4035     "properties": {
4036         "cumalativeBillingForecastValue": {
4037             "type": "number"
4038         },
4039         "cumalitivePastForecastingValue": {
4040             "type": "number"
4041         },
4042         "billingForecastSeries": {
```

```
4043         "type": "object",
4044         "additionalProperties": {
4045             "$ref": "#/components/schemas/BillingForecastValue"
4046         }
4047     },
4048     "failedForecastCharges": {
4049         "type": "array",
4050         "items": {
4051             "type": "string"
4052         }
4053     },
4054     "pastVariance": {
4055         "type": "number"
4056     },
4057     "dailyBurnRate": {
4058         "type": "number"
4059     },
4060     "highestCostDriver": {
4061         "type": "string"
4062     },
4063     "highestCostAcceleration": {
4064         "type": "string"
4065     },
4066     "accelerationRate": {
4067         "type": "number"
4068     }
4069 },
4070 {
4071     "BillingForecastValue": {
4072         "type": "object",
4073         "properties": {
4074             "value": {
4075                 "type": "number"
4076             },
4077             "percentageOfTotal": {
4078                 "type": "number"
4079             },
4080             "chargeLabel": {
4081                 "type": "string"
4082             }
4083         }
4084     },
4085     "BillingForecastIndividualChargesRequestDto": {
4086         "type": "object",
```

```
4087     "properties": {
4088         "chargeIds": {
4089             "type": "array",
4090             "items": {
4091                 "type": "string"
4092             }
4093         },
4094         "forecastSteps": {
4095             "type": "integer",
4096             "format": "int32"
4097         }
4098     },
4099 },
4100 "GcpBillingConfigDto": {
4101     "type": "object",
4102     "properties": {
4103         "billingId": {
4104             "type": "string"
4105         },
4106         "dataset": {
4107             "type": "string"
4108         }
4109     }
4110 },
4111 "GcpCredentialsDto": {
4112     "type": "object",
4113     "properties": {
4114         "projectId": {
4115             "type": "string"
4116         },
4117         "serviceAccountJson": {
4118             "type": "string"
4119         }
4120     }
4121 },
4122 "PersistGcpConnectionRequest": {
4123     "type": "object",
4124     "properties": {
4125         "userId": {
4126             "type": "string",
4127             "format": "uuid"
4128         },
4129         "accountId": {
4130             "type": "string"
```

```
4131     },
4132     "displayName": {
4133         "type": "string"
4134     },
4135     "ingestionPeriod": {
4136         "type": "integer",
4137         "format": "int32"
4138     },
4139     "credentials": {
4140         "$ref": "#/components/schemas/GcpCredentialsDto"
4141     },
4142     "resources": {
4143         "type": "array",
4144         "items": {
4145             "$ref": "#/components/schemas/ResourceSelectionDto"
4146         }
4147     },
4148     "billingConfig": {
4149         "$ref": "#/components/schemas/GcpBillingConfigDto"
4150     }
4151 },
4152 "ResourceSelectionDto": {
4153     "type": "object",
4154     "properties": {
4155         "resourceId": {
4156             "type": "string"
4157         },
4158         "serviceType": {
4159             "type": "string"
4160         },
4161         "resourceType": {
4162             "type": "string"
4163         },
4164         "resourceName": {
4165             "type": "string"
4166         },
4167         "region": {
4168             "type": "string"
4169         },
4170         "tags": {
4171             "type": "object",
4172             "additionalProperties": {
4173                 "type": "string"
4174             }
4175         }
4176     }
4177 }
```

```
4175         }
4176     },
4177     "active": {
4178         "type": "boolean"
4179     }
4180 }
4181 },
4182 "DashboardCreateDTO": {
4183     "type": "object",
4184     "properties": {
4185         "userId": {
4186             "type": "string",
4187             "format": "uuid"
4188         },
4189         "id": {
4190             "type": "string",
4191             "format": "uuid"
4192         },
4193         "displayName": {
4194             "type": "string"
4195         }
4196     }
4197 },
4198 "ChartWidgetDTO": {
4199     "allOf": [
4200         {
4201             "$ref": "#/components/schemas/WidgetDTO"
4202         },
4203         {
4204             "type": "object",
4205             "properties": {
4206                 "id": {
4207                     "type": "string",
4208                     "format": "uuid"
4209                 },
4210                 "widgetType": {
4211                     "type": "string",
4212                     "enum": [
4213                         "KPI",
4214                         "CHART"
4215                     ]
4216                 },
4217                 "displayName": {
4218                     "type": "string"
```

```
4219     },
4220     "startX": {
4221         "type": "integer",
4222         "format": "int32"
4223     },
4224     "startY": {
4225         "type": "integer",
4226         "format": "int32"
4227     },
4228     "width": {
4229         "type": "integer",
4230         "format": "int32"
4231     },
4232     "height": {
4233         "type": "integer",
4234         "format": "int32"
4235     },
4236     "chartType": {
4237         "type": "string",
4238         "enum": [
4239             "gauge_chart",
4240             "line_chart"
4241         ]
4242     },
4243     "chartColour": {
4244         "type": "string",
4245         "enum": [
4246             "chart_1",
4247             "chart_2",
4248             "chart_3",
4249             "chart_4",
4250             "chart_5"
4251         ]
4252     },
4253     "provider": {
4254         "type": "string",
4255         "enum": [
4256             "AWS",
4257             "AZURE",
4258             "GCP"
4259         ]
4260     },
4261     "accountId": {
4262         "type": "string",
```

```
4263         "format": "uuid"
4264     },
4265     "resourceId": {
4266         "type": "string",
4267         "format": "uuid"
4268     },
4269     "metricType": {
4270         "type": "string"
4271     },
4272     "metricName": {
4273         "type": "string"
4274     }
4275 }
4276 }
4277 ]
4278 },
4279 "DashboardDTO": {
4280     "type": "object",
4281     "properties": {
4282         "id": {
4283             "type": "string",
4284             "format": "uuid"
4285         },
4286         "displayName": {
4287             "type": "string"
4288         },
4289         "timeFrom": {
4290             "type": "string",
4291             "format": "date-time"
4292         },
4293         "timeTo": {
4294             "type": "string",
4295             "format": "date-time"
4296         },
4297         "predefinedTime": {
4298             "type": "string",
4299             "enum": [
4300                 "T_5_MIN",
4301                 "T_15_MIN",
4302                 "T_30_MIN",
4303                 "T_1_HOUR",
4304                 "T_6_HOUR",
4305                 "T_12_HOUR",
4306                 "T_24_HOUR",
```

```
4307         "T_7_DAYS",
4308         "T_30_DAYS",
4309         "CUSTOM"
4310     ]
4311 },
4312 "current": {
4313     "type": "boolean"
4314 },
4315 "widgets": {
4316     "type": "array",
4317     "items": {
4318         "oneOf": [
4319             {
4320                 "$ref": "#/components/schemas/ChartWidgetDTO"
4321             },
4322             {
4323                 "$ref": "#/components/schemas/KpiWidgetDTO"
4324             }
4325         ]
4326     }
4327 }
4328 },
4329 },
4330 "KpiWidgetDTO": {
4331     "allOf": [
4332         {
4333             "$ref": "#/components/schemas/WidgetDTO"
4334         },
4335         {
4336             "type": "object",
4337             "properties": {
4338                 "id": {
4339                     "type": "string",
4340                     "format": "uuid"
4341                 },
4342                 "widgetType": {
4343                     "type": "string",
4344                     "enum": [
4345                         "KPI",
4346                         "CHART"
4347                     ]
4348                 },
4349                 "displayName": {
4350                     "type": "string"
```

```
4351     },
4352     "startX": {
4353         "type": "integer",
4354         "format": "int32"
4355     },
4356     "startY": {
4357         "type": "integer",
4358         "format": "int32"
4359     },
4360     "width": {
4361         "type": "integer",
4362         "format": "int32"
4363     },
4364     "height": {
4365         "type": "integer",
4366         "format": "int32"
4367     },
4368     "chargeIds": {
4369         "type": "array",
4370         "items": {
4371             "type": "string"
4372         }
4373     },
4374     "aggregationWindowDays": {
4375         "type": "integer",
4376         "format": "int32"
4377     }
4378 }
4379 }
4380 ]
4381 },
4382 "WidgetDTO": {
4383     "discriminator": {
4384         "propertyName": "widgetType"
4385     },
4386     "properties": {
4387         "widgetType": {
4388             "type": "string"
4389         }
4390     },
4391     "required": [
4392         "widgetType"
4393     ]
4394 },
```

```
4395     "BillingKpiRequest": {
4396       "type": "object",
4397       "properties": {
4398         "chargeIds": {
4399           "type": "array",
4400           "items": {
4401             "type": "string"
4402           }
4403         },
4404         "from": {
4405           "type": "string"
4406         },
4407         "to": {
4408           "type": "string"
4409         },
4410         "aggregation": {
4411           "type": "string"
4412         }
4413       }
4414     },
4415     "BillingKpiResponse": {
4416       "type": "object",
4417       "properties": {
4418         "value": {
4419           "type": "number"
4420         },
4421         "currency": {
4422           "type": "string"
4423         },
4424         "selectedChargeCount": {
4425           "type": "integer",
4426           "format": "int32"
4427         },
4428         "timeLabel": {
4429           "type": "string"
4430         },
4431         "updatedAt": {
4432           "type": "string"
4433         },
4434         "previousValue": {
4435           "type": "number"
4436         }
4437       }
4438     },
```

```
4439     "BillingKpiChargesRequest": {
4440       "type": "object",
4441       "properties": {
4442         "from": {
4443           "type": "string"
4444         },
4445         "to": {
4446           "type": "string"
4447         }
4448       }
4449     },
4450     "BillingChargeResponse": {
4451       "type": "object",
4452       "properties": {
4453         "resourceId": {
4454           "type": "string"
4455         },
4456         "chargeId": {
4457           "type": "string"
4458         },
4459         "service": {
4460           "type": "string"
4461         },
4462         "provider": {
4463           "type": "string",
4464           "enum": [
4465             "AWS",
4466             "AZURE",
4467             "GCP"
4468           ]
4469         },
4470         "resourceName": {
4471           "type": [
4472             "string",
4473             "null"
4474           ]
4475         },
4476         "chargeCost": {
4477           "type": "number"
4478         },
4479         "metadata": {
4480           "type": "object",
4481           "additionalProperties": {
```

```
4483     }
4484   }
4485 }
4486 },
4487 "AzureBillingConfigDto": {
4488   "type": "object",
4489   "properties": {
4490     "storageAccountName": {
4491       "type": "string"
4492     },
4493     "blobContainerName": {
4494       "type": "string"
4495     },
4496     "exportDirectory": {
4497       "type": "string"
4498     },
4499     "exportName": {
4500       "type": "string"
4501     }
4502   }
4503 },
4504 "AzureCredentialsDto": {
4505   "type": "object",
4506   "properties": {
4507     "subscriptionId": {
4508       "type": "string"
4509     },
4510     "tenantId": {
4511       "type": "string"
4512     },
4513     "clientId": {
4514       "type": "string"
4515     },
4516     "clientSecret": {
4517       "type": "string"
4518     }
4519   }
4520 },
4521 "PersistAzureConnectionRequest": {
4522   "type": "object",
4523   "properties": {
4524     "userId": {
4525       "type": "string",
4526       "format": "uuid"
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4527     },
4528     "accountId": {
4529         "type": "string"
4530     },
4531     "displayName": {
4532         "type": "string"
4533     },
4534     "ingestionPeriod": {
4535         "type": "integer",
4536         "format": "int32"
4537     },
4538     "credentials": {
4539         "$ref": "#/components/schemas/AzureCredentialsDto"
4540     },
4541     "resources": {
4542         "type": "array",
4543         "items": {
4544             "$ref": "#/components/schemas/ResourceSelectionDto"
4545         }
4546     },
4547     "billingConfig": {
4548         "$ref": "#/components/schemas/AzureBillingConfigDto"
4549     }
4550 },
4551 "AwsCredentialsDto": {
4552     "type": "object",
4553     "properties": {
4554         "accessKeyId": {
4555             "type": "string"
4556         },
4557         "secretAccessKey": {
4558             "type": "string"
4559         }
4560     }
4561 },
4562 "BillingConfigDto": {
4563     "type": "object",
4564     "properties": {
4565         "bucketName": {
4566             "type": "string"
4567         },
4568         "bucketRegion": {
4569             "type": "string"
4570         }
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4571     },
4572     "exportPrefix": {
4573         "type": "string"
4574     },
4575     "exportName": {
4576         "type": "string"
4577     }
4578 }
4579 },
4580 "PersistAwsConnectionRequest": {
4581     "type": "object",
4582     "properties": {
4583         "userId": {
4584             "type": "string",
4585             "format": "uuid"
4586         },
4587         "accountId": {
4588             "type": "string"
4589         },
4590         "displayName": {
4591             "type": "string"
4592         },
4593         "ingestionPeriod": {
4594             "type": "integer",
4595             "format": "int32"
4596         },
4597         "credentials": {
4598             "$ref": "#/components/schemas/AwsCredentialsDto"
4599         },
4600         "resources": {
4601             "type": "array",
4602             "items": {
4603                 "$ref": "#/components/schemas/ResourceSelectionDto"
4604             }
4605         },
4606         "billingConfig": {
4607             "$ref": "#/components/schemas/BillingConfigDto"
4608         }
4609     }
4610 },
4611 "RegisterRequest": {
4612     "type": "object",
4613     "properties": {
4614         "email": {
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4615         "type": "string"
4616     },
4617     "username": {
4618         "type": "string"
4619     },
4620     "password": {
4621         "type": "string"
4622     }
4623 },
4624 },
4625 "AuthUserResponse": {
4626     "type": "object",
4627     "properties": {
4628         "userId": {
4629             "type": "string",
4630             "format": "uuid"
4631         },
4632         "email": {
4633             "type": "string"
4634         },
4635         "username": {
4636             "type": "string"
4637         }
4638     },
4639 },
4640 "LoginRequest": {
4641     "type": "object",
4642     "properties": {
4643         "email": {
4644             "type": "string"
4645         },
4646         "password": {
4647             "type": "string"
4648         }
4649     },
4650 },
4651 "CreateThresholdRequest": {
4652     "type": "object",
4653     "properties": {
4654         "resourceId": {
4655             "type": "string",
4656             "format": "uuid"
4657         },
4658         "userId": {
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4659         "type": "string",
4660         "format": "uuid"
4661     },
4662     "metric_name": {
4663         "type": "string"
4664     },
4665     "operator": {
4666         "type": "string"
4667     },
4668     "value": {
4669         "type": "number",
4670         "format": "double"
4671     },
4672     "severity": {
4673         "type": "string",
4674         "enum": [
4675             "WARNING",
4676             "CRITICAL"
4677         ]
4678     },
4679     "enabled": {
4680         "type": "boolean"
4681     }
4682 },
4683 "ThresholdResponse": {
4684     "type": "object",
4685     "properties": {
4686         "thresholdId": {
4687             "type": "string",
4688             "format": "uuid"
4689         },
4690         "resourceId": {
4691             "type": "string",
4692             "format": "uuid"
4693         },
4694         "userId": {
4695             "type": "string",
4696             "format": "uuid"
4697         },
4698         "metricName": {
4699             "type": "string"
4700         },
4701         "operator": {
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4703         "type": "string"
4704     },
4705     "value": {
4706         "type": "number",
4707         "format": "double"
4708     },
4709     "severity": {
4710         "type": "string",
4711         "enum": [
4712             "WARNING",
4713             "CRITICAL"
4714         ]
4715     },
4716     "enabled": {
4717         "type": "boolean"
4718     },
4719     "createdAt": {
4720         "type": "string",
4721         "format": "date-time"
4722     },
4723     "updatedAt": {
4724         "type": "string",
4725         "format": "date-time"
4726     }
4727 },
4728 "CloudCredentialsDto": {
4729     "type": "object",
4730     "properties": {
4731         "accessKeyId": {
4732             "type": "string"
4733         },
4734         "secretAccessKey": {
4735             "type": "string"
4736         },
4737         "awsRegion": {
4738             "type": "string"
4739         },
4740         "subscriptionId": {
4741             "type": "string"
4742         },
4743         "tenantId": {
4744             "type": "string"
4745         }
4746     },
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4747     "clientId": {
4748         "type": "string"
4749     },
4750     "clientSecret": {
4751         "type": "string"
4752     },
4753     "projectId": {
4754         "type": "string"
4755     },
4756     "serviceAccountJson": {
4757         "type": "string"
4758     }
4759 }
4760 },
4761 "ResourceDiscoveryDto": {
4762     "type": "object",
4763     "properties": {
4764         "services": {
4765             "type": "array",
4766             "items": {
4767                 "type": "string"
4768             }
4769         },
4770         "credentials": {
4771             "$ref": "#/components/schemas/CloudCredentialsDto"
4772         }
4773     }
4774 },
4775 "ResourceDetailDto": {
4776     "type": "object",
4777     "properties": {
4778         "resourceId": {
4779             "type": "string"
4780         },
4781         "name": {
4782             "type": "string"
4783         },
4784         "resourceType": {
4785             "type": "string"
4786         },
4787         "serviceCategory": {
4788             "type": "string"
4789         },
4790         "region": {
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4791         "type": "string"
4792     },
4793     "tags": {
4794         "type": "object",
4795         "additionalProperties": {
4796             "type": "string"
4797         }
4798     }
4799 },
4800 "CreateBudgetRequest": {
4801     "type": "object",
4802     "properties": {
4803         "userId": {
4804             "type": "string",
4805             "format": "uuid"
4806         },
4807         "scope": {
4808             "type": "string"
4809         },
4810         "scope_id": {
4811             "type": "string",
4812             "format": "uuid"
4813         },
4814         "amount": {
4815             "type": "number"
4816         },
4817         "currency": {
4818             "type": "string",
4819             "enum": [
4820                 "USD",
4821                 "EUR",
4822                 "ZAR"
4823             ]
4824         },
4825         "window_days": {
4826             "type": "integer",
4827             "format": "int32"
4828         },
4829         "enabled": {
4830             "type": "boolean"
4831         }
4832     }
4833 },
4834 },
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4835     "AlertNotificationSilenceRequest": {
4836         "type": "object",
4837         "properties": {
4838             "silenced": {
4839                 "type": "boolean"
4840             }
4841         }
4842     },
4843     "ResourceMetricHistoricalRequestDto": {
4844         "type": "object",
4845         "properties": {
4846             "resourceId": {
4847                 "type": "string",
4848                 "format": "uuid"
4849             },
4850             "metricType": {
4851                 "type": "string"
4852             },
4853             "fromDate": {
4854                 "type": "string",
4855                 "format": "date-time"
4856             }
4857         }
4858     },
4859     "ResourceMetricHistoricalResponseDto": {
4860         "type": "object",
4861         "properties": {
4862             "values": {
4863                 "type": "array",
4864                 "items": {
4865                     "type": "number"
4866                 }
4867             },
4868             "timestamps": {
4869                 "type": "array",
4870                 "items": {
4871                     "type": "string",
4872                     "format": "date-time"
4873                 }
4874             }
4875         }
4876     },
4877     "DownsampledSeriesRequestDto": {
4878         "type": "object",
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4879     "properties": {
4880       "resourceId": {
4881         "type": "string",
4882         "format": "uuid"
4883       },
4884       "metricName": {
4885         "type": "string"
4886       },
4887       "from": {
4888         "type": "string",
4889         "format": "date-time"
4890       },
4891       "to": {
4892         "type": "string",
4893         "format": "date-time"
4894       }
4895     },
4896   },
4897   "NormalizedMetrics": {
4898     "type": "object",
4899     "properties": {
4900       "metricId": {
4901         "type": "string",
4902         "format": "uuid"
4903       },
4904       "resourceId": {
4905         "type": "string",
4906         "format": "uuid"
4907       },
4908       "recordedAt": {
4909         "type": "string",
4910         "format": "date-time"
4911       },
4912       "metricType": {
4913         "type": "string"
4914       },
4915       "metricName": {
4916         "type": "string"
4917       },
4918       "metricValue": {
4919         "type": "number"
4920       },
4921       "unit": {
4922         "type": "string"
```

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4923     },
4924     "currency": {
4925         "type": "string"
4926     },
4927     "periodStart": {
4928         "type": "string",
4929         "format": "date-time"
4930     },
4931     "periodEnd": {
4932         "type": "string",
4933         "format": "date-time"
4934     },
4935     "metricPeriodStart": {
4936         "type": "string",
4937         "format": "date-time",
4938         "writeOnly": true
4939     },
4940     "metricPeriodEnd": {
4941         "type": "string",
4942         "format": "date-time",
4943         "writeOnly": true
4944     }
4945 },
4946 "AiSessionResponseDto": {
4947     "type": "object",
4948     "properties": {
4949         "sessionId": {
4950             "type": "string",
4951             "format": "uuid"
4952         },
4953         "createdAt": {
4954             "type": "string",
4955             "format": "date-time"
4956         },
4957         "lastActivity": {
4958             "type": "string",
4959             "format": "date-time"
4960         },
4961         "currentVersionId": {
4962             "type": "string",
4963             "format": "uuid"
4964         }
4965     }
4966 }
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4967     },
4968     "AiDashboardApplyRequestDto": {
4969         "type": "object",
4970         "properties": {
4971             "mode": {
4972                 "type": "string",
4973                 "enum": [
4974                     "REPLACE_STARTED_DASHBOARD",
4975                     "CREATE_NEW_DASHBOARD"
4976                 ]
4977             },
4978             "startedDashboardId": {
4979                 "type": "string",
4980                 "format": "uuid"
4981             }
4982         },
4983         "required": [
4984             "mode"
4985         ]
4986     },
4987     "AiVersionResponseDto": {
4988         "type": "object",
4989         "properties": {
4990             "versionId": {
4991                 "type": "string",
4992                 "format": "uuid"
4993             },
4994             "sessionId": {
4995                 "type": "string",
4996                 "format": "uuid"
4997             },
4998             "version": {
4999                 "type": "integer",
5000                 "format": "int32"
5001             },
5002             "parentVersionId": {
5003                 "type": "string",
5004                 "format": "uuid"
5005             },
5006             "current": {
5007                 "type": "boolean"
5008             },
5009             "createdAt": {
5010                 "type": "string",
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```
5011         "format": "date-time"
5012     },
5013     "dashboard": {
5014         "$ref": "#/components/schemas/DashboardPlanDto"
5015     }
5016 }
5017 },
5018 "DashboardPlanDto": {
5019     "type": "object",
5020     "properties": {
5021         "title": {
5022             "type": "string"
5023         },
5024         "description": {
5025             "type": "string"
5026         },
5027         "timeFrom": {
5028             "type": "string",
5029             "format": "date-time"
5030         },
5031         "timeTo": {
5032             "type": "string",
5033             "format": "date-time"
5034         },
5035         "predefinedTime": {
5036             "type": "string",
5037             "enum": [
5038                 "T_5_MIN",
5039                 "T_15_MIN",
5040                 "T_30_MIN",
5041                 "T_1_HOUR",
5042                 "T_6_HOUR",
5043                 "T_12_HOUR",
5044                 "T_24_HOUR",
5045                 "T_7_DAYS",
5046                 "T_30_DAYS",
5047                 "CUSTOM"
5048             ]
5049         },
5050         "widgets": {
5051             "type": "array",
5052             "items": {
5053                 "$ref": "#/components/schemas/DashboardPlanWidgetDto"
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```
5054     }
5055   }
5056 }
5057 },
5058 "DashboardPlanWidgetDto": {
5059   "type": "object",
5060   "properties": {
5061     "widgetId": {
5062       "type": "string",
5063       "format": "uuid"
5064     },
5065     "widgetType": {
5066       "type": "string",
5067       "enum": [
5068         "KPI",
5069         "CHART"
5070       ]
5071     },
5072     "displayName": {
5073       "type": "string"
5074     },
5075     "startX": {
5076       "type": "integer",
5077       "format": "int32"
5078     },
5079     "startY": {
5080       "type": "integer",
5081       "format": "int32"
5082     },
5083     "width": {
5084       "type": "integer",
5085       "format": "int32"
5086     },
5087     "height": {
5088       "type": "integer",
5089       "format": "int32"
5090     },
5091     "chartType": {
5092       "type": "string",
5093       "enum": [
5094         "gauge_chart",
5095         "line_chart"
5096       ]
5097     },
```

```
5098     "chartColour": {
5099         "type": "string",
5100         "enum": [
5101             "chart_1",
5102             "chart_2",
5103             "chart_3",
5104             "chart_4",
5105             "chart_5"
5106         ]
5107     },
5108     "provider": {
5109         "type": "string",
5110         "enum": [
5111             "AWS",
5112             "AZURE",
5113             "GCP"
5114         ]
5115     },
5116     "title": {
5117         "type": "string"
5118     },
5119     "accountId": {
5120         "type": "string",
5121         "format": "uuid"
5122     },
5123     "resourceId": {
5124         "type": "string",
5125         "format": "uuid"
5126     },
5127     "metricType": {
5128         "type": "string"
5129     },
5130     "metricName": {
5131         "type": "string"
5132     },
5133     "chargeIds": {
5134         "type": "array",
5135         "items": {
5136             "type": "string"
5137         }
5138     },
5139     "aggregationWindowDays": {
5140         "type": "integer",
5141         "format": "int32"
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5142     }
5143   }
5144 },
5145 "AiDashboardPlanRequestDto": {
5146   "type": "object",
5147   "properties": {
5148     "sessionId": {
5149       "type": "string",
5150       "format": "uuid"
5151     },
5152     "startingDashboardId": {
5153       "type": "string",
5154       "format": "uuid"
5155     },
5156     "message": {
5157       "type": "string",
5158       "minLength": 1
5159     }
5160   },
5161   "required": [
5162     "message",
5163     "sessionId",
5164     "startingDashboardId"
5165   ]
5166 },
5167 "AiDashboardPlanResponseDto": {
5168   "type": "object",
5169   "properties": {
5170     "sessionId": {
5171       "type": "string",
5172       "format": "uuid"
5173     },
5174     "versionId": {
5175       "type": "string",
5176       "format": "uuid"
5177     },
5178     "version": {
5179       "type": "integer",
5180       "format": "int32"
5181     },
5182     "stageAttempted": {
5183       "type": "boolean"
5184     },
5185     "stageSucceeded": {
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5186         "type": "boolean"
5187     },
5188     "assistantMessage": {
5189         "type": "string"
5190     },
5191     "dashboard": {
5192         "$ref": "#/components/schemas/DashboardPlanDto"
5193     }
5194 },
5195 },
5196 "DashboardWindowUpdateDTO": {
5197     "type": "object",
5198     "properties": {
5199         "newTime": {
5200             "type": "string",
5201             "enum": [
5202                 "T_5_MIN",
5203                 "T_15_MIN",
5204                 "T_30_MIN",
5205                 "T_1_HOUR",
5206                 "T_6_HOUR",
5207                 "T_12_HOUR",
5208                 "T_24_HOUR",
5209                 "T_7_DAYS",
5210                 "T_30_DAYS",
5211                 "CUSTOM"
5212             ]
5213         },
5214         "from": {
5215             "type": "string",
5216             "format": "date-time"
5217         },
5218         "to": {
5219             "type": "string",
5220             "format": "date-time"
5221         }
5222     }
5223 },
5224 "ChartWidgetConfigUpdateDTO": {
5225     "allOf": [
5226         {
5227             "$ref": "#/components/schemas/WidgetConfigUpdateDTO"
5228         },
5229         {
```

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5230     "type": "object",
5231     "properties": {
5232       "id": {
5233         "type": "string",
5234         "format": "uuid"
5235       },
5236       "widgetType": {
5237         "type": "string",
5238         "enum": [
5239           "KPI",
5240           "CHART"
5241         ]
5242       },
5243       "displayName": {
5244         "type": "string"
5245       },
5246       "chartType": {
5247         "type": "string",
5248         "enum": [
5249           "gauge_chart",
5250           "line_chart"
5251         ]
5252       },
5253       "chartColour": {
5254         "type": "string",
5255         "enum": [
5256           "chart_1",
5257           "chart_2",
5258           "chart_3",
5259           "chart_4",
5260           "chart_5"
5261         ]
5262       },
5263       "provider": {
5264         "type": "string",
5265         "enum": [
5266           "AWS",
5267           "AZURE",
5268           "GCP"
5269         ]
5270       },
5271       "accountId": {
5272         "type": "string",
5273         "format": "uuid"
```

```
5274         },
5275         "resourceId": {
5276             "type": "string",
5277             "format": "uuid"
5278         },
5279         "metricType": {
5280             "type": "string"
5281         },
5282         "metricName": {
5283             "type": "string"
5284         }
5285     }
5286 }
5287 ]
5288 },
5289 "KpiWidgetConfigUpdateDTO": {
5290     "allOf": [
5291         {
5292             "$ref": "#/components/schemas/WidgetConfigUpdateDTO"
5293         },
5294         {
5295             "type": "object",
5296             "properties": {
5297                 "id": {
5298                     "type": "string",
5299                     "format": "uuid"
5300                 },
5301                 "widgetType": {
5302                     "type": "string",
5303                     "enum": [
5304                         "KPI",
5305                         "CHART"
5306                     ]
5307                 },
5308                 "displayName": {
5309                     "type": "string"
5310                 },
5311                 "chargeIds": {
5312                     "type": "array",
5313                     "items": {
5314                         "type": "string"
5315                     }
5316                 },
5317                 "aggregationWindowDays": {
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5318         "type": "integer",
5319         "format": "int32"
5320     }
5321 }
5322 }
5323 ]
5324 },
5325 "WidgetConfigUpdatedDTO": {
5326     "discriminator": {
5327         "propertyName": "widgetType"
5328     },
5329     "properties": {
5330         "widgetType": {
5331             "type": "string"
5332         }
5333     },
5334     "required": [
5335         "widgetType"
5336     ]
5337 },
5338 "UpdateResourceStatusRequest": {
5339     "type": "object",
5340     "properties": {
5341         "status": {
5342             "type": "string",
5343             "enum": [
5344                 "active",
5345                 "disabled"
5346             ]
5347         }
5348     }
5349 },
5350 "UpdateAccountNameRequest": {
5351     "type": "object",
5352     "properties": {
5353         "name": {
5354             "type": "string"
5355         }
5356     }
5357 },
5358 "WebhookResponse": {
5359     "type": "object",
5360     "properties": {
5361         "webhookId": {
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5362         "type": "string",
5363         "format": "uuid"
5364     },
5365     "webhookName": {
5366         "type": "string"
5367     },
5368     "endpointUrl": {
5369         "type": "string"
5370     },
5371     "eventTypes": {
5372         "type": "array",
5373         "items": {
5374             "type": "string"
5375         }
5376     },
5377     "cloudAccounts": {
5378         "type": "array",
5379         "items": {
5380             "type": "string",
5381             "format": "uuid"
5382         }
5383     },
5384     "webhookStatus": {
5385         "type": "string",
5386         "enum": [
5387             "ACTIVE",
5388             "PAUSED"
5389         ]
5390     }
5391 },
5392 "PagedWebhookDeliveryResponse": {
5393     "type": "object",
5394     "properties": {
5395         "deliveries": {
5396             "type": "array",
5397             "items": {
5398                 "$ref": "#/components/schemas/
5399                 WebhookDeliveryResponse"
5400             }
5401         },
5402         "page": {
5403             "type": "integer",
5404             "format": "int32"
```

```
5405     },
5406     "size": {
5407         "type": "integer",
5408         "format": "int32"
5409     },
5410     "totalElements": {
5411         "type": "integer",
5412         "format": "int64"
5413     },
5414     "totalPages": {
5415         "type": "integer",
5416         "format": "int32"
5417     }
5418 }
5419 },
5420 "WebhookDeliveryResponse": {
5421     "type": "object",
5422     "properties": {
5423         "deliveryId": {
5424             "type": "string",
5425             "format": "uuid"
5426         },
5427         "timestamp": {
5428             "type": "string",
5429             "format": "date-time"
5430         },
5431         "webhookId": {
5432             "type": "string",
5433             "format": "uuid"
5434         },
5435         "eventType": {
5436             "type": "string"
5437         },
5438         "cloudAccountId": {
5439             "type": "string",
5440             "format": "uuid"
5441         },
5442         "cloudAccountName": {
5443             "type": "string"
5444         },
5445         "result": {
5446             "type": "string",
5447             "enum": [
5448                 "PENDING",
```

```
5449         "PROCESSING",
5450         "DELIVERED",
5451         "FAILED"
5452     ]
5453 },
5454 "responseCode": {
5455     "type": "integer",
5456     "format": "int32"
5457 },
5458 "nextRetryAttempt": {
5459     "type": "string",
5460     "format": "date-time"
5461 }
5462 }
5463 },
5464 "SseEmitter": {
5465     "type": "object",
5466     "properties": {
5467         "timeout": {
5468             "type": [
5469                 "integer",
5470                 "null"
5471             ],
5472             "format": "int64"
5473         }
5474     }
5475 },
5476 "TimestampedNumericDataPoint": {
5477     "type": "object",
5478     "properties": {
5479         "value": {
5480             "type": "number"
5481         },
5482         "timestamp": {
5483             "type": "string",
5484             "format": "date-time"
5485         }
5486     }
5487 },
5488 "CloudAccount": {
5489     "type": "object",
5490     "properties": {
5491         "id": {
5492             "type": "string",
```

```
5493         "format": "uuid"
5494     },
5495     "connectionId": {
5496         "type": "string",
5497         "format": "uuid"
5498     },
5499     "connection": {
5500         "$ref": "#/components/schemas/CloudConnection"
5501     },
5502     "accountType": {
5503         "type": "string",
5504         "enum": [
5505             "aws_account",
5506             "azure_subscription",
5507             "gcp_project"
5508         ]
5509     },
5510     "displayName": {
5511         "type": "string"
5512     },
5513     "ingestionPeriod": {
5514         "type": "string"
5515     },
5516     "createdAt": {
5517         "type": "string",
5518         "format": "date-time"
5519     },
5520     "lastUsageIngestion": {
5521         "type": "string",
5522         "format": "date-time"
5523     },
5524     "nextUsageIngestion": {
5525         "type": "string",
5526         "format": "date-time"
5527     },
5528     "lastBillingIngestion": {
5529         "type": "string",
5530         "format": "date-time"
5531     },
5532     "nextBillingIngestion": {
5533         "type": "string",
5534         "format": "date-time"
5535     },
5536     "billingConfigured": {
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```
5537         "type": "boolean"
5538     }
5539 }
5540 },
5541 "CloudConnection": {
5542     "type": "object",
5543     "properties": {
5544         "id": {
5545             "type": "string",
5546             "format": "uuid"
5547         },
5548         "userId": {
5549             "type": "string",
5550             "format": "uuid"
5551         },
5552         "user": {
5553             "$ref": "#/components/schemas/User"
5554         },
5555         "provider": {
5556             "type": "string",
5557             "enum": [
5558                 "AWS",
5559                 "AZURE",
5560                 "GCP"
5561             ]
5562         },
5563         "status": {
5564             "type": "string",
5565             "enum": [
5566                 "active",
5567                 "disabled"
5568             ]
5569         }
5570     }
5571 },
5572 "User": {
5573     "type": "object",
5574     "properties": {
5575         "id": {
5576             "type": "string",
5577             "format": "uuid"
5578         },
5579         "email": {
5580             "type": "string"
```

```
5581     },
5582     "username": {
5583         "type": "string"
5584     },
5585     "passwordHash": {
5586         "type": "string"
5587     },
5588     "createdAt": {
5589         "type": "string",
5590         "format": "date-time"
5591     }
5592 }
5593 },
5594 "ResourceCountResponse": {
5595     "type": "object",
5596     "properties": {
5597         "count": {
5598             "type": "integer",
5599             "format": "int64"
5600         }
5601     }
5602 },
5603 "Resource": {
5604     "type": "object",
5605     "properties": {
5606         "id": {
5607             "type": "string",
5608             "format": "uuid"
5609         },
5610         "accountId": {
5611             "type": "string",
5612             "format": "uuid"
5613         },
5614         "account": {
5615             "$ref": "#/components/schemas/CloudAccount"
5616         },
5617         "resourceType": {
5618             "type": "string"
5619         },
5620         "resourceName": {
5621             "type": "string"
5622         },
5623         "resourceIdentifier": {
5624             "type": "string"
```

```
5625     },
5626     "resourceIdentifierType": {
5627         "type": "string"
5628     },
5629     "region": {
5630         "type": "string"
5631     },
5632     "status": {
5633         "type": "string",
5634         "enum": [
5635             "active",
5636             "disabled"
5637         ]
5638     },
5639     "tags": {
5640         "type": "object",
5641         "additionalProperties": {
5642
5643         }
5644     },
5645     "lastUpdated": {
5646         "type": "string",
5647         "format": "date-time"
5648     },
5649     "createdAt": {
5650         "type": "string",
5651         "format": "date-time"
5652     }
5653 },
5654 "CloudAccountDetailsResponse": {
5655     "type": "object",
5656     "properties": {
5657         "id": {
5658             "type": "string",
5659             "format": "uuid"
5660         },
5661         "displayName": {
5662             "type": "string"
5663         },
5664         "accountType": {
5665             "type": "string",
5666             "enum": [
5667                 "aws_account",
```

```
5669         "azure_subscription",
5670         "gcp_project"
5671     ]
5672 },
5673 "accountEmail": {
5674     "type": "string"
5675 },
5676 "ingestionPeriod": {
5677     "type": "string"
5678 },
5679 "createdAt": {
5680     "type": "string",
5681     "format": "date-time"
5682 }
5683 },
5684 },
5685 "AlertResponse": {
5686     "type": "object",
5687     "properties": {
5688         "alertId": {
5689             "type": "string",
5690             "format": "uuid"
5691         },
5692         "userId": {
5693             "type": "string",
5694             "format": "uuid"
5695         },
5696         "alertType": {
5697             "type": "string",
5698             "enum": [
5699                 "THRESHOLD",
5700                 "BUDGET",
5701                 "ANOMALY"
5702             ]
5703         },
5704         "severity": {
5705             "type": "string",
5706             "enum": [
5707                 "WARNING",
5708                 "CRITICAL"
5709             ]
5710         },
5711         "title": {
5712             "type": "string"
```

```
5713     },
5714     "message": {
5715         "type": "string"
5716     },
5717     "payload": {
5718         "type": "object",
5719         "additionalProperties": {
5720
5721         }
5722     },
5723     "status": {
5724         "type": "string",
5725         "enum": [
5726             "ACTIVE",
5727             "DISABLED"
5728         ]
5729     },
5730     "inAppNotificationsSilenced": {
5731         "type": "boolean"
5732     },
5733     "canonicalKey": {
5734         "type": "string"
5735     },
5736     "createdAt": {
5737         "type": "string",
5738         "format": "date-time"
5739     },
5740     "lastSeen": {
5741         "type": "string",
5742         "format": "date-time"
5743     },
5744     "resolvedAt": {
5745         "type": "string",
5746         "format": "date-time"
5747     }
5748 }
5749 },
5750 "ResourceNameDto": {
5751     "type": "object",
5752     "properties": {
5753         "resourceId": {
5754             "type": "string"
5755         },
5756         "resourceName": {
```

```
5757         "type": "string"
5758     }
5759 }
5760 },
5761 "ResourceMetric": {
5762     "type": "object",
5763     "properties": {
5764         "metricName": {
5765             "type": "string"
5766         },
5767         "metricType": {
5768             "type": "string"
5769         }
5770     }
5771 },
5772 "ResourceMetricsGroupDto": {
5773     "type": "object",
5774     "properties": {
5775         "resourceId": {
5776             "type": "string",
5777             "format": "uuid"
5778         },
5779         "metrics": {
5780             "type": "array",
5781             "items": {
5782                 "$ref": "#/components/schemas/ResourceMetric"
5783             }
5784         }
5785     }
5786 },
5787 "AggregatedMetric": {
5788     "type": "object",
5789     "properties": {
5790         "metricName": {
5791             "type": "string"
5792         },
5793         "resourceId": {
5794             "type": "string",
5795             "format": "uuid"
5796         },
5797         "metricType": {
5798             "type": "string"
5799         },
5800         "sampleCount": {
```

```
5801         "type": "integer",
5802         "format": "int64"
5803     },
5804     "metricValue": {
5805         "type": "number"
5806     },
5807     "unit": {
5808         "type": "string"
5809     },
5810     "periodStart": {
5811         "type": "string",
5812         "format": "date-time"
5813     },
5814     "periodEnd": {
5815         "type": "string",
5816         "format": "date-time"
5817     }
5818 },
5819 },
5820 "MetricDto": {
5821     "type": "object",
5822     "properties": {
5823         "resourceId": {
5824             "type": "string",
5825             "format": "uuid"
5826         },
5827         "metricType": {
5828             "type": "string"
5829         },
5830         "metricName": {
5831             "type": "string"
5832         },
5833         "metricValue": {
5834             "type": "number"
5835         },
5836         "unit": {
5837             "type": "string"
5838         },
5839         "periodStart": {
5840             "type": "string",
5841             "format": "date-time"
5842         },
5843         "periodEnd": {
5844             "type": "string",
```

```

5845         "format": "date-time"
5846     },
5847     "sampleCount": {
5848         "type": "integer",
5849         "format": "int64"
5850     }
5851 }
5852 },
5853 "AiVersionSummaryDto": {
5854     "type": "object",
5855     "properties": {
5856         "versionId": {
5857             "type": "string",
5858             "format": "uuid"
5859         },
5860         "version": {
5861             "type": "integer",
5862             "format": "int32"
5863         },
5864         "parentVersionId": {
5865             "type": "string",
5866             "format": "uuid"
5867         },
5868         "title": {
5869             "type": "string"
5870         },
5871         "createdAt": {
5872             "type": "string",
5873             "format": "date-time"
5874         },
5875         "current": {
5876             "type": "boolean"
5877         }
5878     }
5879 }
5880 }
5881 }
5882 }

```

B.3 Intelligence

1 {

```
2  "openapi": "3.1.0",
3  "info": {
4    "title": "FastAPI",
5    "version": "0.1.0"
6  },
7  "paths": {
8    "/forecast-chronos": {
9      "post": {
10        "summary": "Root",
11        "operationId": "root_forecast_chronos_post",
12        "requestBody": {
13          "content": {
14            "application/json": {
15              "schema": {
16                "$ref": "#/components/schemas/ForecastRequest"
17              }
18            }
19          },
20          "required": true
21        },
22        "responses": {
23          "200": {
24            "description": "Successful Response",
25            "content": {
26              "application/json": {
27                "schema": {
28                  "$ref": "#/components/schemas/ChronosForecastResponse"
29                }
30              }
31            }
32          },
33          "422": {
34            "description": "Validation Error",
35            "content": {
36              "application/json": {
37                "schema": {
38                  "$ref": "#/components/schemas/HTTPValidationError"
39                }
40              }
41            }
42          }
43        }
44      }
45    }
46  }
```

```
44     "security": [  
45         {  
46             "APIKeyHeader": []  
47         }  
48     ]  
49 }  
50 },  
51 "/health": {  
52     "get": {  
53         "summary": "Health",  
54         "operationId": "health_health_get",  
55         "responses": {  
56             "200": {  
57                 "description": "Successful Response",  
58                 "content": {  
59                     "application/json": {  
60                         "schema": {  
61                             }  
62                         }  
63                     }  
64                 }  
65             }  
66         }  
67     }  
68 }  
69 },  
70 "components": {  
71     "schemas": {  
72         "ChronosForecastResponse": {  
73             "properties": {  
74                 "forecast": {  
75                     "items": {  
76                         "type": "number"  
77                     },  
78                     "type": "array",  
79                     "title": "Forecast"  
80                 },  
81                 "timestamps": {  
82                     "items": {  
83                         "type": "string"  
84                     },  
85                     "type": "array",  
86                     "title": "Timestamps"  
87                 },  
88             }  
89         }  
90     }  
91 }
```

```
88     "q1": {
89         "items": {
90             "type": "number"
91         },
92         "type": "array",
93         "title": "Q1"
94     },
95     "q3": {
96         "items": {
97             "type": "number"
98         },
99         "type": "array",
100         "title": "Q3"
101     }
102 },
103 "type": "object",
104 "required": [
105     "forecast",
106     "timestamps",
107     "q1",
108     "q3"
109 ],
110 "title": "ChronosForecastResponse"
111 },
112 "ForecastRequest": {
113     "properties": {
114         "forecast_horizon": {
115             "type": "integer",
116             "title": "Forecast Horizon"
117         },
118         "timestamps": {
119             "items": {
120                 "type": "string"
121             },
122             "type": "array",
123             "title": "Timestamps"
124         },
125         "values": {
126             "items": {
127                 "type": "number"
128             },
129             "type": "array",
130             "title": "Values"
131         },
132     },
133     "type": "object",
134     "title": "Forecast Request"
```

```
132     "model": {
133       "anyOf": [
134         {
135           "type": "string"
136         },
137         {
138           "type": "null"
139         }
140       ],
141       "title": "Model"
142     },
143   },
144   "type": "object",
145   "required": [
146     "forecast_horizon",
147     "timestamps",
148     "values"
149   ],
150   "title": "ForecastRequest"
151 },
152 "HTTPValidationError": {
153   "properties": {
154     "detail": {
155       "items": {
156         "$ref": "#/components/schemas/ValidationError"
157       },
158       "type": "array",
159       "title": "Detail"
160     }
161   },
162   "type": "object",
163   "title": "HTTPValidationError"
164 },
165 "ValidationError": {
166   "properties": {
167     "loc": {
168       "items": {
169         "anyOf": [
170           {
171             "type": "string"
172           },
173           {
174             "type": "integer"
175           }
176         ]
177       }
178     }
179   }
180 }
```

```
176         ]
177     },
178     "type": "array",
179     "title": "Location"
180 },
181 "msg": {
182     "type": "string",
183     "title": "Message"
184 },
185 "type": {
186     "type": "string",
187     "title": "Error Type"
188 },
189 "input": {
190     "title": "Input"
191 },
192 "ctx": {
193     "type": "object",
194     "title": "Context"
195 }
196 },
197 "type": "object",
198 "required": [
199     "loc",
200     "msg",
201     "type"
202 ],
203 "title": "ValidationError"
204 }
205 },
206 "securitySchemes": {
207     "APIKeyHeader": {
208         "type": "apiKey",
209         "in": "header",
210         "name": "X-API-Key"
211     }
212 }
213 }
214 }
```